

	Flight Dispatch Manual and Procedures
	CONTENTS

TABLE OF CONTENTS

TABLE OF CONTENTS	1
RECORD OF AMENDMENTS	4
LIST OF EFFECTIVE PAGES	6

SECTION 1. INTRODUCTION

1.1	This Manual	1
1.2	Operations Manual	1
1.3	Procedure Updates	1
1.4	Flight Dispatch General Directory	2
1.5	Dispatch White Boards	2
1.6	Abbreviations	3

SECTION 2. DISPATCH ADMINISTRATION

2.1	Organization Structure	1
2.2	Dispatch Leave Policy	1
2.3	Time Keeping	2
2.4	Dispatch Roster	2
2.5	International Telephone Calls	2
2.6	Office Etiquette	3

SECTION 3. DOCUMENTATION, RECORD RETENTION & SYSTEMS

3.1	General	1
3.2	Documents Available in Flight Dispatch	1
3.3	Record Retention	1
3.4	Automated Systems	2

SECTION 4. QUALIFICATION & RESPONSIBILITIES

4.1	General	1
4.2	Training	1
4.3	Qualification	1
4.4	Responsibilities	2
4.4.1	Supervisor Flight Dispatch	2
4.4.2	Flight Dispatch Training Supervisor	4
4.4.3	Flight Dispatch Officer	4
4.5	Appendix A - Recurrent Training Syllabus	6

SECTION 5. DISPATCH GENERAL

5.1	Shift Procedures General	1
5.2	Shift Hand Over (FDP-01)	1
5.3	Message Processing	2
5.4	Arrival/En-route Traffic Delay (FDP-02)	2
5.5	Air TEST & Local Training Flights (FDP-03)	2
5.6	Critical Sheets & Daily Report	3
5.7	Backup System & Historic Flight Plan (FDP-04)	3
5.8	Special Weather & Storm Warning (FDP-05)	3
5.9	Gulf to London Operations - Night Curfew	4
5.10	Systems Failure (FDP-06)	4
5.11	Non-Standard Configurations Flight Planning (FDP-07)	4

Date:	01/04/08	Page	1 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	CONTENTS

5.12	Emergencies (FDP-08)	5
5.13	Communications (FDP-09)	5
5.14A	Appendix A - AD HOC FLIGHT CHECKLIST	8
5.14B	Appendix B – ACARS VHF Coverage Maps	9

SECTION 6. FLIGHT DISPATCH DAILY FUNCTION

6.1	Flight Plan Input sheets Preparation (FDP-10)	1
6.2	Alternates Display on OFP (FDP-11)	1
6.3	Preparation of OFP Request (FDP-11)	2
6.4	ATS Flight Plan (FDP-12)	2
6.5	Non-Standard Flights (FDP-13)	3
6.5.1	Gulf to/from Europe Operations (FDP-13.1)	3
6.5.2	Gulf to/from Far East Operations (FDP-13.2)	4
6.5.3	Kathmandu Operations with B767/A340 (FDP-13.3)	4
6.5.4	Sanaa Operations with A320 (FDP-13.4)	5
6.5.5	Khartoum Fuel Situation (FDP-13.5)	5
6.5.6	Handling Diversion Flights (FDP-13.6)	5
6.5.7	Fuel Police Change for A330/A340 (FDP-13.7)	5
6.6	Fuel Tankering Consideration (FDP-14)	6
6.7	Handling Payload Restriction (FDP-15)	7
6.8	Fuel Docket	7

SECTION 7. ALL FLEET OPERATIONS

7.1	Airport Operating Minimums (AOM)	1
7.2	Take-Off Alternate Airport Requirement (FDP-11)	1
7.3	Destination & Destination Alternates Planning Minima Policy. FDP-11	1
7.4	Destinations & Alternates Weather Minima Guidance Table	1
7.5	Weather Information Sources & Meteorological Interpretation	2
7.6	Rescue and Fire Fighting (RFF) Categories	3
7.7	MEL/CDL Affecting Operations (FDP-16)	3
7.8	RNP, RVSM & TCAS II Requirements (FDP-16)	3
7.9	High Speed and Maximum Range OFP (FDP-17)	4
7.10	Flight Plan Wind Shear	5
7.11	Appendix A-Destination/Alternates WX Minima Guidance Table	6

SECTION 8. A320 OPERATIONS

8.1	Operations to Indian Sub Continent	1
8.2	Payload Consideration to Indian Sub Continent (FDP-14)	1
8.3	Sanaa Operation (FDP-13.4)	1
8.4	Taxi-in Fuel	2

SECTION 9. B767 OPERATIONS

9.1	B767 Range with Centre Tank unusable	1
9.2	Kathmandu Operations (FDP-13.3)	1
9.3	B767 Taxi-in Fuel	1

SECTION 10. A340/A330 OPERATIONS

10.1	A340/A330 Nose Wheel Steering (NWS) Limitation	1
10.2	Taxi-in Fuel	2
10.3	Contingency Fuel Policy Change	2

Date:	01/04/08	Page	2 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	CONTENTS

SECTION 11 ETOPS (FDP-18)

11.1	General	1
11.2	Definitions	1
11.3	Gulf Air ETOPS Fleet	3
11.4	ETOPS En-route Weather Minima	4
11.5	ETOPS Flight Planning	4
11.6	Appendix A - ETOPS ERA Weather Minima Guidance Table	4

SECTIONS 12. FLIGHT DISPATCH PROCEDURES

12.1	Overview	1
FDP-01	Shift Hand Over	3
FDP-02	Arrival and En-route Traffic Delay	5
FDP-03	Air Test & Local Training Flights	11
FDP-04	Backup System & Historic Flight Plan	13
FDP-05	Special Weather & Volcanic Activities	17
FDP-06	System Failure	21
FDP-07	Non-Standard Configuration Flight Planning	23
FDP-08	Emergency Response	25
FDP-09	Communication	27
FDP-10	Flight Plan Input Sheets Preparation	33
FDP-11	OFP Request Procedure	39
FDP-12	ATS Flight Plan	53
FDP-13	Non-Standard Flights	55
FDP-13.1	Gulf to/from Europe Operations	56
FDP-13.2	Gulf to/from Far East	57
FDP-13.3	Kathmandu Operations with B767	58
FDP-13.4	Sanaa Operations with A320	60
FDP-13.5	Khartoum Fuel Situation	61
FDP-13.6	Diversion Flights	62
FDP-13.7	Fuel Policy Change for A330/A340 on Approved Sectors	64
FDP-13.8	TOW Calculation Program	65
FDP-14	Fuel Tankering Procedures	67
FDP-15	Handling Payload Restrictions	70
FDP-16	MEL/CDL Affecting Operations	72
FDP-17	High Speed and Maximum Range OFP	76
FDP-18	ETOPS Procedure	80
FDP-19	Pilot Self Services	87
FDP-20	Mfs2 Warning	90

Date:	01/04/08	Page	3 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	CONTENTS

LIST OF EFFECTIVE PAGES – FLIGHT DISPATCH MANUAL AND PROCEDURES (FDMP)

ITEM	DATE	REV	PAGE	ITEM	DATE	REV	PAGE
Contents	01/04/08	0	1/3	5.14A	01/04/08	0	8
Revision	01/04/08	0	4	5.14B	01/04/08	0	9/10
LEP	01/04/08	0	5/6	6.1	01/04/08	0	1
1.1	01/04/08	0	1	6.2	01/04/08	0	1
1.2	01/04/08	0	1	6.3	01/04/08	0	1/2
1.3	01/04/08	0	1	6.4	01/04/08	0	2
1.4	01/04/08	0	2	6.5	01/04/08	0	2
1.5	01/04/08	0	2	6.5.1	01/04/08	0	3
1.6	01/04/08	0	3/4	6.5.2	01/04/08	0	3
2.1	01/04/08	0	1	6.5.3	01/04/08	0	4
2.2	01/04/08	0	1/2	6.5.4	01/04/08	0	4
2.3	01/04/08	0	2	6.5.5	01/04/08	0	4
2.4	01/04/08	0	2	6.5.6	01/04/08	0	4
2.5	01/04/08	0	2	6.5.7	01/04/08	0	4
2.6	01/04/08	0	3	6.6	01/04/08	0	4/5
3.1	01/04/08	0	1	6.7	01/04/08	0	5
3.2	01/04/08	0	1	6.8	01/04/08	0	6
3.3	01/04/08	0	1	7.1	01/04/08	0	1
3.4	01/04/08	0	2	7.2	01/04/08	0	1
4.1	01/04/08	0	1	7.3	01/04/08	0	1
4.2	01/04/08	0	1	7.4	01/04/08	0	1/2
4.3	01/04/08	0	1/2	7.5	01/04/08	0	2
4.4.1	01/04/08	0	2/4	7.6	01/04/08	0	3
4.4.2	01/04/08	0	4	7.7	01/04/08	0	3
4.4.3	01/04/08	0	4/5	7.8	01/04/08	0	3/4
4.5	01/04/08	0	6	7.9	01/04/08	0	4/5
5.1	01/04/08	0	1	7.10	01/04/08	0	5/6
5.2	01/04/08	0	1	7.11	01/04/08	0	7/24
5.3	01/04/08	0	2	8.1	01/04/08	0	1
5.4	01/04/08	0	2	8.2	01/04/08	0	1
5.5	01/04/08	0	3	8.3	01/04/08	0	1
5.6	01/04/08	0	3	8.4	01/04/08	0	2
5.7	01/04/08	0	3	9.1	01/04/08	0	1
5.8	01/04/08	0	3/4	9.2	01/04/08	0	1
5.9	01/04/08	0	4	9.3	01/04/08	0	1
5.10	01/04/08	0	4	10.1	01/04/08	0	1
5.11	01/04/08	0	4	10.2	01/04/08	0	1
5.12	01/04/08	0	5	10.3	01/04/08	0	1
5.13	01/04/08	0	5/7				

Date:	01/04/08	Page	5 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	CONTENTS

LIST OF EFFECTIVE PAGES – FDMP

ITEM	DATE	REV	PAGE	ITEM	DATE	REV	PAGE
11.1	01/04/08	0	1	FDP-14	01/04/08	0	67/70
11.2	01/04/08	0	1/3	FDP-15	01/04/08	0	71/72
11.3	01/04/08	0	3/4	FDP-16	01/04/08	0	73/76
11.4	01/04/08	0	4	FDP-17	01/04/08	0	77/80
11.5	01/04/08	0	4	FDP-18	01/04/08	0	81/86
11.6	01/04/08	0	4/10	FDP-19	01/04/08	0	87/88
12.1	01/04/08	0	1	FDP-20	01/04/08	0	89/90
FDP-01	01/04/08	0	3/4				
FDP-02	01/04/08	0	5/10				
FDP-03	01/04/08	0	11/12				
FDP-04	01/04/08	0	13/16				
FDP-05	01/04/08	0	17/20				
FDP-06	01/04/08	0	21/22				
FDP-07	01/04/08	0	23/24				
FDP-08	01/04/08	0	25/26				
FDP-09	01/04/08	0	27/32				
FDP-10	01/04/08	0	33/38				
FDP-11	01/04/08	0	39/52				
FDP-12	01/04/08	0	53/54				
FDP-13	01/04/08	0	55/66				
FDP-13.1	01/04/08	0	56/56				
FDP-13.2	01/04/08	0	57/57				
FDP-13.3	01/04/08	0	58/59				
FDP-13.4	01/04/08	0	60/60				
FDP-13.5	01/04/08	0	61/61				
FDP-13.6	01/04/08	0	62/63				
FDP-13.7	01/04/08	0	64/64				
FDP-13.8	01/04/08	0	65/66				

Date:	01/04/08	Page	6 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	INTRODUCTION

1.0 INTRODUCTION

1.1 THIS MANUAL

This manual is the Flight Dispatch Manual and Procedures (FDMP), and contains guidelines for Flight Dispatch to carry out 'day to day' flight planning tasks. The business process of the department is in conjunction with the automated Flight Planning system: MFS2 Phoenix. There are multiple references to this system, with data input keywords. It is issued on the authority of Manager Flight Dispatch (MFD).

It is to be used in conjunction with following:

- Operations Manuals (OM)
- CAR-OPS1 of DGCAM Bahrain
- Aeroplane Flight Manuals
- JEPPESEN Airway Manuals.
- Dispatch Procedure Updates (Memo)
- Instructions on input sheets
- Fleet Office Notices related to dispatch work
- Aeronautical Information Circulars / Supplements (NOTAM)

Dispatchers must be fully aware of the contents of these manuals and know where to find any procedures or data required for Gulf Air flight planning.

The Shift Supervisor or FDP is to be consulted whenever a situation arises for which no specific instruction has been issued.

1.2 Operations Manual (OM)

This manual contains instructions relating to the Administration, Organisation and Policy for the Operations Division as approved by BAH-CAA. There is a general section and an Appendix for each aircraft type.

Subjects of interest to Dispatch found in this manual are: Operations Manual Part A, Section 8 containing such information as: Airport Operating Minima (AOM), details of Fuel Policy, Operational Flight Plan (OFF) explanation, ETOPS, etc.

1.3 Procedure Updates

A 'FDMP Update' issued by MFD will normally advise new instructions or changes to any of the Flight Dispatch Manual and Procedures.

Temporary new procedures and/or revisions to FDMP that lasts less than 6 months will be issued as bulletins and distributed and filed in Flight Dispatch share folder.

A permanent new and/or revised procedures that lasts 6 months or more, will be issued as a revision to the existing 'FDMP / FDP' and incorporated in the manual.

Date:	01/04/08	Page	1 of 4	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	INTRODUCTION

1.4 Flight Dispatch General Directory

This directory contains reference information prepared by other sections for Dispatch use, such as fuel indices, VHF station frequencies, telephone lists etc.

It will also contain copies of Notices to Crew or memos issued by Fleet Management when they contain Dispatch related information. This information will normally be circulated via e-mail and then placed in the General Directory under the 'NOTICES / MEMOS' tab, for easy reference.

1.5 Dispatch White Boards

The large Dispatch white board is used to advise of 'Warnings' such as Tropical Cyclone / Volcanic Activity / Military Exercise / MEL / CDL restrictions and special requirements for filing ATS plans.

The small board near the attendance sheets is for administrative notes or to advice of other short-term activities affecting GF operations.

Date:	01/04/08	Page	2 of 4	FDP
Authority:	MFD	Issue	3	Rev 00

1.6 ABBREVIATIONS

ACARS	Airborne Communications Addressing and Reporting System
ACC	Area Control Centre
ATFCM	Air Traffic Flow and Capacity Management
AFM	Aircraft Flight Manual
AFTN	Aeronautical Fixed Telecommunication Network
AGL	Above Ground Level
AIC	Aeronautical Information Circular
AIP	Aeronautical Information Publication
AIS	Aeronautical Information System
ANM	ATFCM Notification Message
ANTR	Air Navigation Technical Regulations
AOM	Airport Operating Minima
ASR	Airport Surveillance Radar
ATC	Air Traffic Control
ATS	Air Traffic Services
B-RNAV	Basic RNAV
CAA	Civil Aviation Authority
CAR	Civil Aviation Requirements
CDL	Configuration Deviation List
CDR	Conditional Route
CFMU	Central Flow Management Unit
CRAM	Conditional Route Availability Message
CVSM	Conventional Vertical Separation Minima
DA	Decision Altitude
DGCAM	Director General of Civil Aviation and Meteorology
DH	Decision Height
ETOPS	Extended Range Twin Engine Operations
FCOM	Flight Crew Operating Manual
FMGC	Flight Management Guidance Computer
FDMP	Flight Dispatch Manual & Procedures
FDP	Flight Dispatch Procedures
FMGS	Flight Management Guidance System
FMS	Flight Management System
FSR	Flight Safety Report
FWZ	Flugwerkzeuge
GPS	Global Positioning System
GPWS	Ground Proximity Warning System
ICAO	International Civil Aviation Organization
IFR	Instrument Flight Rules
IMC	Instrument Meteorological Conditions
IOC	Integrated Operations Control
JAA	Joint Aviation Authority
Km	Kilometres
LVP	Low Visibility Procedures
LHR	London Heathrow
M	Meter
MAA	Max Authorized Altitude
MCA	Minimum Crossing Altitude
MDA	Minimum Descent Altitude

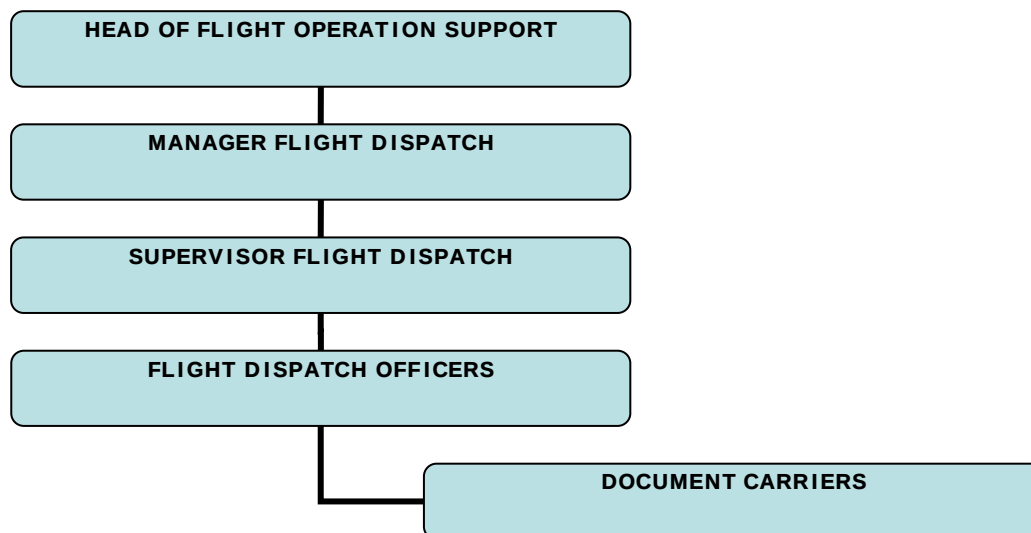
Flight Dispatch Manual and Procedures

INTRODUCTION

MDH	Minimum Descent Height
MEA	Minimum En-route Altitude
MEL	Minimum Equipment List
MET	Meteorology
Mfs2	Modular Flight Planning System Version 2
MMEL	Master Minimum Equipment List
MMO	Mach Maximum Operation
MNPS	Minimum Navigation Performance Specifications
MOCA	Minimum Obstruction Clearance Altitude
MORA	Minimum Off Route Altitude
MSA	Minimum Safe Altitude
NM	Nautical Mile
NOTAM	Notices to Airmen
P-RNAV	Precision RNAV
OAC	Oceanic Area Control
OCA	Oceanic Control Area
OCC	Operations Control Centre
OM	Operations Manual
OTP	On Time Performance
QFE	Height above airport (or runway threshold) elevation
QNE	Altimeter setting 29.92 Hg or 1013.2 Mb.
QNH	Altitude above sea level based local airport pressure
QRH	Quick Reference Handbook
RNAV	Area Navigation
RNP	Required Navigation Performance
RTE	Radio Telephony
RTOW	Regulated Take Off Weight
RVSM	Reduced Vertical Separation Minima
SAM	Slot Allocation Message
SELCAL	Selective Call System
SIGMET	Significant Meteorological report
SM	Statute Mile
SPECI	Special Weather Report
SRM	Slot Revised Message
TCAS	Traffic-Alert and Collision Avoidance System
TOW	Take Off Weight
VFR	Visual Flight Rules
VMC	Visual Meteorological Condition
WSI	Weather Services International

2.0 DISPATCH ADMINISTRATION

2.1 ORGANIZATION STRUCTURE



2.2 DISPATCH LEAVE POLICY

- A. Staff Regulations state, "Annual leave may not be divided into more than two period of leave per year. In exceptional cases approval may be given for leave to be taken in more than two periods but only when there is a valid reason."
- B. In normal circumstances the minimum amount of leave to be taken is five days; however Department Heads may allow one day to be taken due to emergencies or in extremely difficult circumstances.
- C. At the end of the year leave balances should not exceed five days and the Company reserves the right to delay, advance or schedule leave for employees.
- D. Dispatch Policy will be flexible as long as the roster requirements are satisfied that includes having people available in Bahrain for call out. Staff must get permission before leaving Bahrain on days off. Short trips away should be kept to a minimum and are only to be done by applying for leave.
- E. The H.R copy of the Dispatch roster will show a shift pattern for leave periods and this pattern will be used by H.R staff to determine the number of leave days (working days) taken. The pattern when on leave will be a six days on / two days off rotation.
- F. As staff numbers limit the number of staff on leave at the same time, leave requests must be submitted using the electronic 'LEAVE APPLICATION FORM', at least one month before the beginning of the month in which the leave is

Date:	01/04/08	Page	1 of 4	FDP
Authority:	MFD	Issue	3	Rev 00

requested. Management will determine priority after considering the following factors:

- a) When leave was requested, late notice and frequent changes will result in a lower priority.
- b) When leave was last taken, how much and how often leave has been taken.
- c) For those times of the year that are popular for leave, people who had the period off in the previous year will normally have lower priority.
- d) The amount of leave required to be used by the end of the year.
- e) Leave balance must be reduced as notified by personnel department.
- f) To resolve disputes; job performance, attendance and seniority may be used to assess priority for leave.

2.3 TIME KEEPING

All staffs are to record their time worked on the 'Employee Attendance / Exception Report' forms provided. The Automatic Time recorder (ATR) must be used for sign-in and sign-out of duty. In case ATR is forgotten, attendance and reason must be submitted to MFD through the GF Intranet.

If illness prevents you from attending assigned shift or training, contact the Duty Shift Supervisor as soon as possible and advise the length of absence.

If you are going to be late for your shift advise the Duty Shift Supervisor who will arrange coverage until you arrive.

2.4 DISPATCH ROSTER

A Dispatch roster will be issued each calendar month. Swapping of shifts should be kept to a minimum. It is important that we have people in Bahrain for call out, and swaps are not to be made to allow time for short trips outside Bahrain.

2.5 INTERNATIONAL TELEPHONE CALLS

If contact with airports using SITA/fax or email is not successful, the Supervisor's phone may be used for international calls rather than delaying the flight. Log details in the logbook provided.

2.6 OFFICE ETIQUETTE

The stresses of shift work can sometimes leave people short tempered and there will be times when an extra effort is required to act in a professional manner to fellow staff and others you need to have dealings with.

It is also very important that everyone on the shift fully understands what is going on and therefore all work related discussions must be in English unless all staffs on duty are of the same Nationality.

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	Flight Dispatch Manual and Procedures
	DOCUMENTATIONS, RECORD RETENTION & SYSTEMS

3.0 DOCUMENTATIONS, RECORD RETENTION & SYSTEMS

3.1 GENERAL

All Flight Dispatchers must familiarize themselves with the various volumes of the Company Operations Manual and documents listed on paragraph 3.2 below.

Revision services for documents in hard copies are being provided by the service providers and are maintained by Flight Dispatch, except JEPPESEN Airway Manuals that is amended by the Flight Services Office & library Department.

Manuals in electronic format are up to date and can be accessed from 'O' drive of the company server.

3.2 DOCUMENTS AVAILABLE IN FLIGHT DISPATCH

1. JEPPESEN Airway Manual for Gulf Air network (updated by aircraft library)
2. FCOM 2 & 3 for A320/330 & A340 (electronic file)
3. Operations Manual Part A, B & C (electronic file)
4. B767 Flight Planning & Performance Manual (electronic file)
5. CAROPS1 (electronic file)
6. MEL/CDL Manuals for each fleet.
7. Operations Manual Volume 3 – RTOW Charts for all fleet.
(Electronic version available in the Public Folder)
8. Airport Operating Minima Guidance Table
9. Flight Dispatch Manual and Procedures Part 1 & 2

Note: Documents on item 6, 7, 8 & 9 are maintained by Flight Dispatch staff as assigned by Manager Flight Dispatch.

3.3 RECORD RETENTION

All Flight Plan documents and NOTAMS' that are used for the preparation of flights shall be retained in dispatch for a period of 3 months to comply with CAR-OPS 1, as per OMA 2.1.7.2. These include all operational email, telex, WX information and NOTAM.

The documentation of this file is a requirement for accident and occurrence reporting related to preparation of the Operational Flight Plan.

All training records for Flight Dispatch staff shall be maintained by MFD.

Date:	01/04/08	Page	1 of 2	FDP
Authority:	MFD	Issue	3	Rev 00

3.4 AUTOMATED SYSTEMS

Systems available in Flight Dispatch are:

- Mfs2 (Flight Planning System)
- OFP Input Sheet Preparation Program.
- AIMS
- Air Ops
- Sabre View (FALCON)
- VTM (DCS/CARGO)
- RIO
- SITATEX (BAHOKGF & BAHONGF)
- SITA Printers (BAHOKGF & BAHONGF)
- AFTN Printer (OBBIZKZX)

All Flight Dispatch Personnel must have access to the listed systems to retrieve relevant information to process the assigned dispatch work. As 'MFS2' is the Flight Planning System, Dispatchers have read access to the various data base of the Flight Planning System, except the OFP release page (FPROR screen) where change of ETD, Format, Address and Comment can be updated.

The MFS2 database is maintained by MFS2 and Navigation Services.

The rest of the systems are used to retrieve load, flights information and crew names for OFP preparation and a means for communication.

	Flight Dispatch Manual and Procedures
	QUALIFICATION & RESPONSIBILITIES

4.0 QUALIFICATION & RESPONSIBILITIES

4.1 GENERAL

Airlines are engaging Flight Dispatchers for obtaining ATC and MET briefing and the preparation of Flight Plans in accordance with company policy as approved by the controlling local authority. In order to ensure safety of aircraft operations it is necessary to have a uniform and desired level of proficiency for Flight Dispatcher Personnel.

4.2 TRAINING

Recurrent Training:

Flight Dispatch personnel will be given recurrent training every 2 years, to refresh the basic knowledge in aviation and be familiarized with new regulations and changes in the industry.

Competency Check:

MFD to conduct a minimum of one assessment competency proficiency check per staff every 12 months. The check will include a written performance review and action plan. MFD will work along side this staff members in an on duty environment . A maximum of 3 staff members may be reviewed per duty.

Familiarization Flight:

Each Dispatcher should complete a familiarization flight by occupying a jump seat in the cockpit and to gain visual experience in Dispatcher's areas of responsibilities.

For Training and Certification Requirements for Flight Dispatcher Personnel refer to OMD Section 14.3.

4.3 QUALIFICATION

The qualifications and requirements for a Flight Dispatch Officer as per CAR-FCL Supplement 5.1 are:

- 1) Age: Minimum 21 years.
- 2) Knowledge:
 - a) Air Law
 - Rules and regulations relevant to the performance of duties as a Flight Dispatcher;
 - Appropriate ATS practise and procedures.
 - b) Aircraft general knowledge
 - Aircraft operations, systems and instruments;
 - Operating limitations of aeroplanes and power plants;

Date:	01/04/08	Page	1 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	QUALIFICATION & RESPONSIBILITIES

- MEL/CDL
- c) Flight performance calculation and planning procedures
 - OFP; fuel consumption and endurance calculations;
 - Alternate airport selection;
 - En-route speed selection and effect;
 - Extended range operation
 - Filing of ATS plans;
- d) Meteorology
 - Interpretation and application of met reports;
(Significant Charts, TAF, METAR)
 - Movement of pressure systems
 - Structure of fronts
 - Storm warnings
- e) Navigation
 - Principles of air navigation and instrument flight rules;
 - RNP, RVSM, MNPS, TCAS II
- f) Operational procedures
 - Use of aeronautical information (NOTAM, AIC, AIP)
 - Procedures relating to aircraft accidents and incidents
- g) Principles of flight (category of aircraft)
- h) Radio communication
- i) Experience

A Flight Dispatch Officer must have basic subjective knowledge in aviation from a training school recognized by CAA/FAA; or can have similar experience in the aviation industry (as CPL holder, Air Traffic Controller, etc.). After completion of the training, must work in the Dispatch under supervision until assessed in the specific duties and responsibilities in terms of the company policy and BAH-CAA-Oman requirements.

4.4 RESPONSIBILITIES

4.4.1 SUPERVISOR FLIGHT DISPATCH

- 1) Monitors Flight Planning/Dispatch documentation and associated procedures for Gulf Air head office and out stations staff to ensure correct interpretation and compliance to the highest standards to support the safe, efficient and economical operation of all company flights.

Date:	01/04/08	Page	2 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	QUALIFICATION & RESPONSIBILITIES

- 2) Ensure correct staff numbers for anticipated workload, discuss staff currency in the various Dispatch functions, and allocate duties.
- 3) Supervise all personnel in Flight Dispatch during the shift and record duties and any training in the Daily log.
- 4) In the event that an Officer does not attend his shift, the Shift Supervisor will decide if anyone has to stay back until a replacement is found.
- 5) If there is a system failure to receive latest NOTAM via Mfs2, the shift Supervisor is to nominate someone to check NOTAMS received via SITA during the shift and action as required. In these cases the Shift Supervisor is to do a NOTAM hand-over and record details in either the NOTAM Log or the Dispatch log.
- 6) The 'M' Shift Supervisor is to brief Flight Dispatch Manager each morning on any significant occurrences such as delays, PIREPS, strike, airport closure, etc.
- 7) Monitor Storm Warning, Volcanic activities and hazard areas; and take action as required.
- 8) In the event of any MEL/CDL that affects performance of aircraft, check its implication on assigned sector for corrective measure. Advice IOC, Maintenance Control and Load Control of any load penalty. Inform Aircraft Performance Sections if RTOW Charts required.
- 9) In the case of fuel shortage or no fuel at destination/alternate airport, take the necessary action to minimize its effect on operation.
- 10) Any fault or error by the Computer Flight Planning System and correspondence to Mfs2 help desk to be passed to Manager Navigation and Aircraft Performance Section for further follow up and action.
- 11) Monitor all operational messages for aircraft change, delay, etc, issued by IOC and action taken by Flight Dispatchers.
- 12) Be able to handle pressure and lead the staffs as a team during disruptions, Flight Planning System fault, equipment failure, short notice route restrictions, etc.
- 13) Take the appropriate steps to have any faulty equipment repaired as soon as possible and enter Help Desk docket number in the log.
- 14) Ensure the engineering 'CRITICAL ASSOCIATED DEFECTS' sheets are received by email, printed, and placed on the clipboard on the briefing counter.
- 15) Ensure Company and Dispatch procedures are complied with.
- 16) To take appropriate action on any new problems that may occur in Flight Dispatch that are not covered in FDMP

Date:	01/04/08	Page	3 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	QUALIFICATION & RESPONSIBILITIES

- 17) Suggest to Management any new ideas that can improve the work situation
- 18) Delays due to Flight Planning, ATC or other related cause to be in the Logbook.
- 19) Arrange replacements for staff that leave the shift due sickness or other reason.
- 20) In the absence of MFD, the SFD shall be assume managements of Flight Dispatch. This may be done whilst on shift or working business hours.

4.4.2 FLIGHT DISPATCH TRAINING SUPERVISOR

- 1) Prepare and update refresher and competency check for Flight Dispatch Personnel in accordance with Company Policy and DGCAM Bahrain requirements.
- 2) Prepare and maintain training materials and syllabus, and assist Manager Flight Dispatch in maintaining the training file for all Flight Dispatch Staff.
- 3) Take the duties and responsibilities of Flight Dispatch Supervisor item 4.4.1 above, when assigned on shift duty.

4.4.3 FLIGHT DISPATCH OFFICER

- 1) Act under the direct control of the Flight Dispatch Supervisor to whom they are responsible on shift and from whom they receive their assigned duties.
- 2) As allocated on shift, responsible for the duty assigned in Flight Dispatch. This shall include the production, collection and checking of Flight Operations Documentation, including the Operational Flight Plan and Pilot Briefing documents, for all standard and non-standard, schedule and charter flights including ETOPS flights. This task shall be done quickly and accurately whilst working under pressure and prepared in order that safety standards are maintained and regulatory compliances; punctuality and cost efficiency requirements are fully met.
- 3) Responsible for the filing in a timely and accurate manner, of all ATS flight plan with relevant air traffic authorities.
- 4) Check aircraft allocation, sectors and ETD for all flights from AIOPS and amend input sheets when necessary, paying particular attention to filing ATS plan. Always inform the Supervisor of action taken with change messages.
- 5) Continuously monitor, analyze and take appropriate action with regard to actual and forecast weather bulletins, all NOTAMS, as well as any aircraft

Date:	01/04/08	Page	4 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	QUALIFICATION & RESPONSIBILITIES

technical defects which may effect either planning or progress of any flight on the Gulf Air network. Special attention shall be given to ETOPS flights.

- 6) As allocated on shift. Monitor all communications, including Radio (VHF/HF), Phone, fax, email, SITA and any other interdepartmental communication Mediums.
- 7) In accordance with Company Standard Operating Procedures, maintain all records required by the Company in an orderly and neat fashion including the daily log of events.
- 8) Assist with the maintenance of Flight Dispatch dept. manuals.
- 9) Assist with the provision of 'on the job' training for new joiners, when required.

NOTE: Staff who have been on leave for over 10-days are required to report to work 01 hour before commencement of the first shift back to work. During this hour the staff member shall familiarize himself.

Date:	01/04/08	Page	5 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	QUALIFICATION & RESPONSIBILITIES

4.5 APPENDIX A

GENERAL RECURRENT TRAINING SYLLABUS

- Company Policy and CAR regulations.
- MET & NOTAM access procedures.
- JEPPESEN Manuals.
- Computerized Flight Planning.
- Weather minima interpretation.
- Review of Accident / Incident reporting procedures.
- Exercises on Take off Performance, one for each type / version of aircraft.
- Exercises in Flight Planning, Using Simplified graphs, FCOM tables, range tables.
- CRM (Joint Crew Resource Management)

ETOPS RECURRENT TRAINING SYLLABUS

- ETOPS and Regulations
- Criteria to obtain approval
- Area of operation
- Diversion strategies
- ETOPS fuel requirements
- Weather minima
- Minimum Equipment List
- Dispatch requirements
- Flight Watch

Date:	01/04/08	Page	6 of 6	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	DISPATCH GENERAL

5.0 DISPATCH GENERAL

5.1 SHIFT PROCEDURES – GENERAL

1. All staff on duty should be at their work stations at all times during the shift unless required to leave because of operational / administration reasons in which case permission must be obtained from the Shift Supervisor and a contact number must be left.
2. All staff must be aware of significant matters that need to be brought to the attention of the shift supervisor for further consideration. All related paperwork will need to be put aside for use by management for any follow up investigation that may be required. Examples are:
 - Non receipt of Flight Plans or ATS not filed
 - Aircraft diversions or deviations
 - Volcanic or Cyclone Activity
 - Fuel non-availability
 - System or equipment problems / limitations and unscheduled MFS2 outage
 - Delays due de-fuelling or off-load of baggage
 - Airport closures, route limitations or significant military activity
 - PIREPS
 - Significant NOTAMS, significant weather
 - ATC strikes or any other event that affects flight planning for the shift or future shifts.
 - Details of any movement messages that give Dispatch as delay reason.
3. Temporary RTOW charts are stored in the green box file located at the front counter. Advance copies of normal charts may also be there pending placement in Operations manuals, Volume 3. These charts are only valid if there is a reference to them in the Stop Press, Weekly NOTAM or Shift Log. If Performance Section delivers charts and there has not been time for an entry in the Stop Press, details are to be entered into the Shift Log.
4. Flight Planning Document Storage: Night shift is to collect all flight planning documentation for the previous day, bundle them up and store in the area provided.

5.2 SHIFT HANDOVER

A proper communication and hand over of the work between each shift shall take place.

Dispatchers are to remain behind at the end of each shift until the new Dispatcher has arrived; they may leave earlier if released by the Shift Supervisor. If staff is late, the Supervisor may give permission to leave and complete the hand-over himself.

Refer to Flight Dispatch Procedures: FDP-01

Date:	01/04/08	Page	1 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	DISPATCH GENERAL

5.3 MESSAGE PROCESSING - GENERAL

1. To reduce communication costs, e-mail is the preferred method of communicating; If SITA is used send with priority indicator 'QD' or 'QK'. OFP and operational messages to be sent via email, if this is not possible to be sent via SITA on 'QU' priority.
2. Messages to other sections / departments should be addressed to the section manager, i.e. all messages concerning Navigation Services or Performance Section should be addressed to MNAPS.
3. Action should be taken for all operational messages received in Dispatch by contacting the section or person concerned by phone, re-transmitting the message via e-mail or SITATEXT, placing in appropriate in-tray, file or referring it to Dispatch management.
4. Charter requests from other stations are to be sent to BAHSPGF and details of action taken entered into the log.
5. OPCON OO/OW messages are to be brought to the attention of the Shift Supervisor. If this is not practical at the time, the Shift Supervisor is to be advised later of action taken with any change messages.

5.4 ARRIVAL / EN-ROUTE TRAFFIC DELAYS.

Traffic delays at arrival airports and on flight planned routes, may be announced by an Air Traffic Control Unit or Area Control Center for various reasons.

NOTAMS and messages from concerned authorities must be closely monitored and Flight Dispatch requires action when such information received. Normally an airport delay due to special activity of VVIP movements is up 30 minutes though the total duration for the warning may be more in the NOTAM. Some authorities announce a complete closure of the airport for a certain period, which is required to plan the arrival after opening.

En-route traffic delays are normally broadcasted well in advance in an area where there is a proper ATC coordination between the controlling units of the states. The European Union has adopted this and use Brussels EUROCONTROL Center for all this kinds of message to flow to one center to be analyzed and the effect to be transmitted to all operators for early flight planning. All kinds of message from Brussels (i.e. CRAM, ANM, CDR, etc) are received in Flight Dispatch via SITA on 'BAHOKGF'.

Refer to Flight Dispatch Procedures: FDP-02

Date:	01/04/08	Page	2 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	DISPATCH GENERAL

5.5 AIR TEST & LOCAL TRAINING FLIGHTS

It is not a company requirement to prepare a computerized flight plan for Air Test and Training Flights that are to be conducted in the Local Area, i.e. origin and destination of the flight are the same.

However, Flight Dispatch shall do the filing of the ATS Flight Plan for Local flights.

ATS Flight Plan filing is normally done either by auto release with the OFP input or by copying from the actual OFP, but since there is no OFP for Local Flights, manual filing is required by updating the entire items in the ATS plan.

Test flights are required at the request of Maintenance Control or Maintenance Planning, for maintenance reasons. These flights will be co-ordinated between the concerned fleet or HOFO, and one of the mentioned maintenance departments.

Training Flights will be coordinated by Crew Planning and published by IOC.

Refer to Flight Dispatch Procedures: FDP-03

5.6 CRITICAL SHEETS & DAILY STATUS REPORT

Maintenance Control issue Critical Associated Defect Reports for all GF aircraft. This report is received by email daily after 03:00 UTC, except Thursday and Friday which is the GAMCO weekend.

An Aircraft Daily Status Report is issued summarizing a 12 hours aircrafts defect, emphasizing on defects on APU, ETOPS status, Performance/OPS effect, VHF/HF status, RVSM/RNP10 & TCAS, etc.

Dispatcher must print these reports every time it is received and keep it on the clipboard found on the counter for crew reference.

5.7 BACKUP SYSTEM & HISTORIC FLIGHT PLAN

In Microsoft Outlook Public Folders, historical OFP's are stored. In the events of a complete failure to access the Flight Planning System, the CP shall be approached to approve usage of historical OFPs'.

Refer to Flight Dispatch Procedures: FDP-04

5.8 SPECIAL WEATHER, STORM WARNING & VOLCANIC ACTIVITY

Weather is one of the main factors that affect flight planning. During the actual flight it may cause a change from the planned route or diversion to alternate depending on the intensity of the actual weather and limitation of weather minima.

Date:	01/04/08	Page	3 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	DISPATCH GENERAL

As it is one of the major factors affecting the aircraft during flight, it shall be monitored and pre-flight action shall be taken for the appropriate weather report or forecast or any combination at the planning stage.

The effect of weather may cause: delay, cancellation of flight, diversion, carrying extra fuel, payload penalty, crew duty time limitation, etc.

Volcanic activity is the other factor that must be monitored to avoid planning the flight through volcanic ash. It is broadcasted on SIGMET and NOTAM when issued by the controlling authority. It contains the latitude and longitude, height and wind direction to determine the movement of the ash.

Flight Dispatch must monitor the Meteorological Weather Information and Volcano Advisory Report of Gulf Air's network, for safe and economical planning of flights in accordance with company policy.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-05.

5.9 GULF TO LONDON HEATHROW OPERATIONS - NIGHT CURFEW

London Heathrow (LHR) applies a night curfew between 23:00 and 06:00. Flights, which are scheduled to arrive before 06:00, have been allocated a night slot, and can arrive at any time during the curfew.

At the planning stage, the Dispatcher shall closely monitor the estimated arrival time at LHR, for those flights that are scheduled to arrive after 06:00. A tail wind may reduce the flight time and estimate the aircraft to arrive during the curfew. In this event the Dispatcher shall advise IOC.

5.10 AUTOMATED SYSTEM FAILURES

Multiple system failures may disrupt flight schedules as it affects preparation of Flight Plans. The OFP is an essential flight document, and we are unable to dispatch the flight with out it, as it is a BAH-CAA requirement and safety matter. Thus any system failure that can jeopardize the function of Dispatch must be addressed to the responsible agent for prompt action.

Refer to Flight Dispatch Procedures: FDP-06.

5.11 Non-Standard Configurations Flight Planning

This item refers to the MFS2 flight planning entry in the case of Landing Gear Extended Operations for all Fleets and 3 Engine Operations for A340.

The aircraft performance data for Non-Standard Configurations is stored in the Flight Planning System and the OFP can be prepared using its own entry format depending on the case.

Refer to Flight Dispatch Procedures: FDP-07

Date:	01/04/08	Page	4 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
طيران الخليج GULF AIR	DISPATCH GENERAL

5.12 Emergency Response

In the event of accident or incident, involving GF aircraft, the Supervisor shall collect all of the Flight Planning documents for the flight and handover to the Crisis Management Team.

Refer to Flight Dispatch Procedures: FDP-08

5.13 Communications

A. Facilities

The following aircraft communications facilities are available

- VHF
- HF
- ACARS
- SATCOM

The use of VHF and HF is the most common method of communication and cost effective as there is no extra communication cost involved. But any operational message such as passing OFP data to aircraft should use ACARS if contact can be established.

B. VHF

Aircraft must be within VHF range to enable contacting an aircraft. Aircraft must be in visual range of the transmitter and receiver antennas. Even though communication is very clear once it is within range, other radio transmission equipment using the same band might affect it.

C. HF

Direct contact to and from aircraft from Flight Dispatch is not possible for HF communication. Dispatch staff can call FALCON Radio, the main aircraft communication center for the company, and ask to contact the flight.

When HF communications are required the following information need to be available for the radio station:

- Aircraft Call Sign
- Registration
- SELCAL
- Geographical Location

D. ACARS

All GF aircraft have ACARS equipment fitted with the exception of three. Aircraft's system can be accessed either by ground stations in VHF range only or both by ground stations on VHF range and satellite link.

Date:	01/04/08	Page	5 of 10	FDP
Authority:	MFD	Issue	3	Rev
				00

	Flight Dispatch Manual and Procedures
	DISPATCH GENERAL

All A340 and A330 have SATCOM and can be contacted by ground stations having ACARS relying facility on a VHF range, and by satellite link without range restriction.

All A320 and B767 except the one listed below can be contacted via ground station in VHF range.

Gulf Air aircraft that have no ACARS communication facility are the following B767:

- A40-GI
- A40-GJ
- A40-GK

All current destinations have ACARS relay system except the following airports:

Bangalore	BLR
Bangkok	BKK
Cochin	COK
Imam Khomeini	IKA
Islamabad	ISB
Lahore	LHE
Mashhad	MHD
Peshawar	PEW
Shiraz	SYZ

The Hermes system is the tool that can be used for receiving and sending message to/from aircraft that is equipped with serviceable Acars system.

Siata can be used for transmission of ACARS message as well.

Example: ACARS Message when transmitted from SITATEX:

QU QXSXMXS BAHOKGF
BAHOKGF 250612
AGM
AN A40-BZ* / FI GFxxx / MA yyyA
- Text Starts

- Very Important - the A40 is an O NOT a ZERO. Add actual aircraft registration letters.
- Where 'xxx' is Flight number
- 'yyy' is any code number that will identify message. Day of the year is convenient.
- '- ' is a Dash and two Spaces.

1. If your transmission has been successful you should quickly receive a telex from QXSXMXS that will be similar to the above but AGM will

Date:	01/04/08	Page	6 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	DISPATCH GENERAL

become MAS and at the end yyyA will become yyyS. Following your last line there will be an additional line something like:
DT QXS KEF1 060741 S21A

- If your transmission is not successful then you will receive a rejection message similar to:

**ZCZC BAN227 XBC482
QU BAHOKGF BAHOTGF
.QXSXMXS 060735
MAS
AN A4O-LA/FI GF033/MA 310F
- NO ACK
ADDRESSEE: .A4O-LA**

- If you receive **241 - Message too old**, we have discovered that this could be because the difference between the GMT on your SITATEX exceMfs2 by +/- 5 minutes that of QXSXMXS. Adjust your SITATEX GMT, and try again. You can check if this is causing the problem by comparing the rejection message transmission time (high - lighted in above example), and the difference will exceed 5 minutes.

Refer to **Appendix-B** for ACARS VHF stations and coverage range.

E. SATCOM

This communication is through a link with a number of satellites to establish communication with aircraft at any position. The use of normal telephone contacts will enable you to reach the aircraft at any position, but is very expensive.

All A340 and A330 have SATCOM and Flight Dispatch can establish contact using this system for operational reasons, if unable to communicate by any other means.

Refer to Flight Dispatch Procedures: FDP-09

Date:	01/04/08	Page	7 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	DISPATCH GENERAL

5.14 APPENDIX A:

CHARTER / AD HOC FLIGHT CHECKLIST

FLT NO.: _____ SECTOR: _____ Date: _____

1. Operations Control

Check that Over-flight / Diplomatic approval / Military clearances have been obtained.

2. Navigation Services & Aircraft Performance

- ROUTES AVAILABILITY, Terrain Clearance
- DEST ALT's and ERA's (if applicable) NOMINATED
- TEMP RTOW CHARTS AVAILABLE (if required)
- FUEL INDEX

3. Flight Dispatch

	DESCRIPTION	REMARK / RESULT
1	DEST / ALT / ERA & En-route FIR's NOTAM availability.	Any affecting performance to Route Planning.
2	DEST / ALT / ERA weather availability	
3	Handling Agent Name: Tel/Fax - Email - SITA - OPERATIONAL MSG's SITA - BP delivery AFTN -	
4	AGENT TO FILE ATC - Y / N If 'N' AFTN address -	
5	OFP SPEC REQ's: NON SCHEDULE RMK/	Check fuel saving in MFS2. Performance Limitations?
6	Other - TRIP KIT - FUEL availability	Is clearance number required on ATS plan

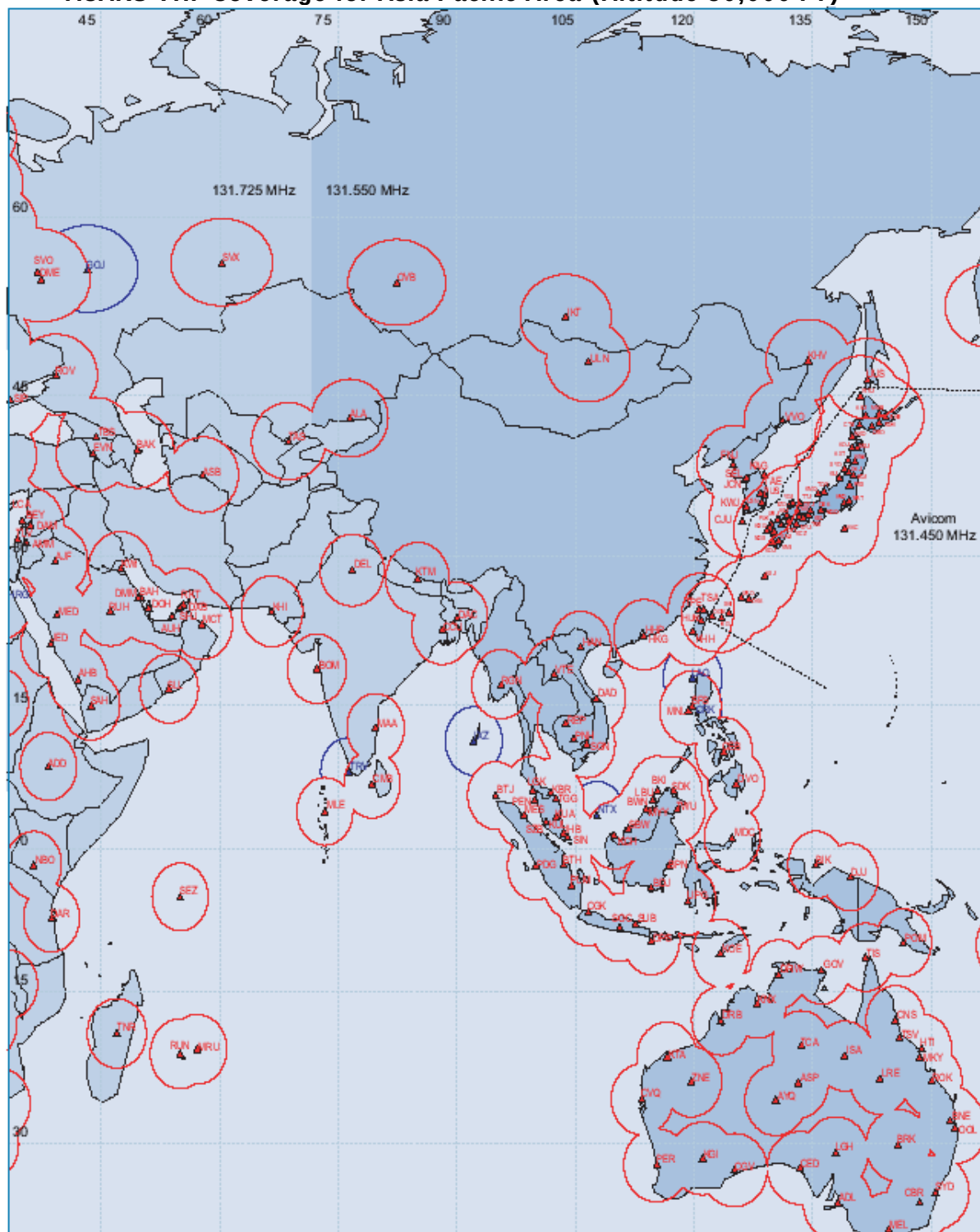
PREPARED BY: _____

DATE: _____

Date:	01/04/08	Page	8 of 10	FDP
Authority:	MFD	Issue	3	Rev
				00

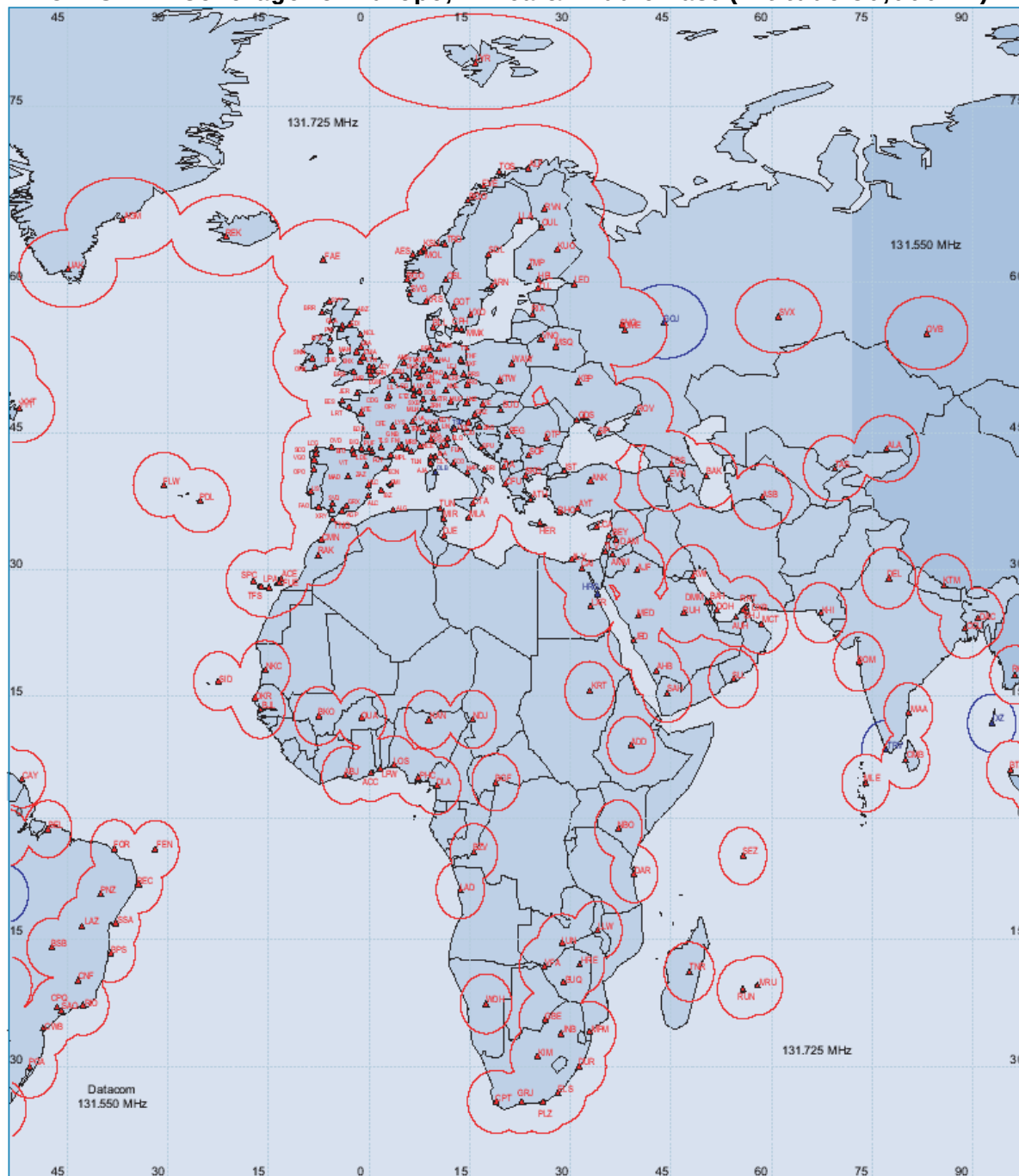
5.14 APPENDIX – B

ACARS VHF Coverage for Asia Pacific Area (Altitude 30,000 FT)



Date:	01/04/08	Page	9 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

ACARS VHF Coverage for Europe, Africa & Middle East (Altitude 30,000 FT)



	Flight Dispatch Manual and Procedures
	FLIGHT DISPATCH DAILY FUNCTIONS

6.0 FLIGHT DISPATCH DAILY FUNCTIONS

6.1 FLIGHT PLAN INPUT SHEETS PREPARATION

General:

The Input sheet is a document that contains Sector, Flight Leg Record (FLR), Registration, Load, ETD, Flight Number, Airports Fuel Price Index, and additional options for name of Dispatcher, Operating Crew and Remark column.

Input sheets are prepared by the previous Day shift Dispatcher, and it covers all flights departure between 0001-2400 UTC. Depending on fuel Tankering sectors and crew turn around, the 2400 UTC time limit might continue up to the following day.

Input Sheets Program:

Flight Dispatch is using this in-house developed Input Sheet program to read the Daily Movement Plan from Air Ops, read Crew Names from AIMS and available Remark from the database, and display all information on the Input sheets for easy reference to the Dispatcher.

Refer to Flight Dispatch Procedures: FDP-10.

6.2 ALTERNATES DISPLAY ON OFP

For pilot planning considerations, even when the destination weather is above the required minima, the standard flight plan format will show two alternate summaries, but the fuel requirement will be based the selected single alternate only.

These alternates are stored in the Flight Planning System for reference.

Refer to Flight Dispatch Procedures: FDP-11

6.3 PREPARATION OF OFP REQUEST

Flight Dispatch is responsible for preparing a flight plan that is safe and economical according to the company policy and BAH-CAA requirements. A number of checks are required before submission of a Flight Plan request. All pertinent information: NOTAM Briefs, Weather, Significant Chart, MEL/CDL, Fuel Index for Tankerin, and additional memos and procedures shall be taken into consideration before a request for an OFP is made.

Computerized Flight Plans are not fail safe. Various reasons such as the possibility of human error may occur. Although most wrong entries to the system

Date:	01/04/08	Page	1 of 8	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	FLIGHT DISPATCH DAILY FUNCTIONS

are rejected when figures exceed limitations, it is important that staff shall check their work at all times.

Refer to Flight Dispatch Procedures: FDP-11.

6.4 ATS FLIGHT PLAN

General:

For the departure of all flights, a clearance from the ATC authority is required. The submission can be either be a Repetitive Flight Plan (RPL) for flights operating over a defined period of time, or filing an ATS Plan for a flight on daily basis. If there is a change of equipment or routing or delay more than the permitted time, a revised ATS Plan is required to be submitted to the local ATC unit.

Flight Dispatch is responsible for filing ATS Flight Plan for all flights except those departing European airports.

The Handling Agent will file for flights from Europe, and if this is not possible due to various reasons, Flight Dispatch will file.

Conflicting Call Sign:

It is possible for multi sector flights to operate on two different aircraft. If they are in the same airspace at the same time, there will be confusion and a threat to safety.

Example: GF972/AUH-BAH-BEY

In this case, the BAH/BEY sector may have been changed to another aircraft as the AUH/BAH sector is running late.

To avoid this conflict, flights operating to more than one airport in the Gulf using the same call sign must add the suffix 'A' and 'B' as described in the procedure.

Refer to Flight Dispatch Procedures: FDP -12.

6.5 NON-STANDARD FLIGHTS

Non-standard flights are Non-ETOPS sectors that require additional checks and procedures when the OFP is prepared in Flight Dispatch.

The Following are factors used to categorize Non-Standard Flights:

- Special states requirement to over fly FIR
- Additional daily check for route availability
- Known area of operation for Tropical Storm/Cyclone and Volcanic Activity

Date:	01/04/08	Page	2 of 8	FDP
Authority:	MFD	Issue	3	Rev 00

6.5.1 GULF TO /FROM EUROPE OPERATIONS

A network of Conditional Routes (CDR) exists in Europe to allow flights to be planned on airways that have varying hours of availability.

A full explanation of the system can be found in the JEPPESEN Manual, Europe En-route section, and the pages in that section contain the availability hours and flight level allocation for the segment. JEPPESEN charts indicate CDR type after the airway designator.

A 'Conditional Route Availability Message' (CRAM) is published each day to advise operators status of conditional routes for use if they are available as per the times and conditions published. It must be studied before using a route that contains any CDR segments.

CRAM received via SITA line on BAHOKGF address or can be accessed from its website.

Please use the below link to access the site for CRAM, ANM, ATC Validation, etc.

http://www.cfm.eurocontrol.be/j_cia_public/cia_public/pages/msgCram.jsf

1. Gulf to London Heathrow OFP include 20 minutes extra holding fuel due to landing traffic.
2. Normal OFPs' are on best route that complies with published routing schemes. Airport representatives may request a plan on another route to get a better slot allocation when there is significant delay on Best Route.
3. Verify final routing on CFMU validation site from the 'Flight Page' in mfs2.

Refer to Flight Dispatch Procedures: FDP-13.1

6.5.2 GULF TO /FROM FAR EAST

Due to presence of known Tropical Cyclone and Volcanic activity in the Far East and Arabian Sea area, a close monitor and avoidance of affected routes is required.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.2

Date:	01/04/08	Page	3 of 8	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	FLIGHT DISPATCH DAILY FUNCTIONS

6.5.3 KATHMANDU OPERATIONS WITH B767

There is an aircraft performance limitation on operation to Kathmandu due to high airport elevation and obstructions due to mountains all around the airport. Aircraft LGW/TOW need to be monitored when OFP is prepared.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.3

6.5.4 SANAA OPERATIONS WITH A320

Schedule operations to Sanaa are with the A320, and due to high airport elevation and obstructions due mountains around the airport, Take-Off Weight (TOW) can be restricted when the temperature is high. When forecast temperature for the day and time of departure is available, the RTOW Charts must be checked and a weight adjustment should be made during OFP preparation.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.4

6.5.5 KHARTOUM FUEL SITUATION

The fuel situation in Khartoum is unreliable so fuel availability has to be checked on daily basis. This may affect revenue load and proper coordination with Khartoum and departure airport station's agents is required.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.5

6.5.6 HANDLING DIVERSION FLIGHTS

Flights can divert en-route or at destination alternates for various reasons. Flight Dispatch must react quickly when a diversion occurs. A computer flight plan for the crew and ATS plan to the respective ATC unit shall be released.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.6

6.5.7 FUEL POLICY CHANGE FOR A330 /A340 ON APPROVED SECTORS

Refer to section 10.5 (Contingency Fuel Policy Change).

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.7

6.6 FUEL TANKERING PROCEDURES

Due to fuel price difference and the significant amount of saving involved, Flight Dispatch Officer should consider up-lifting extra fuel for the next sectors with out payload penalty when it is feasible. The MFS2 Flight Planning System program can analyze whether extra fuel tanker has a positive saving or not by mathematical analysis of factors that affects operating cost.

Date:	01/04/08	Page	4 of 8	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	FLIGHT DISPATCH DAILY FUNCTIONS

When multi sector tanker flight plan is processed, payload figures taken from the system at the time of planning may not be accurate. This inaccuracy of payload on the flight plan may restrict the subsequent airports loads, if the OFP shows Maximum Limited Take-off Weight (TOW) or Landing Weight (LGW). Dispatchers are to be vigilant on such sectors and reduce the Tankering fuel to leave room for extra payload at the next airports. This applies particularly to multi sector Gulf-Gulf to Indian sub-continent and to Cairo flights.

Refer FLIGHT DISPATCH PROCEDURE: FDP-14.

6.7 HANDLING PAYLOAD RESTRICTION

General:

Payload can be restricted for various reasons. Flight Dispatch must inform Load Control when there is payload penalty or discuss the matter with IOC for any other option to avoid offloading; such as change of equipment or tech stop.

The following are some of the reasons for payload restriction:

- Fuel Shortage
- Destination/Alternate weather
- MEL/CDL Performance penalty
- En-route restriction
- Airport Restriction
- WAT Limitation

RTOW Restrictions due WAT Limits:

WAT stands for Weight, Altitude and Temperature and all three can affect RTOW, especially with their combined effect. A good example in the Gulf is AUH airport that has a 4,100M Runway and 88 ft elevation, but RTOW begins to be restricted with temperatures above 40 Degree Centigrade.

RTOW reductions occur on Gulf to Manila sector in summer with A340 aircraft. When temperatures reach 40 degrees check RTOW charts and include QNH correction if applicable for that a/c type.

The following GF destinations have elevations greater than 1,000 feet and should be checked, especially when temperatures are high.

AMM / DAM / IKA / RUH / SAH / THR / KTM

Refer to FLIGHT DISPATCH PROCEDURE: FDP-15

Date:	01/04/08	Page	5 of 8	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	FLIGHT DISPATCH DAILY FUNCTIONS

6.8 FUEL DOCKETS FOR FLIGHTS ORIGINATING BAHRAIN

For BAH departure flights, when the crew reports to the Briefing room, Dispatcher will hand over Fuel Docket to the crew with EZFW as retrieved from DCS system, if it is ready. If the flight is on PSS, Dispatcher will check ZFW in DCS system and update in Mfs2 as per the following procedure:

- DCS ZFW update:
 - Check DCS ZFW around 2 hrs before ETD
 - If DCS ZFW is more than 1000kg or less than 3000kg against the OFP, insert the new ZFW in mfs2 and recalculate OFP
 - If the ZFW is not ready in DCS system, crew should be informed for their late planning of fuel after reaching at the aircraft.
 - Print and attach all DCS ZFW information with the flight paper work
 - Signed copy of OFP to be attached and filed along the paper work

Date:	01/04/08	Page	6 of 8	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

7.0 ALL FLEET OPERATIONS

7.1 AIRPORT OPERATING MINIMUMS (AOM)

'AIRPORT OPERATING MINIMUM' (AOM) policy is contained in Operations Manual, Section 8.1.3. The OM is the master document and should be thoroughly read and understood. Gulf Air flights are to be carried out using instrument Flight Rules, and any Visual Flights Rules need approval from BAH-CAA through VPO.

Gulf Air's method of determining the airport minima are approved by BAH-CAA.

Factors affecting minima are:

- Aeroplane Capability (Flight Manual)
- State Operating Minima (Minima established by local CAA authority)
- Operator's Minima (DGCAM requirement)
- Crew Minima (Qualification & experience)

7.2 TAKE-OFF ALTERNATE AIRPORT REQUIREMENT

When performance or weather condition restricts flights to return to departure airport, a take off alternate must be nominated.

Refer full explanation in Operations Manual Part A - Section 8.1.2.3.2.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-11

7.3 DESTINATION AND DESTINATION ALTERNATES PLANNING MINIMA POLICY

For full explanation and company minima, refer Operations Manual Part A - Section 8.1.2.3.3/4.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-11

7.4 DESTINATIONS & ALTERNATES PLANNING MINIMA GUIDANCE TABLE

Tables have been produced to assist Dispatchers in determining destinations and alternates minimums. They contain minima for at least the first and second listed company alternates for each scheduled destination.

The airport approach facility chosen to determine the minima is the one with the least restriction that satisfies the AOM requirements.

This minima in the table is to be used in conjunction with Airport Minima in the JEPPESEN Airway Manual and NOTAM affecting the navigation aids for landing and take-off.

Date:	01/04/08	Page	1 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

When NOTAMs regarding down grade of minima or unserviceable navigational aids are received, or the wind is not favourable for the given runway direction the JEPPESEN approach plate must be referred to in order that the new minima may be determined.

Flight Dispatch must monitor JEPPESEN revision and change of GF Destination/Alternate minima should be advised to MFD for corrective action.

Refer to Appendix-A for Destinations & Alternates WX planning minima table.

7.5 WEATHER INFORMATION SOURCES & METEOROLOGICAL INTERPRETATION

The weather information for a flight is comprised of the following:

- Airport forecasts for departure, en-route, destination and destination alternates;
- Significant Weather Chart for the route;
- Upper wind analysis charts for the route.

This information is available in the mfs2 flight planning system.

Additional weather information such as TAF and METAR for Gulf Airports is received regularly through the AFTN network.

In case the mfs2 system is not accessible, TAF/METAR & Sig WX Chart can be collected from BAH MET office for use to all flights.

Mfs2 WX system is accessible for TAF/METAR request via SITA as follows:

QU BAHAPYF
.BAHOKGF 100730
FF LOWMYBYX
071005 OBBIGFAX
RQM/FTOBBI OTBD (TAF BAH & DOH airports)

Or

RQM/SAOBBI OTBD (METAR for BAH & DOH airports)

Dispatchers are to refer to OMA 8.1.6 for explanation and interpretation of Significant Charts, SIGMET (significant meteorological report), SPECI (Special WX report), TAF, and METAR and associate weather trend conditions.

Date:	01/04/08	Page	2 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

7.6 Rescue and Fire Fighting (RFF) Categories

ICAO (Annex 14) recommends ten categories of RFF protection based on aircraft length and maximum fuselage width.

The respective RFF Categories for airports can be found in the AIRPORT DIRECTORY section of the JEPPESEN Airway Manuals.

The minimum RFF Categories required at an airport for Gulf Air fleet are:

- A340 - 8
- A330 - 7
- B767 - 7
- A321 - 6
- A320 - 5

Refer to OMA 8.1.2.1 for latest RFF category listing and temporary downgrading.

7.7 MEL /CDL AFFECTING OPERATIONS

Operation of every system or component of an aircraft is not required when the remaining operative system enables safe operations. Because of this, an aircraft can operate with some of these systems/components being inoperative provided the aircraft manufacturer and Bahrain-CAA authorize it.

The Minimum Equipment List is designed to satisfy these conditions and is approved by the CAA of Bahrain for use on all Gulf Air aircraft types. All items that are related to airworthiness of the aircraft and are not listed on the MEL are required to be serviceable.

Maintenance Control shall advise aircraft defects with performance restrictions to Flight Dispatch. Such information must be documented and filed, and any advice should be accompanied by email or telex for reference. The same must be noted on the White Board for easy reference and follow up.

Refer Flight Dispatch Procedures: FDP-16.

7.8 RNP, RVSM & TCAS II REQUIREMENTS

7.8.1. RNP REQUIREMENTS

RNP (Required Navigation Performance) is the required navigation accuracy in the lateral and longitudinal position of the aircraft in a defined airspace. That is the accuracy should be within a given limit for 95 % of the flying time.

Gulf Air is presently approved for RNP 5 and RNP 10 operation based on radio navigation aid infrastructure.

Date:	01/04/08	Page	3 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

RNP5 is currently mandatory on most of the Gulf Region ATS structure and the entire European (ECAC) ATS route network.

RNP 10 is also being introduced in the sub-Indian continent, Asia and Australian route structure based on areas of operation.

Gulf Air aircraft compliance with the various RNP requirements is indicated by the letter 'R' in item 10 of the ATS plan.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-16

7.8.2. RVSM REQUIREMENTS:

Reduced Vertical Separation Minimum (RVSM) provides a 1000 ft vertical separation between RVSM approved aircrafts. The RVSM airspace is between FL 290 & FL 410 inclusive and refer to the relevant Jeppsen chart for actual FL indication.

Gulf Air's fleet is equipped and approved for operation in RVSM airspace. The letter 'W' must be shown in item 10 of the ATS plan to indicate compliance with RVSM airspace.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-16

7.8.3. TCAS II REQUIREMENTS:

Traffic/Aircraft Collision Avoidance System II (TCASII/ACASII) provides the pilot a Traffic Advisory (TA) and Resolution Advisory (RA) by detecting other aircraft in the vicinity, and recommends manoeuvring or manoeuvring by it to avoid conflict with the other aircraft.

All Gulf Air's fleet is fitted with the TCAS II system and it is to be used at all times when airborne in accordance with the procedures laid down in OM-B for the applicable aircrafts. (Reference OM-A 8.3.6)

Refer to FLIGHT DISPATCH PROCEDURE: FDP-16

7.9 HIGH SPEED AND MAXIMUM RANGE OFP

The MFS2 Flight Planning System is capable of being set to different criteria for the calculation of OFP to meet the demand of the operation. The following criteria are available:

- Minimum cost (ECON);
- Minimum fuel (MINF);
- Minimum time (MINT); and
- Preferred speed (USER)

Date:	01/04/08	Page	4 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

The company fuel policy for flight plan calculation is Minimum Cost. When a flight is required to have a higher speed to gain en-route time for operational reason (crew duty time, curfew or delay recovery), then Dispatcher can select either the Cost Control or ETA Control button in order for the system to pick higher Cost Index (CI) to provide new reduced flight time.

High Speed OFP:

Communicate with IOC/Rostering to get required Target Time for crew duty time/delay recovery plan.

Factors to be considered during High Speed Flight Planning:

- En-route significant weather
- Wind shear
- MEL/CDL
- Satisfactory tail wind component
- En-route fixed speed/Mach technique restriction

MAXIMUM RANGE OFP:

Maximum Range OFP is a selection of Minimum Fuel 'MINFUEL' calculation that attempt to provides less fuel burn that enable to trade the weight to be used to uplift extra payload on sectors limited by fuel tank capacity.

A good example for this is the A320 operations on Gulf to CCU sectors. If a distant alternate is nominated due to weather and the MFS2 system selects higher CI, due to environmental factors, then possibilities for payload restriction may occur due aircraft fuel tank capacity limitation.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-17

7.10 FLIGHT PLAN WIND SHEAR

The wind shear figure (SR) shown on GF flight plans is knots per 1,000 ft and there is no maximum or minimum figure. Turbulence can be expected in mfs2 flight plan as follows:

- 1 on flt plan = 5 kt per 1,000 ft.
- 2 on flt plan = 10 kt per 1,000 ft.
- 3 on flt plan = 15 kt per 1,000 ft.
- etc. to maximum of 9 = 45 kt per 1,000 ft.

Note that there is no industry standard formula for this calculation.

Date:	01/04/08	Page	5 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
طيران الخليج GULF AIR	ALL FLEET OPERATIONS

Dispatchers are to highlight flight plan wind shear figures that indicates 5 and above for flights from Bahrain and for departure from other airports to put a comment on the OFP for the pilot.

Date:	01/04/08	Page	6 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

7.11 APPENDIX - A

DESTINATIONS & ALTERNATES WX MINIMA GUIDANCE TABLE

The following tables contain lists of destination alternate minima for at least the first and second company alternates for each destination.

The values for 'LANDING' minima in the table are for the runway and approach aid listed, and in most cases the runway chosen has the lowest minima of all the runways and approach types for that airport.

As per Operation Manual - Part A, section 8.1.2.3.4 – item 1, the destination airport minima is the approved AOM minima that is published on the JEPPESEN instrument approach chart or on a separate JEPPESEN JAA-9X minima page.

The following guidelines have been applied when extracting the specific minima from the JEPPESEN manuals :

- All navigational equipment for the airport are assumed serviceable as stated on the JEPPESEN approach charts.
- VOR/DME and NDB are used for non-precision minima unless the LOC minima is a significant benefit.
- When JEPPESEN publishes the State minima 'FOR FILING AS AN ALTERNATE', usually on the front or back of the airport diagram, the figures used in the tables. E.g. Saudi, Taiwan and Australia. These will have an asterisk (*) under the figures.
- When RVR and Visibility are published together (R550 V800), use the 'V' figure.
- CAT II RVR system minima : CAT C = 300, CAT D = 350.
(Refer to OMA 8.1.3.6.4.1)

Planning Minima for En-route and Destination Alternates: (Reference OM-A 8.1.2.3.4)

TYPE OF APPROACH	PLANNING MINIMA/RVR
CAT II & CAT III	CAT I (Note 1)
CAT I	Non-Precision (Notes 1 and 2)
Non-precision	Non-Precision (Notes 1 and 2) plus 200 ft/1,000m
Circling	Visual

Note 1: RVR

Note 2: The ceiling must be at or above MDH

Warning: For any downgrade of navigation aid and/or if the wind is not favorable for the runway heading given on the table, refer to the relevant JEPPESEN approach chart to determine new minima.

APPLICATION OF AERODROME FORECASTS (TAF & TREND) TO PRE-FLIGHT PLANNING (ICAO Annex 3 refers)									
APPLICATION OF INITIAL PART OF TAF (For Aerodrome planning minima see CAR-OPS 1.297) a) Applicable time period: From the start of the TAF validity period up to the time of the applicability of the first subsequent 'FM' or 'BCMG' or, if no FM or BCMG is given, up to the end of the validity period of the TAF. b) Application of the forecast: The prevailing weather conditions forecast in the initial part of the TAF should be fully applied with the exception of the mean wind and gusts (and x-winds) which should be applied in accordance with the policy in the column 'BCMG AT and FM' in the table below. This may however be overruled temporarily by a "TEMPO" or "PROB" if applicable, add to the table below.									
APPLICATION OF FORECAST FOLLOWING CHANGE INDICATORS IN TAF AND TREND.									
TAF or TREND for Aerodrome planned as:	FM (alone) and BCMG AT: Deterioration and improvement.	BECMG (alone), BECMG FM, BECMG TL, BECMG FM, TL. In case of;		TEMPO (alone), TEMPO FM, TEMPO TL, TEMPO FM.TL, PROB 30/40. Deterioration Improvement			In any case	PROB & TEMPO	Deterioration and improvement.
		Deterioration	Improvement	Transient/showery conditions in connection with short-lived weather phenomena e.g., thunderstorms, showers.	Persistent conditions in connection with e.g., haze, mist, fog dust/sandstorm, continuous precipitation.	Applicable			
Destination at ETA +/- 1hr.	Applicable from the time of start of the change.	Applicable from the time of start of the change.	Applicable from the time of end of the change.	Not applicable.		Applicable			
Takeoff Alt at ETA +/- 1Hr.									
Destination Alt at ETA +/- 1 Hr.	Mean wind should be within required limits.	Mean wind should be within required limits.	Mean wind should be within required limits.	Mean wind and gusts exceeding required limits may be disregarded.		Mean wind should be within limits			Deterioration may be disregarded; improvement should be disregarded including mean wind gusts.
Enroute Alt at ETA +/- 1 Hr. (See CAR-OPS AMC 1.255)	Gusts may be disregarded.	Gusts may be disregarded.	Gusts may be disregarded.			Gusts may be disregarded.			
ETOPS Enroute at earliest/latest +/- 1Hr.	Applicable from the start of the change. Mean wind should be within limits. Gusts exceeding X-wind limits should be fully applied.	Applicable from the start of the change. Mean wind should be within limits. Gusts exceeding X-wind limits should be fully applied.	Applicable from the end of the change. Mean wind should be within limits. Gusts exceeding X-wind limits should be fully applied.	Applicable if below applicable landing minima. Mean wind should be within required limits. Gusts exceeding crosswind limits should be fully applied.		Applicable if below applicable landing minima. Mean wind should be within required limits. Gusts exceeding crosswind limits should be fully applied.	Should be disregarded.		

Note 1: "Required Limits" are those contained in the Operations Manual

Note 2: If promulgated aerodrome forecasts do not comply with the requirements of ICAO Annex 3, operators should ensure that guidance in the application of these reports is provided.

MIDDLE EAST REGION

* = STATE PUBLISHED FOR FILING AS ALTERNATE

** = LVP NOT AUTHORIZED

AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
ABU DHABI OMAA	RWY 31	CAT II	C	200	550		300
			D	200	550		350
BAHRAIN OBBI	RWY 30R	CAT I	C	310	1000		550
			D	310	1400		600
BANDAR ABBAS OIKB	RWY 21L	CAT I	C	620	1200		800
			D	620	1600		800
DAMMAM OEDF	RWY 16R	CAT I	C	600	3200		350
			D	600	3200		350
				*	*		
DOHA OTBD	RWY 34	CAT II	C	200	550		300
			D	200	550		350
DUBAI OMDB	RWY 12L	CAT II	C	200	550		350
			D	200	550		350
ESFAHAN OIFM	RWY 26R	CAT I	C	380	1600		800
			D	380	2000		800
JEDDAH OEJN	RWY 34C	CAT II	C	600	3200		350
			D	600	3200		350
				*	*		
KUWAIT OKBK	RWY 33R	CAT II	C	200	550		300
			D	200	550		350
MADINAH OEMA	RWY 17	CAT I	C	600	3200		800
			D	600	3200		800
				*	*		
MASHHAD OIMM	RWY 31R	CAT I	D	370	1000		800
			D	370	1400		800

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

MIDDLE EAST REGION

* = STATE PUBLISHED FOR FILING AS ALTERNATE

** = LVP NOT AUTHORIZED

AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
MUSCAT OOMS	RWY 08	CAT I	C	380	1000		600
			D	380	1400		600
RAS AL KHAIMAH OMRK	RWY 34	CAT I	C	400	1000		550
			D	400	1400		550
RIYADH OERK	RWY 15L	CAT I	C	600	3200		550
			D	600	3200		550
				*	*		
SALALAH OOSA	RWY 25	CAT I	C	400	1000		600
			D	400	1400		600
SHARJAH OMSJ	RWY 30	CAT II	C	200	550		300
			D	200	550		350
SHIRAZ OISS	RWY 29L	CAT I	C	650	1200		800
			D	650	1600		800
TEHRAN OIIE	RWY 29R	CAT I	C	440	1200		800
			D	440	1400		800
ADEN OYAA	RWY 08	CAT I	C	350	1400		800
			D	350	1600		800
				(LOC)			
DJIBOUTI HDAM	RWY 27	CAT I	C	270	1800		1250
			D	270	2000		1350
SANAA OYSN	RWY 18	CAT I	C	600	4000		800
			D	600	4000		800
				*	*		

Date:	01/04/08	Page	10 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

NORTH AFRICA & MEDITERRANEAN REGION

* = STATE PUBLISHED FOR FILING AS ALTERNATE

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AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
ALEXANDRIA BORG EL ARAB HEBA	RWY 32	CAT I	C	380	1000		600
			D	380	1400		650
ALEXANDRIA INTL HEAX	RWY 04	VORDME	C	770	2600	570	1600
			D	770	2800	570	1800
AMMAN / MARKA OJAM	RWY 24	CAT I	C	400	1000		650
			D	400	1400		650
					(LOC)		
AMMAN / QUEEN A. OJAI	RWY 26L 259 (262)	CAT I	C	310	1000		550
			D	310	1400		550
					(LOC)		
ANKARA / ESENBOGA LTAC	RWY 21L	CAT II	C	200	550		300
			D	200	550		350
ASWAN HESN	RWY 35	CAT I	C	430	1000		800
			D	430	1400		800
					(LOC)		
ATHENS LGAV	RWY 03L	CAT II	C	200	550		300
			D	200	550		300
BEIRUT OLBA	RWY 03	CAT I	C	650	1200		800
			D	650	1600		800
					(LOC)		
CAIRO HECA	RWY 05R	CAT I	C	330	1000		600
			D	330	1400		650
CASABLANCA GMMN	RWY 35	CAT II	C	200	550		300
			D	200	550		350
DAMASCUS OSDI	RWY 05R	CAT II	C	200	550		300
			D	220	600		350
FES / SAISS GMFF	RWY 27	CAT I	C	400	1400		700
			D	400	1600		700

Date:	01/04/08	Page	11 of 24	FDP
Authority:	MFD	Issue	3	Rev
				00

NORTH AFRICA & MEDITERRANEAN REGION

* = STATE PUBLISHED FOR FILING AS ALTERNATE

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AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
ISTANBUL ATATURK LTBA	RWY 36R	CAT II	C	220	600		350
			D	220	600		350
ISTANBUL SABIHA-OKCEN LTFJ	RWY 06	CAT I	C	710	1400		600
			D	710	1800		600
KHARTOUM HSSS	RWY 18	CAT I	C	300	1400		800
			D	300	1600		800
LARNACA LCLK	RWY 22	CAT I	C	370	1600		800
			D	370	1800		800
LUXOR HELX	RWY 20	CAT I	C	370	1000		650
			D	370	1400		650
PAFOS LCPH	RWY 29	CAT I	C	340	1400		700
			D	340	1600		700
PORT SUDAN HSPN	RWY 35	CAT I	C	310	1600		800
			D	310	1800		800
RABAT / SALE GMME	RWY 22	CAT I	C	300	1000		600
			D	300	1400		600
RODOS LGRP	RWY 25	CAT I	C	1090	1400		650
			D	1090	1800		800
SHARM EL SHEIK HESH	RWY 04L	CAT I	C	510	1200		650
			D	510	1600		650
TANGER GMTT	RWY 28	CAT I	C	520	1200		650
			D	520	1600		650
THESSALONIKI LGTS	RWY 16	CAT II	C	200	700		450
			D	200	700		450

INDIAN SUB CONTINENT REGION

* = STATE PUBLISHED FOR FILING AS ALTERNATE

** = LVP NOT AUTHORIZED

AIRPORT ICAO NAME	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
AHMEDABAD VAAH	RWY 23	CAT I	C	550	2400		800
			D	550	2800		800
BANGALORE VOBG	RWY 27	CAT I	C	410	2800		800
			D	410	3200		800
BANGALORE VOBL	RWY 27	CAT I	C	510	2000		600
			D	510	2800		600
CALCUTTA KOLKATA VECC	RWY 19L	CAT I	C	410	2000		800
			D	410	2400		800
CHENNAI – MADRAS VOMM	RWY 07	CAT I	C	500	2000		600
			D	500	2800		600
COCHIN VOCI	RWY 27	CAT I	C	590	2400		900
			D	590	2800		900
COLOMBO VCBI	RWY 04	ILS	C	360	1000		600
			D	360	1400		600
DELHI VIDP	RWY 28	CAT IIIB	C	230	600		75
			D	230	600		75
DHAKA VGZR	RWY 14	ILS	C	410	2400		1000
			D	410	2400		1000
HYDERABAD VOHS	RWY 27	ILS	C	410	2000		550
			D	410	2400		550
ISLAMABAD OPRN	RWY 30	ILS DME	C	320	1000		800
			D	320	1400		800

INDIAN SUB CONTINENT REGION

* = STATE PUBLISHED FOR FILING AS ALTERNATE

** = LVP NOT AUTHORIZED

AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
KARACHI OPKC	RWY 25R	ILS	C	390	1000		600
			D	390	1400		600
KATHMANDU VNKT (DAY TIME)	RWY 02	VOR DME	C	1010	2500	810	1500
			D	1010	2800	810	1800
LAHORE OPLA	RWY 36R	CAT II	C	220	600		350
			D	220	600		350
MALE VRMM	RWY 36	CAT I	C	470	1800		1000
			D	470	2000		1000
MUMBAI - BOMBAY VABB	RWY 27	CAT I	C	520	2000		800
			D	520	2800		800
					(LOC)		
PESHAWAR OPPS	RWY 35	VOR DME	C	570	2000	370	1000
			D	570	2400	370	1400
TRIVANDRUM VOTV	RWY 32	CAT I	C	510	2400		800
			D	510	3200		800
					(LOC)		

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

ASIA REGION

* = STATE PUBLISHED FOR FILING AS ALTERNATE

** = LVP NOT AUTHORIZED

AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/F	VIS/M
BANGKOK VTBS	RWY 01L	CAT I **	C	440	1000		550
			D	440	1400		550
					LOC		
CHIANG MAI VTCC	RWY 36	CAT I	C	430	1600		900
			D	430	1800		900
DEN PASAR / BALI WRRR	RWY 27	CAT I	C	390	1400		800
			D	390	1600		800
HONG KONG VHHH	RWY 07R	CAT II	C	200	550		350
			D	200	550		350
JAKARTA / SOEKARNO WIII	RWY 07R	CAT I	C	330	1000		550
			D	330	1400		550
JAKARTA / HALIM WIHH	RWY 24	CAT I	C	410	1000		650
			D	410	1400		650
JOHOR BARU WMKJ	RWY 16	CAT I	C	510	1800		900
			D	510	2000		900
KAOSHIUNG RCKH	RWY 09	CAT I	C	600	3200		800
			D	600	3200		800
				*	*		
KUALALUMPUR WMKK	RWY 14L	CAT I	C	400	1000		550
			D	420	1400		550
LAPU LAPU RPVM	RWY 22	CAT I	C	360	2000		900
			D	360	2000		900
MACAU VMMC	RWY 34	CAT II	C	200	800		350
			D	200	800		350

Date:	01/04/08	Page	15 of 24	FDP
Authority:	MFD	Issue	3	Rev
				00

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AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
PENANG WMKP	RWY 04	CAT I	C	530	1200		650
			D	530	1600		650
PHUKET VTSP	RWY 27	CAT I	C	610	1800		1000
			D	610	2000		1000
RAYONG / U- TAPHAO VTBU	RWY 18	CAT I	C	300	1200		1200
			D	300	1600		1200
SINGAPORE WSSS	RWY 02L	CAT II	C	200	550		300
			D	200	550		350
SONGKHLA / HAT YAI VTSS	RWY 26	CAT I	C	350	1600		800
			D	350	1800		800
SURABAYA WRSJ	RWY 10	ILS	C	500	1000		550
			D	500	1400		550
TAIPEI RCTP	RWY 05	ILS II	C	600	3200		350
			D	600	3200		350
				*	*		
YANGON VYYY	RWY 21	CAT I	C	510	2000		1200
			D	510	2000		1200
SUBIC BAY RPLB	RWY 07	CAT I	C	660	660		1600
			D	2000	2000		1600
PAMPANGA CLARK INT RPLC	RWY 20L	CAT I	C	330	2000		800
			D	330	2000		800
					LOC		

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AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
PENANG WMKP	RWY 04	CAT I	C	530	1200		650
			D	530	1600		650
PHUKET VTSP	RWY 27	CAT I	C	610	1800		1000
			D	610	2000		1000
RAYONG / U- TAPHAO VTBU	RWY 18	CAT I	C	300	1200		1200
			D	300	1600		1200
SINGAPORE WSSS	RWY 02L	CAT II	C	200	550		300
			D	200	550		350
SONGKHLA / HAT YAI VTSS	RWY 26	CAT I	C	350	1600		800
			D	350	1800		800
SURABAYA WRSJ	RWY 10	ILS	C	500	1000		550
			D	500	1400		550
TAIPEI RCTP	RWY 05	ILS II	C	600	3200		350
			D	600	3200		350
				*	*		

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

ASIA REGION

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AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
SHANGHAI PUDONG ZSPD	RWY 17L	CAT II **	C	450	1200		550
			D	450	1200		550
SHANGHAI HONGQIAO ZSSS	RWY 36	CAT I	C	290	800		550
			D	290	1200		550
					(LOC)		
NINJING LUKOU ZSNJ	RWY 24	CAT I	C	430	1000		550
			D	430	1400		600
					(LOC)		
HANGZHOU XIZOSHAN ZSHC	RWY 25	CAT I	C	340	1000		550
			D	340	1400		600
					(LOC)		
BEIJING ZBAA	RWY 36R	CAT I **	C	340	1000		550
			D	340	1400		550
					(LOC)		
HANGZHOU XIZOSHAN ZSHC	RWY 25	CAT I	C	340	1000		550
			D	340	1400		600
					(LOC)		

Date:	01/04/08	Page	18 of 24	FDP
Authority:	MFD	Issue	3	Rev
				00

EUROPE REGION

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AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNAT		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
AMSTERDAM EHAM	RWY 06	CAT IIIB	C	200	550		75
			D	200	550		75
BASLE- MULHOUSE LFSB	RWY 16	CAT IIIB	C	260	650		125
			D	270	650		125
BIRMINGHAM EGBB	RWY 15	CAT IIIB	C	200	550		75
			D	200	550		75
BRUSSELS EBBR	RWY 25L	CAT IIIB	C	200	550		150
			D	200	550		150
COLOGNE EDDK	RWY 32R	CAT IIIB	C	200	550		75
			D	200	550		75
DUBLIN EIDW	RWY 28	CAT IIIA	C	200	550		200
			D	200	550		200
DUSELDORF EDDL	RWY 23L	CAT IIIB	C	200	550		75
			D	200	550		75
EAST MIDLANDS EGNX	RWY 27	CAT IIIB	C	200	550		75
			D	200	550		75
FRANKFURT EDDF	RWY 25R	CAT IIIB	C	200	550		75
			D	200	550		75
LONDON HEATHROW EGLL	RWY 09L	CAT IIIB	C	200	550		75
			D	200	550		75
LONDON STANSTED EGSS	RWY 23	CAT IIIB	C	200	550		75
			D	200	550		75

EUROPE REGION

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AIRPORT ICAO	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
LYON- SATOLAS LFFL	RWY 36L	CAT IIIB	C	200	550		125
			D	200	550		125
MANCHESTER EGCC	RWY 24R	CAT IIIB	C	200	550		75
			D	200	550		75
MUNICH EDDM	RWY 08L	CAT IIIB	C	200	550		75
			D	200	550		75
PARIS CHARLES DE GAULE LFPG	RWY 08L	CAT IIIB	C	200	550		125
			D	200	550		125
PARIS ORLY LFPO	RWY 06	CAT IIIB	C	200	550		125
			D	200	550		125
SHANNON EINN	RWY 24	CAT II	C	200	550		300
			D	200	550		350
ZURICH LSZH	RWY 16	CAT IIIA	C	200	550		200
			D	200	550		200

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

	ESCAPE ROUTES EN-ROUTE AIRPORT SUITABILITY	
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To comply with the requirements of CAR OPS 1.500, effective immediately we shall check the suitability of the en-Route diversion airports along the escape routes of all flight plans over high terrain area.

For easy selection of the diversion airports and as guidance, a table for each escape route chart has been prepared. Please cross check the diversion airports in the table against the escape routes chart.

The A320 & B767 have identical escape routes diversion charts due to approximate similar oxygen system. (13 & 12 minutes respectively)

Likewise, the A330 & A340 have identical charts due to similar oxygen system. (22 minutes)

Procedure:

Run a TEST flight plan and if the route is via high terrain area, the OFP displays the reference to the escape routes chart.

If any one of the required airport is not suitable and no alternative airport available for the intended segment on the escape route, change the OFP via non-escape route.

If there is no non-escape route for the sector, flight shall be delayed until the WX improves. Refer the Flight Leg Record (FPIFL) and List of Routes (FPLRT) to check availability and detail of routes.

Only routes in the 'FPIFL' shall be used unless informed by Navigation Services.

The diversion airport WX minima shall fulfil the conditions in OMA 8.1.2.3.4 & 8.1.2.3.5; i.e.

Planning Minima - En-route and Destination Alternates

TYPE OF APPROACH	PLANNING MINIMA/RVR
CAT II & CAT III	CAT I - (Note 1)
CAT I	Non-Precision (Notes 1 and 2)
Non-precision	Non-precision (Notes 1 and 2) plus 200 ft/1,000m
Circling	Visual

Note 1: RVR

Note 2: The ceiling must be at or above MDH

8.1.2.3.5. Planning Minima for En-route Alternate Aerodromes

An aerodrome shall not be selected as an en-route alternate aerodrome unless the appropriate weather reports or forecasts, or any combination thereof, indicate that, during a period commencing one hour before and ending one hour after the expected

Date:	01/04/08	Page	21 of 24	FDP
Authority:	MFD	Issue	3	Rev
				00

	Flight Dispatch Manual and Procedures
	ALL FLEET OPERATIONS

time of arrival at the aerodrome, the weather conditions will be at or above the planning minima in accordance with the Table in 8.1.2.3.4. above.

ESCAPE ROUTES ENROUTE DIVERSION AIRPORTS GUIDE TABLE

AREA OF OPERATIONS		
CHART REF	ROUTE SEGMENTS	ENROUTE ALTERNATES

A340/330 IRAN-TURKEY AREA		
CHART 1		OSAP/LCLK, OITT, OIIE
CHART 2A	ERZ.UL851.VAN VAN.UG81.BONAM.UL124.UMH ERZ.UR660.DASIS	LTAC, OSAP/LCLK OSAP/LCLK, OITT OITT
CHART 2 B		LTAC, OSAP/LCLK, OITT
CHART 3		OITT, OIIE, OKBK, OMSJ/OMRK/OMAA/OBBI

A320/B767 NORTH PAKISTAN AREA		
CHART 4		OOMS/OMSJ, OPKC/OOMS, OPLA, OPPS/OPRN

A320/B767 IRAN-GULF AREA		
CHART 1		GULF/OISS, OIIE
CHART 5		GULF/OISS, OIIE, OIMM
CHART 6		OIKB/OMRK/OMSJ/OMDB, OIZH, OIIE, OIMM

A320/B767 IRAN-TURKEY AREA		
CHART 2		OSAP/LCLK, OITT, OIIE/OIFM
CHART 3A	CRM.UL852/UW704 or ORKAN.UL615.ARKIN or TBN.UL851.DORUK or SONAD.UW704.ERZ or ARKIN.UL615.ERZ.UL851.DORUK or ARKIN.UL852.VAN DORUK.UL851.VAN or XXX.UL852.VAN VAN.UG81.BONAM HOCAM.UR660.DASIS.UL333.TBZ DASIS.UL223.UMH XXX.UL125.ZAJ	LTAC LTAC LTAC LTAC LTAC, LCLK LCLK OSAP/LCLK OSAP/LCLK OSAP/LCLK OITT OITT OIEE
CHART 3B		LTAC, OSAP/LCLK,
CHART 8	UL223 GULF, W3 OISS	GULF/OISS, OKBK, OIIE, OITT, LCLK

**Note: OSAP/LCLK means OSAP or LCLK. LCLK, OITT means LCLK & OITT.
: XXX – LAT/LONG**

Date:	01/04/08	Page	22 of 24	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
طيران الخليج GULF AIR	ALL FLEET OPERATIONS

ESCAPE ROUTES DIVERSION AIRPORTS WX MINIMA

Note: ** = LVP NOT AUTHORIZED

*** = NO NON-PRECISION APRCH (+200ft/1000m used for alternate)

(2) = 2 SEPARATE RWY

AIRPORT ICAO NAME	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER				
			MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
LCLK LARNACA (ESENBOGA)	22	CAT I	C	370	1600	200	800
			D	370	1800	200	800
LTAC ANKARA	03L	CAT II	C	200	550	100	300
			D	200	550	100	350
OSAP ALEPPO	27	CAT II	C	250	600	170	450
			D	260	650	190	450
OIIE IMAM K.	29R	CAT I	C	430	1000	225	600
			D	430	1400	225	600
OIMM MASHHAD	31R	CAT I	C	370	1000	265	1200
			D	370	1400	265	1200
OISS SHIRAZ	29L	CAT I	C	650	1200	226	600
			D	650	1600	226	600
OITT TABRIZ	30R ***	CAT I	C	455	1650	255	650
			D	455	1650	255	650
OIZH ZAHEDAN	35	CAT I	C	620	1800	616	2800
			D	620	2000	616	3200
OPKC KARACHI	25R	CAT I	C	390	1000	230	600
			D	390	1400	230	600
OPLA LAHORE	36R	CAT II	C	220	600	100	350
			D	220	600	100	350
OPPS PESHAWAR	35	NP	C	660	2200	460	1200
			D	660	2600	460	1600
OPRN ISLAMABAD	30	CAT I	C	400	1000	310	800
			D	400	1400	310	800

Date:	01/04/08	Page	23 of 24	FDP
Authority:	MFD	Issue	3	Rev
				00

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	Flight Dispatch Manual and Procedures
	A320 OPERATIONS

8.0 A320 OPERATIONS

8.1 A320 OPERATIONS TO INDIAN SUB CONTINENT

Gulf Air A320 aircraft do not have ETOPS approval, so all operations shall be with in 60 minutes of an en-route diversion adequate airport at the approved single engine speed in still air, ISA conditions. Therefore the following airports must be adequate for operations to destinations in the Indian Sub Continent.

MCT / KHI / BOM

Check ERA NOTAMS to ensure these airports are 'Adequate'; if not check if other airports can substitute; if this is not possible, advise IOC that the flight cannot operate with A320.

The southerly routes across the Arabian Sea to south of BOM are ETOPS routes; and caution is advised especially for sectors scheduled to be operated with B767 on ETOPS route.

Seasonal development of Tropical Storm/Cyclone in Arabian see may force change of routing; thus special care shall be given on route selection.

8.2 INDIAN SUB CONTINENT PAYLOAD CONSIDERATIONS

Problems arise on A320 tankering sectors that operate Gulf port to Gulf port then to the Sub Continent and return, e.g. AUH - MCT - BOM - MCT, because at the time of planning, normally 3 hours before STD first Gulf port, we do not have accurate PLD figures for 2nd and/or last Gulf departure port (e.g. from MCT), and if too much fuel is tankered into MCT, i.e. up to MLGW in BOM, and there is an increase in payload in MCT, baggage or fuel will have to be offloaded in MCT

Refer to FLIGHT DISPATCH PROCEDURE: FDP-14

8.3 SANAA OPERATION

Due to high elevation and terrain around the airport, landing and take-off performance is limited at Sanaa airport. This will affect the payload uplift from SAH airport and appropriate coordination is required at planning stage of the Computer Flight Plan.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.6

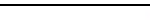
Date:	01/04/08	Page	1 of 2	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures	
	A320 OPERATIONS	

8.4 TAXI IN FUEL

On a tankering leg, to avoid fuel shortage at next sector due to fuel burn for taxi-in, an extra 200kg of fuel is automatically included on the total fuel uplift from the first airport payload permitting. This fuel figure is not shown on the fuel detail format on the OFP, but added with the FOD (Fuel Over Destination).

Date:	01/04/08	Page	2 of 2		FDP
Authority:	MFD	Issue	3	Rev	00

 طيران الخليج GULF AIR	Flight Dispatch Manual and Procedures
	B767 OPERATIONS

9.0 B767 OPERATIONS

9.1 B767 RANGE WITH CENTRE TANK UNUSABLE

Defects that render the centre tank fuel unusable reduce the amount of usable fuel in the remaining tank to 36,000 KGS. The corresponding reduced range will limit the aircraft to operating on the following sectors:

GULF - SAUDI
GULF - IRAN
GULF - AMM / DAM / BEY / LCA
GULF - KHI / BOM / DEL / TRV
GULF - CAI,

Additional sectors and allowable payload can be analyzed by running a TEST OFP by restricting the ramp fuel uplift to 36,000. The same can be done by using the following additional keywords with the TEST OFP inputs:

9.2 KATHMANDU OPERATIONS

Kathmandu is one of the destinations where aircraft performance is limited due to high airport elevation and obstructions due to airport surrounded by mountains.

The landing weight must be restricted when tankering or a full load is booked from Gulf airport. At the same time for departure flight from Kathmandu, a take of alternate must be selected regardless of departure weather condition. The Flight Planning System has 'DAC' as a stored takeoff alternate.

Refer to FLIGHT DISPATCH PROCEDURE: FDP-13.3

9.3 TAXI-IN FUEL

On a tankering sector, the flight planning system uplift an extra 100 kg of fuel for taxi-in on landing at next airport payload permitting. This fuel is part of FOD on the tankering sector Flight Plan.

Date:	01/04/08	Page	1 of 2	FDP
Authority:	MFD	Issue	3	Rev 00

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	Flight Dispatch Manual and Procedures
	A340 / A330 OPERATIONS

10.0 A340 / A330 OPERATIONS

10.1 A340 / A330 NOSE WHEEL STEERING (NWS) LIMITATION

There is a NWS angle limit of these aircraft that results in the need for larger turning pad dimensions. Refer to Navigation Services & Aircraft Performance Sections for the latest table of company airports that identifies runways or taxiways that are not suitable for the operation of these aircraft when required. Normally, Performance Sections will advise RWY condition when turning pad is affecting aircraft operations.

10.2 TAXI-IN FUEL

On tankering sector, Taxi-in fuel for A340 is 400kg and for A330 is 300kg.

10.3 CONTINGENCY FUEL POLICY CHANGE

Refer OMA Section 8.1.7.3.3 for Gulf Air contingency fuel policy.

Further to the standard 5 % trip fuel contingency, Gulf Air is authorized by Bahrain CAA to Use 3% contingency fuel for selected sectors and aircraft type.

Refer FDP-13.7.

Date:	01/04/08	Page	1 of 2	FDP
Authority:	MFD	Issue	3	Rev 00

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	Flight Dispatch Manual and Procedures	
	ETOPS	

11.0 ETOPS

11.1 GENERAL

ETOPS (Extended Twin Operations) is the acronym created by ICAO (International Civil Aviation Organization) to describe the operation of twin-engined aircraft over a route that contains a point further than one hour's flying time, at the approved one-engine inoperative cruise speed (under standard condition in still air) from an adequate airport.

11.2 DEFINITIONS

A. ETOPS:

Aviation regulations restrict two engine aircraft to flying on routes beyond 60 minutes at the approved one-engine inoperative cruise speed, under standard atmospheric condition in still air, from an adequate airport, for Performance Class A aero planes.

Performance Class A aero planes is with either:

Passenger seating configuration of 20 or more

Or

MTOW 45,360kg or more.

B. Adequate Airport:

An airport is adequate if it satisfies the requirement listed in the Operations Manual, Part A - Section 8.5.2.1. That is after considering the following criteria at the expected time of use:

- Minimum runway length 2500 meters or a length that is sufficient to satisfy aircraft performance limitation at the expected landing weight.
- Terrain: Highest MSA not above 10,000 feet AMSL.
- Airport Operational Hours
- Fuel availability (Jet A-1 or acceptable as per AFM).
- At least one letdown aid for instrument approach (Such as ILS, VOR, NDB) and runway light for night operation.
- Minimum RFF category of 4.

Date:	01/04/08	Page	1 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

	Flight Dispatch Manual and Procedures
	ETOPS

C. Suitable Airport:

An adequate airport with weather report, or forecast or any combination thereof, indicating that the weather conditions are at or above operating minima, as specified in OMA section 8.1.3.

D. ETOPS En-route Alternate:

Nominated suitable airports for particular ETOPS flights at the planning stage providing the minimum forecast weather conditions shall satisfy the table in OMA Section 8.5.6.1 for a valid period commencing one hour before the earliest ETA up to one hour after the latest ETA at the airport.

E. Threshold Distance:

The threshold distance is the approved distance covered in still air and ISA conditions in 60 minutes; at the selected one-engine-out diversion speed schedule at the associated cruise altitude.

The threshold distance approved for the B767 is **430 nm**.

F. ETOPS Area of Operations:

The Area of Operations is an area in which it is authorized to conduct a flight under ETOPS rules, and is defined by the maximum diversion distance from an adequate airport or set of adequate airports. It is represented by circles centered on the adequate airport, at a radius of the approved maximum diversion distance.

G. ETOPS Entry Point (EEP)

The extended range entry point is the point on an outbound route, which is 1-hour flying time at the approved single engine inoperative cruise speed under standard conditions from an adequate airport.

H. ETOPS Segment

The ETOPS segment start at the EEP and finishes at the ETOPS Exit Point and remains within 1- hour flying time at the approved single engine inoperative cruise speed under standard conditions from an adequate airport. An ETOPS route may and can have several successive ETOPS Segments well separated from each other.

I. ETOPS Exit Point

The first position on an ETOPS route, after the ETOPS Entry Point, at which the aircraft returns and remains within the declared 60-minute distance of an adequate airport(s). It marks the end of an ETOPS segment.

Date:	01/04/08	Page	2 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

J. Equal Time Point (ETP)

It is a point on the route, which is located at the same flying time from two suitable diversion airports.

K. Critical Point (CP)

The Critical Point is that point on the route, which is critical to the ETOPS fuel requirements, if a diversion has to be initiated, from that point. The CP is usually, but not always the last ETP within the ETOPS segment. Therefore, the CP should be computed carefully.

L. Critical Fuel Scenario

The Critical Fuel Scenario takes into consideration a normal flight and three different failure scenarios at Critical Point. The scenario resulting in the highest fuel requirement is referred to as the ETOPS Critical Fuel Scenario. With this FOB, the crew is best equipped to meet all the situations.

The three failure cases are:

- Engine Failure
- Depressurization
- Engine Failure & depressurization

The OFP fuel calculation from the critical point to the en-route diversion alternate is planned at both the single –engine and all-engine operating cause at 10,000 feet.

11.3 GULF AIR ETOPS FLEET FACTS

- Gulf air is currently approved and operates ETOPS for the B767ER fleet.
(Ref OMA 8.5.3.1)
- One-engine-inoperative Cruise Speed (Single-engine Speed) for the B767ER is:

330 KIAS / M.80
(Ref OMA 8.5.4.2)
- Aircraft Reference Gross Weight at threshold point from departures airport is:

170.000 kgs.
(Ref OMA 8.5.4.3)
- Maximum Diversion Distance at the approved speed for B767ER is:

	Flight Dispatch Manual and Procedures
	ETOPS

B767	
75 minutes	533 nm
60 minutes	430 nm

(Ref OMA 8.5.4.4)

- ETOPS Areas of Operation:

(Refer OMA 8.5.5.)

Note: Dispatchers must read and fully comply with OMA 8.5 - ETOPS Section.

11.4 ETOPS ENROUTE WEATHER MINIMA

For an airport to be selected as ETOPS en-route alternate at the planning stage, the weather condition, 1 hour before the earliest arrival time to 1 hour after the latest arrival time, should satisfy the minima in OMA 8.5.6.1.

That is:

Approach Facility Configuration	Alternate Airport Weather Minima	
	Ceiling	Visibility
At least 1 Nav Aid Providing precision or non-precision app	Add 400ft with DH or MDH	Add 1500m with authorised landing minima
2 Nav Aid providing precision or non-precision app for 2 separate RWY	Add 200ft to the higher of the 2 DHs or MDHs	Add 800m with the higher of the 2 landing minima

11.5 ETOPS Flight Planning

Refer to **FLIGHT DISPATCH PROCEDURE: FDP-18**

11.6 ETOPS ERA Weather Minima Guidance Table

Refer to **APPENDIX – A.**

Date:	01/04/08	Page	4 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

11.6 APPENDIX - A

ETOPS ERA'S WEATHER MINIMA GUIDANCE TABLE

The appendixes of this section contain ETOPS planning minima tables to be used as a guide when assessing suitability of en-route alternate airports (ERA) during the required validity periods specified in the OFP.

Normal en-route minima are also specified in the tables, primarily to allow for assessment of any TEMPO and PROB 30/40 deterioration included in the forecast.

ETOPS En-route Alternate Airport (ERA) PLANNING minima are for pre-flight ETOPS planning only; the destination minima apply in-flight.

The following tables are copies of those provided in the OM-A, section 8.5.6.1.

Approach Facility Configuration	Alternate Airport Weather Minima	
	Ceiling	Visibility
For airports with at least one operational navigation facility, Providing a precision or non-precision approach procedure	A ceiling derived by adding 400 ft. to the authorised DH or MDH	A visibility derived by adding 1500 m to the authorised landing minima
For airports with at least two operational facilities providing a precision or non-precision approach to at least two separate runways	A ceiling derived by adding 200 ft. to the higher of the two authorised DHs or MDHs	A visibility derived by adding 800 m to the higher of the two authorised landing minima

Warning: For any downgrade of navigation aid or if the wind is not favorable for the runway heading given on the table, refer to the relevant JEPPESEN approach chart to determine new minima.

APPLICATION OF AERODROME FORECASTS (TAF & TREND) TO PRE-FLIGHT PLANNING (ICAO Annex 3 refers)									
APPLICATION OF INITIAL PART OF TAF (For Aerodrome planning minima see CAR-OPS 1.297)									
a) Applicable time period: From the start of the TAF validity period up to the time of the applicability of the first subsequent 'FM' or 'BCMG' or, if no FM or BCMG is given, up to the end of the validity period of the TAF. b) Application of the forecast: The prevailing weather conditions forecast in the initial part of the TAF should be fully applied with the exception of the mean wind and gusts (and x-winds) which should be applied in accordance with the policy in the column 'BECMG AT and FM' in the table below. This may however be overruled temporarily by a "TEMPO" or "PROB" if applicable, add to the table below.									
APPLICATION OF FORECAST FOLLOWING CHANGE INDICATORS IN TAF AND TREND.									
TAF or TREND for Aerodrome planned as:	BECMG (alone), BECMG FM, BECMG TL, BECMG FM.. TL. In case of:		TEMPO (alone), TEMPO FM, TEMPO TL, TEMPO FM.TL, PROB FM.TL, PROB 30/40.			PROB & TEMPO			
	FM (alone) and BECMG AT:	Deterioration and improvement.	Deterioration	Improvement	Transient/showery conditions in connection with short-lived weather phenomena e.g., thunderstorms, showers.	Persistent conditions in connection with e.g., haze, mist, fog dust/sandstorm, continuous precipitation.	In any case		
Destination at ETA +/- 1hr.	Applicable from the time of start of the change.	Applicable from the time of start of the change.	Applicable from the time of start of the change.	Applicable from the time of end of the change.	Not applicable.	Applicable			
Takeoff Alt at ETA +/- 1Hr.	Mean wind should be within required limits.	Mean wind should be within required limits.	Mean wind should be within required limits.	Mean wind should be within required limits.	Mean wind and gusts exceeding required limits may be disregarded.	Mean wind should be within limits			
Destination Alt at ETA +/- 1 Hr. (See CAR-OPS AMC 1.255)	Gusts may be disregarded.	Gusts may be disregarded.	Gusts may be disregarded.	Gusts may be disregarded.		Gusts may be disregarded.			Deterioration may be disregarded; improvement should be disregarded including mean wind gusts.
ETOPS Enroute at earliest/latest +/- 1Hr.	Applicable from the start of the change. Mean wind should be within limits. Gusts exceeding X-wind limits should be fully applied.	Applicable from the start of the change. Mean wind should be within limits. Gusts exceeding X-wind limits should be fully applied.	Applicable from the start of the change. Mean wind should be within limits. Gusts exceeding X-wind limits should be fully applied.	Applicable from the end of the change. Mean wind should be within limits. Gusts exceeding X-wind limits should be fully applied.	Applicable if below applicable landing minima. Mean wind should be within required limits. Gusts exceeding crosswind limits should be fully applied.	Applicable if below applicable landing minima. Mean wind should be within required limits. Gusts exceeding crosswind limits should be fully applied.	Should be disregarded.		

Note 1: "Required Limits" are those contained in the Operations Manual

Note 2: If promulgated aerodrome forecasts do not comply with the requirements of ICAO Annex 3, operators should ensure that guidance in the application of these reports is provided.

	Flight Dispatch Manual and Procedures
طيران الخليج GULF AIR	ETOPS

Note: Ceiling is not required for Precision Approach Landing Minima at planning stage. However it is shown in the table below for reference only.

GULF REGION – ETOPS PLANNING MINIMA

AIRPORT ICAO NAME	RUNWAY	TYPE OF APPROACH	ETOPS PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
ABU DHABI OMAA	RWY 31	ILS CAT II	C	500	1800	100	300
			D	500	1800	100	300
MUSCAT OOMS	RWY 08	ILS CAT I	C	610	2100	210	600
			D	630	2100	230	600
SHARJAH OMSJ	RWY 30	ILS CAT I	C	600	2050	200	550
			D	600	2050	200	550

AFRICA REGION-ETOPS PLANNING MINIMA

AIRPORT ICAO NAME	RUNWAY	TYPE OF APPROACH	ETOPS PLANNING WEATHER MINIMA				
			A/C CAT	ALTERNATE		LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
ADEN OYAA	RWY 08	ILS CAT I	C	600	2300	200	800
			D	610	2300	210	800
ADDIS ABABA HAAB	RWY 25L	ILS CAT I	C	600	2300	200	800
			D	600	2300	200	800
DAR-ES- SALAAM -HTDA	RWY 05	ILS CAT I	C	600	2200	200	700
			D	600	2200	200	700
DJIBOUTI HDAM	RWY 27	ILS CAT I	C	600	2750	200	1250
			D	600	2850	200	1350
NAIROBI HKJK	RWY 06	ILS CAT I	C	600	2050	200	550
			D	600	2050	200	550
MOMBASA HKMO	RWY 21	ILS CAT I	C	600	2050	200	550
			D	600	2050	200	550

Date:	01/04/08	Page	7 of 10	FDP
Authority:	MFD	Issue	3	Rev 00

INDIAN SUB CONTINENT REGION – ETOPS PLANNING MINIMA

Note: **(2)** = 2 SEPARATE RWY

AIRPORT ICAO NAME	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	PLANNING		IN-FLIGHT/ LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
AHMEDABAD VAAH	RWY 23	ILS CAT I	C	650	2300	250	800
			D	650	2300	250	800
BANGALORE VOBL	RWY 27	ILS CAT I	C	610	2100	210	600
			D	610	2100	210	600
CHENNAI VOMM	RWY 07	ILS CAT I	C	650	2100	250	600
			D	650	2100	250	600
COCHIN VOCI	RWY 27	ILS CAT I	C	690	2400	290	900
			D	690	2400	290	900
COLOMBO VCBI	RWY 04	ILS CAT I	C	630	2100	230	600
			D	640	2100	240	600
DHAKA VGZR	RWY 14	ILS CAT I	C	630	2500	230	1000
			D	640	2500	240	1000
HYDRABAD VOHS	RWY 27	ILS CAT I	C	600	2050	200	550
			D	600	2050	200	550
KARACHI OPKC	RWY 25L/25R (2)	ILS CAT I	C	430	1400	230	600
			D	430	1400	230	600
KOLKATA VECC	RWY 19L	ILS CAT I	C	630	2300	230	800
			D	630	2300	230	800
MALE VRMM	RWY 36	ILS CAT I	C	730	2500	330	1000
			D	740	2500	340	1000
MUMBAI VABB	RWY 27	ILS CAT I	C	610	2300	210	800
			D	610	2300	210	800
NAGPUR VANP	RWY 32	ILS CAT I	C	640	2300	240	800
			D	640	2300	240	800
TRIVANDRUM VOTV	RWY 32	ILS CAT I	C	650	2300	250	800
			D	650	2300	250	800

ASIA REGION – ETOPS PLANNING MINIMA

Note: ** = LVP NOT AUTHORIZED
(2) = 2 SEPARATE RWY

AIRPORT ICAO NAME	RUNWAY	TYPE OF APPROACH	PLANNING WEATHER MINIMA				
			A/C CAT	PLANNING		IN-FLIGHT/ LANDING	
				CEIL/FT	VIS/M	CEIL/FT	VIS/M
BANGKOK VTBS	RWY 21R/ 21L (2)	ILS CAT I **	C	400	1350	200	550
			D	400	1350	200	550
CHIANG MAI VTCC	RWY 36	ILS CAT I	C	770	2400	370	900
			D	780	2400	380	900
DEN PASAR BALI WRRR	RWY 27	ILS CAT I	C	650	2200	250	650
			D	650	2200	250	650
JAKARTA SOEKARNO WIII	RWY 25R/ 25L (2)	ILS CAT I	C	410	1400	210	600
			D	410	1400	210	600
KUALALUMPUR WMKK	RWY 14R/14L (2)	ILS CAT I	C	400	1350	200	550
			D	400	1350	200	550
PENANG WMKP	RWY 04	ILS CAT I	C	690	2150	290	650
			D	690	2150	290	650
PHUKET VTSP	RWY 27	ILS CAT I	C	860	2500	460	1000
			D	870	2500	470	1000
RAYONG U- TAPHAO VTBU	RWY 18	ILS CAT I	C	600	2200	200	700
			D	600	2200	200	700
SINGAPORE WSSS	RWY 02L/02C (2)	ILS CAT I	C	400	1350	200	550
			D	400	1350	200	550
YANGON VYYY	RWY 21	ILS CAT I	C	670	2700	270	1200
			D	670	2700	270	1200

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	Flight Dispatch Manual and Procedures
	Flight Dispatch Procedures

12.0 FLIGHT DISPATCH PROCEDURES

12.1 Overview

The following pages are the Flight Dispatch Procedures (FDP) contained in Section 12 that extracts from the Operations Procedures Manual pertaining to Flight Dispatch. Although this duplicates the location of the procedures, the Operations Procedures Manual is not in a hard copy format. It contains multiple procedures from different departments in Operations including the Flight Dispatch Procedures that can be identified by the pre-fix 'FDP-'.

These procedures are the responsibility of Manager Flight Dispatch. In the event that a procedure had been amended, it is the MFD responsibility to ensure that the procedure within the FDMP and the OPM are identical.

Date:	01/04/08	Page	1 of 91	FDP
Authority:	MFD	Issue	2	Rev 00

	Flight Dispatch Manual and Procedures	
	Flight Dispatch Procedures	

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Date:	01/04/08	Page	2 of 91	FDP
Authority:	MFD	Issue	2 Rev	00

	Operations Procedures Manual	FDP-01 Shift Hand Over
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Title: Shift Hand Over

Type: Flight Dispatch

Purpose: To ensure incoming staff on each shift have a proper hand over of the daily operation from previous shift staff.

Scope: Flight Dispatch

References:

Introduction:

Due to the nature of the work, it is necessary to have a proper handover between the off going and the incoming shifts. This will ensure smooth handling of pending operational matters by the shift coming on duty.

Shift Hand over Procedure:

1. The incoming Dispatcher must verify weather data and become familiar with the overall weather affecting the areas of operation.
2. The departing Dispatcher must be ready to brief the following:
 - a) Weather conditions for departure, en-route, terminal areas; cyclonic or Volcanic Activity
 - a) Any pertinent log entries from previous shifts; any new instructions issued by e-mail or Memo; any changes or additions to the 'Warnings Board'.
 - b) Significant changes to NOTAM bulletins or new NOTAM issued since incoming Dispatcher's last shift that affect departure, destination / destination alternate, ETOPS en-route alternate airports and areas affecting Gulf Air routes.
 - c) Engineering messages (MEL/CDL), IOC change messages, aircraft deviation or Pilot Reports (PIREPS) received.
 - d) Any items like; system outage, equipment failure, etc; that may affect Dispatch operations.
 - e) NOTAM monitoring hand-over.
 - f) New Supervisor is to record the names of staff on duty in the Daily Log, allocate shift duties, and brief staff of any special requirements for the shift.

Date:	01/04/08	Page	3 of 91	FDP-01
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-01 Shift Hand Over
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Abbreviations:	CDL Configuration Deviation LIST
ETOPS	Extended Range Operations with two engine aeroplanes
MEL	Minimum Equipment List
NOTAM	Notice to Air Men
PIREPS	Pilot Reports

Responsibilities: Flight Dispatch Staff

Date:	01/04/08	Page	4 of 91	FDP-01
Authority:	MFD	Issue	2 Rev	00

	Operations Procedures Manual	FDP-02 Arrival / En-route Traffic DELAYS
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Title:	Arrival / En-route Traffic delays
Type:	Flight Dispatch
Purpose:	To alert all concerned parties and take necessary action for affected flights at planning stage.
Scope:	Flight Dispatch, IOC, Handling Agent, Operating crew, Navigation Services
References:	NOTAM, ATC/ACC, EUROCONTROL

Introduction:

Local Air Traffic Control (ATC) or Area Control Centre (ACC) may announce a delay at Gulf Air destination airport or en-route due to traffic congestion. This type of delay may cause major disruption, such as fuel shortage for a flight operating on affected sector, followed by a diversion, unless Flight Dispatch takes proper action at the time of planning the OFP. NOTAM or Message regarding traffic delay at airports is expected to be received in Flight Dispatch well ahead of departure time for proper action.

Procedure:

Flight Dispatch is responsible for taking the following steps at time of planning the OFP.

- Monitor Traffic Delays for arrival/departure or en-route traffic. These notices may be announced by NOTAM, or from Brussels EUROCONTROL, or by phone from either Local ATC authorities or through Gulf Air agents.
- Advise the information to IOC
- Advise Navigation Services to post it in the NOTAM brief if it is necessary

1. Arrival Delays at airports:

The following are possible causes of delays at airports that must be monitored:

- Special activities (VVIP)
- ATS staff shortage or slow down strike
- Airport congestion
- Weather
- Work in progress
- Down grade of Navigation facilities, etc

A. Delay at alternate airports:

- a) When delay message at primary alternates airport received, if the time window falls between GF arrival times, consider nominating the second alternate

Date:	01/04/08	Page	5 of 91	FDP-02
Authority:	MFD	Issue	2	Rev 00

- b) If the second alternate is very far and the duration of the delay at 1st alternate is not lengthy, add extra Company Fuel (CF) in mfs2 and run a TEST Flight Plan.
- c) Select the most economical OFP out of (a) and (b) above after comparison.
- d) If the delay period is not reliable, avoid nominating the alternate.

Advise crew reason for extra holding fuel with **'Edit Info' entry'**.

B. Delay at Destination airports:

- a) Check the duration of the delay from NOTAM if issued or as advised by airports authority.
- b) Discuss the matter with IOC for the length of delay and options for holding the flight on the ground before departure.
- c) If decided to proceed on time, the Flight Plan must carry extra fuel for holding over destination for the period of possible delay time.
- d) Add the required additional Company Fuel (CF) in mfs2 as per the NOTAM.
- e) Add comment on the Flight Plan for crew information.

Note: Company Fuel for lengthy type of notice is already stored in the database by Navigation Services.

2. En-route Delays and slot Allocation for Flights Departing Europe:

- a) Check Slot Allocation Message (SAM) originating from EUROCONTROL by the Flow Management Division (FMD) that is responsible for monitoring the traffic demand and capacity. These messages are received in Flight Dispatch via SITA on 'BAHOKGF' address.
- b) Follow up Slot Revised Message (SRM) for possible change of the original Calculated Take-Off Time (CTOT). If there is a change by more than 5 minutes, a new slot would be broadcast as New Calculated Take-Off Time (NEWCTOT)
- c) If no improvement and the delay time are significant, see the Regulation Type on 'SAM/SRM' message to identify the route or area of restriction.
- d) Create alternative route by avoiding the congested area using 'Edit Rtg' in mfs2.
- e) Go to 'Varient' and compare
- f) Select the best routes that satisfies the conditions
- g) Analyse the extra cost of the new OFP; and liaise with IOC if the penalty of the delay against the extra cost for the new OFP on new routing is feasible for change of plan.

Date:	01/04/08	Page	6 of 91	FDP-02
Authority:	MFD	Issue	2	Rev 00

- h) Communicate with Operating Crew, via ACARS or the ground staff, to advise the alternative plan you have. This may help the crew to start fuelling the aircraft early if extra fuel required.
- i) If it is decided to use the new OFP, advise ground staff the change and the requirement to re-file ATS flight plan.

3. En-Route Slot Delay from Other Areas:

- a) Area Control Centre (ACC) may issue a message to hold departing traffic when there is traffic congestion on the planned route.
- b) If the delay time on the ground is significant and may affect operations, check for alternative plan by avoiding affected area.
- c) Advise IOC to discuss the operational effects.
- d) If alternative route is available and a new routing is decided, release the new OFP.
- e) Advise the change to crew and handling agent.
- f) Re-file ATS Flight Plan.

Note:

EUROCONTROL some times suggests alternative routing for specific flight and sector to avoid the restricted area. Flight Dispatch must check the suggested route against other restrictions to make sure its compliance with other states preferable routing schem. (See example 3 below).

Example: 1. SAM

QN ATHIAXH ATHKKGF ATHWHOA ATHWJOA BAHOKGF
 .BRUEA7X 140955
 -TITLE SAM
 -ARCID GFA042
 -IFPLID BB65584178
 -ADEP LGAV
 -ADES OBBI
 -EOBD 050914
 -EOBT 1155
 -CTOT 1355
 -REGUL LCWN14
 -TAXITIME 0012
 -REGCAUSE CE 81

2. SRM

QN ATHIAXH ATHKKGF ATHWHOA ATHWJOA BAHOKGF
 .BRUEA7X 141150
 -TITLE SRM
 -ARCID GFA042
 -IFPLID BB65584178
 -ADEP LGAV
 -ADES OBBI
 -EOBD 050914
 -EOBT 1155
 -NEWCTOT 1330
 -REGUL LCWN14
 -TAXITIME 0012
 -REGCAUSE CE 81

3. Route suggestion

QN ATHKKGF ATHWHOA ATHWJOA BAHOKGF
 .BRUEA7X 140821
 -TITLE RRP
 -ARCID GFA042
 -IFPLID BB65584178
 -ADEP LGAV
 -ADES OBBI
 -EOBD 050914
 -EOBT 1155
 -ORGRTE N0468F390 VARIX UL995 RDS UM601 EVENO UR19 LCA
 UR655 BALMA/N0469F390 R655 CAK J222 LATEB/N0470F390 J222
 BASEM R785 ZELAF/N0470F410 UR785 RASLI UP559 KEDAT R219 KFA
 A791 BAH DCT
 -PTOT 1215
 -RRTEREF LGAVOBBIO
 -NEW RTE N0470F410 VARIX UL995 RDS UN136 DAL UW79 AYT UN135
 VESAR UB15 BALMA/N0512F410 R655 CAK J222 BASEM DCT
 ZELAF/N0470F410 UR785 RASLI DCT
 -RESPBY 0851
 -REASON OUTREG

Abbreviation:	ACC	Area Control Centre
	ADEP LGAV	Airport Departure
	ADES OBB I	Airport Destination
	ARCID	Aircraft Identification
	ATC	Air Traffic Control
	ATS	Air Traffic Service
	CTOT	Calculated Take-Off Time
	EOBD	Estimated Off-Block Date
	EOBT	Estimated Off-Block Time
	IFPLID	Flight Plan Identifier
	NEWCTOT	New Calculated Take-Off Time
	NOTAM	Notice to Air Men
	OPF	Operational Flight Plan
	ORGRTE	Original Route
	PTOT	Provisional Take-Off Time
	REGCAUSE	Reason of Regulation
	REGUL	Regulation Identifier
	RESON OUTREG	Reason Out of Regulation
	RESPBY	Latest time for Response
	RRTEREF	Re-route Reference
	SAM	Slot Allocation Message
	SRM	Slot Revised Message
	TAXITIME	Taxi Time

Responsibilities: Flight Dispatch Manager, IOC, Operating crew,
Navigations Services, Handling Agents

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	Operations Procedures Manual	FDP-03 Air TEST & Local Training Flights
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Title: Air TEST & Local Training Flights

Type: Flight Dispatch

Purpose: To Submit Traffic Clearance Request to ATS Reporting Unit

Scope: Flight Dispatch, ATC, IOC, Operating Crew,

References: CAR-OPS 1.300, OMA 8.1.10.9.1, Item 36, 8.1.14.3, ICAO DOC-4444

Procedure:

1. Introduction

Prior to departure, an ATS Flight Plan for the Air TEST or Local Training Flight shall be submitted to the Air Traffic Service or Briefing Office at the departure airport.

2. ATS Flight Plan Filing

- Since it is not a company requirement to prepare a computerized Flight Plan for Local Area Flights, the Dispatcher shall fill all the required items on the ATS Flight Plan and file it to the respective ATC unit.
- Check AIROPS or Fleet Office Instruction for availability of Training or Air TEST flight in the local area from given airport.
- Retrieve sample ATS Flight Plan, for Local Area - for the same fleet type, from the SITATEX file.
- The flight time required for Training Flight normally mentioned on the instruction letter from Fleet Office. Use that time to determine and amend the time and endurance, on item 16 & 19, of the ATS Plan. If it is an Air TEST flight, contact the operating captain or engineering to get an idea of the Air Test time.
- Check entire item on the ATS Flight Plan and amend all as required and file at least one hour before departure time.
- Give copy of the ATS Plan to operating crew. Departure from other airport to be transmitted to the respective airport handling agent via SITA or email.
- Any delay or time change more than 1 hour (depending on the ATC authority requirement), the ATS Plan to be retransmitted to the local ATC Office.

Example: From the sample ATS Flight Plan below, the highlighted information is based on information from AIROPS/Engineering or Fleet Office and the non-highlighted parts of the ATS portion is from the aircraft registration.

Date:	01/04/08	Page	11 of 91	FDP-03
Authority:	MFD	Issue	2	Rev 00

The ATC address is the departure airport 4 letters code followed by AFTN address.

FF **OBBIZPZX** OBBIZKZX
170105 OBBIZKZX
 (FPL-GFA**9100**-IX
 -A343/H-SRIXWEY/S
 -**OBBI0630**
 -N0469F390 LOCAL AREA ONLY
 -**OBBI0100 OTBD**
 -EET/**OBBI0100** REG/A4OLG SEL/RSAC COM/TCAS
 -E/**0230** P/TBN R/VES S/M J/LF
 -**RMK/TEST FLIGHT WITHIN LOCAL AIRSPACE**

Or

-RMK/TRAINING FLIGHT WITHIN LOCAL AREA

Note: If FL is given by operating crew, it can be amended on the above plan.

Abbreviations:	AIROPS	Airlines Operations Control System
	ATC	Air Traffic Control
	ATS	Air Traffic Service
	FL	Flight Level
	ICAO	International Civil Aviation Organization

Responsibilities: Manager Flight Dispatch, IOC, Manager Fleet Training
 Maintenance Control Centre

	Operations Procedures Manual	FDP-04 Backup System & Historic Flight Plan
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Title: **Backup System & Historic Flight Plan**

Type: Flight Dispatch

Purpose: To avoid disruption or cancellation of flights during MFS2 Flight Planning System failure

Scope: Flight Dispatch, IOC, All Scheduled Flights

References:

Procedure:

1. Introduction

During lengthy unscheduled maintenance or complete failure of the Flight Planning System, all scheduled flights without OFP for the sector will be affected. If the period of the failure is extended, it may disrupt the whole schedule for period of outage unless the option to use old OFPs' is arranged with the approval of Fleet Office.

2. Backup System

A. Mfs:2 Backup:

When there is a failure to access mfs:2 system in BAH-GF server, mfs:2 Vienna can be accessed as follows:

Click the shortcut on the desktop 'mfs2-VIE-Backup' (via Remote Desktop Connection)

- Insert User Name: remoteGFA01
- Insert Password: mfs24GFA
- Log on to: FWZ-EXT
- Go to 'Start'
- Click 'mfs2 Flightplanning'

The Input Sheet Preparation Program will be available as a back-up in Flight Dispatch to retrieve flights detail and crew name from Airops and AIMS for preparation of the daily input sheet.

When there is a complete failure to produce computer flight plan and/or unable to get updated NOTAM and WX information from Mfs2 system, see the procedure below:

B. NOTAM Backup:

The previous day NOTAM for the sector can be used from the 'Comm Page' of the mfs2:

- Select the same flight number and sector used recently
- Open 'Comm Page'
- Select 'NOTAM' from the Briefing window to be send
- Check new relevant NOTAM from BAH CAA to be send

Date:	01/04/08	Page	13 of 91	FDP-04
Authority:	MFD	Issue	2	Rev 00

New NOTAM should be checked from BAH CAA NOTAM that is received in Flight Dispatch SITATEX system (BAHONGF). The same should be checked against the route and airports. All relevant NOTAM for the sector should be sent for the crew.

C. WX Backup:

TAF for the required airports are available from BAH Met Office and the same should be delivered to crew if it is not given by the handling agent at departure airport.

Under normal condition Met information is provided by all airports except from BAH:

- Check with handling agent/airport/crew if Met information is provided locally
- If not, fax copy of the WX that is received from BAH Met office.

3. Use of Historic Flight Plan

A. When there is a complete failure to produce OFP from Mfs2, historical OFP's can be used subject to approval by the Chief Pilot (CP).

B. Once approval is obtained the following precautions are to be taken on selecting OFPs':

- The OFPs' must be of the same fleet type and engine type as the operating flight.
- The same sector and route shall be selected.
- Recent OFPs' to be used for better Upper Wind information
- As much as possible, get a plan with payload close to the operating flight.
- The following factors are required to be considered for recommending extra fuel:
 - o En-route weather
 - o Destination/alternate/en-route airports WX
 - o NOTAM
 - o Aircraft performance degradation

After system restored, fresh OFP should be calculated and Summary of plan should be released to all affected flight.

Such system failure and restoration should be advised to all concerned party including IOC as practical as possible.

Date:	01/04/08	Page	14 of 91	FDP-04
Authority:	MFD	Issue	2	Rev 00

The following sections should be advised system failure that may cause schedule disruption due to flight plan preparation:

- VPFO
- VPIOC & Ops Control
- VP Training
- CPs
- Navigation Services & Aircraft Performance
- MCC
- Affected stations and flights
- FWZ

Responsibilities: Manager Flight Dispatch, Fleet Office

Date:	01/04/08	Page	15 of 91	FDP-04
Authority:	MFD	Issue	2	Rev 00

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	Operations Procedures Manual	FDP-05 Special Weather & Volcanic Activity
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Title:	Special Weather & Volcanic Activities
Type:	Flight Dispatch
Purpose:	To ensure Gulf Air Flights are dispatched on a safe route and airport by monitoring Special Weather and Volcanic Activities in the area of operations
Scope:	Flight Dispatch, Operating Flights, IOC, Navigation Services
References:	OMA Section 8.1.6.3, 8.3.9

Procedure:

1. Introduction:

Special weather reports includes warning at airports, such as Microburst/Wind shear/FOG etc, and en-route Significant Weather Warning due to development of Tropical Storm and low pressure activity.

Weather data (Airports WX report and en-route charts) is accessible from the flight planning system.

Volcanic activity is monitored by respective Volcanic Observatory Centres and broadcasted through NOTAM/SIGMET report.

2. Airport Weather Warning:

- Warning at airport will be received in the flight planning system when such data are missing or below the required minima.
- This type of weather may affect landing and departing traffic depending on the intensity of the weather and the landing/take-off limitation of the aircraft type.
- Dispatch Officer on duty must advise the captain about the airport weather warning, when received.
- If aircraft is still on ground, the Dispatcher must check the company weather minima and take appropriate action when preparing the OFP
- If the weather for destination is below minima, inform IOC; check for any change of ETD if weather improvement is not predicted.
- When the airport weather report indicates that the wind is exceeding the limitation of the aircraft type for landing, distant alternate should be selected for fuel calculation.

Refer to the respective aircraft manual for wind limitations:

➤ B767 OM – Volume 1 - **Company Limitations**-L.20.3

Date:	01/04/08	Page	17 of 91		FDP-05
Authority:	MED	Issue	2	Rev	00

	Operations Procedures Manual	FDP-05 Special Weather & Volcanic Activity
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➤ A320/330/340 - FCOM 3.1.20 AIRPORT OPERATIONS

- If there is any payload restriction, Dispatcher to pass the information to Load Control for the respective airport.

3. En-route Storm Warning

- There should be a hand over of Significant Weather information if this may affect Gulf Air operations.
- Go through the SIGWX chart accessed from mfs2 system to analyse low-pressure area, or FRQCB, or Tropical Cyclone.
- If any significant weather is reported, get detailed and updated information from the nearest meteorological service provider. Met Office from India, Bangkok, Singapore and Hong Kong are the most common airports that monitor and provide Storm Warning development in their respective area.
- Analyse the information collected and check if the best route for the day is affected.
- If available routes are affected, edit routing to avoid.
- Put a note on the remark column for the crew and release the new OFP.
- Keep the flying crew updated of any current development of en-route weather when latest information is available.

4. Volcanic Activity

- Dispatcher shall not dispatch flight through known volcanic ash area, as it is hazardous to aircraft and must be avoided at all time.
- In the event of a broadcast of Volcano Activity Report, via NOTAM, or SIGMET, or AIREP, Dispatcher must check the area and avoid using route through the activity.
- Dispatcher must monitor Active Volcanic Regions that may affect Gulf Air routes. Currently Known regions for volcanic activities are:

Indian Ocean: Andaman Island, Barren Island
Indonesia: Bali, Jakarta area
Philippines: Mt. Pinatubo, Mayon

Date:	01/04/08	Page	18 of 91	FDP-05
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-05 Special Weather & Volcanic Activity
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Abbreviation:	AIREP	Air Report
	FRQCB	Frequent Cumulonimbus Cloud
	FLR	Flight Leg record
	METAR	Meteorological Report
	NOTAM	Notice to Air Men
	SIGMET	Significant Meteorological Information
	SPECI	Special Report
	TAF	Terminal Area Forecast

Responsibilities: Manager Flight Dispatch, Meteorological Service Providers, Operations Control, Navigation Services, Operating Crew

Date:	01/04/08	Page	19 of 91	FDP-05
Authority:	MFD	Issue	2	Rev 00

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	Operations Procedures Manual	FDP-06 Systems Failure
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Title: **Systems Failure**

Type: Flight Dispatch

Purpose: To ensure disruption is kept at a minimum when systems in Flight Dispatch fail.

Scope: Flight Dispatch, IOC, Operating Flights, Navigation Services

References:

1. Introduction

It is necessary to have communication with all concerned sections through all available means and all systems are normal to process Flight Dispatch work properly. The flight documents given by Flight Dispatch (OFP, NOTAM, Weather) are a CAA-Bahrain and company requirement.

2. Procedures:

Flight Dispatch has a responsibility for ensuring that all flight documents are transmitted and collected by operating crew for dispatching flights. Therefore the required basic systems must function at all time.

- The duty Dispatcher must take all precaution and advise concerned agents in the event of system outages to avoid flight disruptions.
- Duty Dispatcher must prepare early, all required documents and OFPs' for flights during pre-scheduled system outage.
- The following failure should be reported to the appropriate section for necessary action:

Unscheduled Flight Planning System failure
Interface system to mfs2 failure (Airops, Load, AIMS, MEL)
Falcon, Cargo, SITATEX, E-mail, Hermes, Telephone

MFS2 contact address:

- Telephone 9-01-43-1-522-6166-50
- Email: mfs2support@fwz.aero

GF-IT Help Desk: Tel ext – 8500

- If Flights are going to be affected and you are unable to produce OFPs', follow procedure FDP-04 to use Historic Flight Plan.
- Obtain a docket number from GF-IT for internal system failure

Date:	01/04/08	Page	21 of 91	FDP-06
Authority:	MFD	Issue	2	Rev 00

- Check other alternative means of receiving/transmitting data and flight documents; i.e. to communicate by telephone, using Fax machine, etc.
- Log down all events of system failure and consequences, and hand over to next shift staff for proper follow up.

Abbreviations:

GF-IT	Gulf Air IT
MEL	Minimum Equipment List
Mfs2	Modular Flight Planning System version 2
NOTAM	Notice to Air Men
SITA	Societe International Telecommunications Aéronautiques

Responsibilities: Manager Flight Dispatch, MFS2 Help Desk, GF-IT

	Operations Procedures Manual	FDP-07 Non-Standard Configuration Flight Planning
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Title:	Non-Standard Configuration Flight Planning
Type:	Flight Dispatch
Purpose:	To prepare and provide OFP for flights in Non-Standard Configuration
Scope:	Flight Dispatch, Navigation Services & Aircraft Performance Sections, Fleet Office, IOC, Engineering, CAA-Bahrain, Operating Crew
References:	Operations Manual, Aircraft manual, Navigation Service & Aircraft Performance Section

Procedure:

Introduction:

This procedure applies to dispatching a flight in a non-standard configuration. The flight plan can be prepared for an aircraft in the following condition:

- 1. Landing Gear Extended Operation for all fleet**
- 2. Three Engine ferry for A340**
- 3. One Engine en-route Analysis operations for A340**

1. Landing Gear Extended (GEAR DOWN) Operation:

Check with Navigation Services and Aircraft Performance section for necessary additional procedures and safety precaution.

Insert the appropriate MEL in mfs2 for consideration of aircraft gear down operation.

Revenue load can be taken on gear down operation.

NOTE: Fleet office approval, escape route, RTOW and other limitation should be checked thoroughly.

2. A340 Three engine Ferry:

The A340 three engine ferry flight is based on non-revenue load, i.e., no passenger and cargo. Because the A340 takeoff performance with one-engine out is considerably restricted, special attention should be given to the maximum takeoff weight before releasing a computer flight plan.

Date:	01/04/08	Page	23 of 91	FDP-07
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-07 Non-Standard Configuration Flight Planning
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Check with Navigation Service & Aircraft Performance Sections for additional procedure and requirements.

3. A340 One Engine En-Route Re-Analysis Operations

(Under Development)

Abbreviation:	ACARS	Aircraft Communication and Addressing System
	IATA	International Air Transport Association
	NAVAID	Navigation Aid
	OFP	Operational Flight Plan
	WPT	Way Point

Responsibilities: Manager Flight Dispatch, Navigation Service & Performance Section, Fleet Office, Operating Crew.

Date:	01/04/08	Page	24 of 91	FDP-07
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-08 Emergency Response
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Title:	Emergency Response
Type:	Flight Dispatch
Purpose:	To be aware of the responsibilities and task required by Duty Flight Dispatcher in the event of accident or incident
Scope:	Flight Dispatch, Crises Management Centre, Flight involved in accident/incident
References:	GF Corporate Emergency Response Manual

Procedure:

- In the event of an aircraft emergency situation, the duty Dispatch Officer must collect all the required information and put aside to be delivered to the Crisis Management Centre (CMC).
- Staff is not allowed to pass any kind of information related to the emergency to outsiders as Gulf Air Public Relation will do the announcement to concerned parties and agencies accordingly.
- For full description, and checklist during incident or accident of Gulf Air Aircraft, refer the Corporate Emergency Response Manual, **section 4.3**.

Onboard Emergency and Survival Equipment:

Boeing 767

Number of rafts – 6 (Yellow colour)
 Type of Rafts – Raft/slide self-stabilising with locator lights
 Pyrotechnics – Day/Night Flares
 Automatic Survival Beacon (ASB) – Rescue 99
 Emergency Medical and Water supplies:

First Aid kit
 Two Pints of drinking water per raft
 Water purification tablets
 Seasickness tablets
 Survivor Manual

Airbus A320

Number of rafts – 4 (Yellow colour)
 Type of Rafts – Raft/slide self-stabilising with locator lights
 Pyrotechnics – Day/Night Flares
 Automatic Survival Beacon (ASB) – Rescue 99

Date:	01/04/08	Page	25 of 91	FDP-08
Authority:	MFD	Issue	2	Rev 01

	Operations Procedures Manual	FDP-08 Emergency Response
---	---------------------------------------	--

Emergency Medical and Water supplies:

- 1 x First Aid kit
- Two Pints of drinking water per raft
- 50 x Water purification tablets
- 100 x Seasickness tablets
- 12 x Ammonia inhalants
- Survivor Manual

Airbus A330

Number of rafts – 6 (Yellow colour)
Type of Rafts – Raft/slide self-stabilising with locator lights
Pyrotechnics – Day/Night Flares
Automatic Survival Beacon (ASB) – Rescue 406
Emergency Medical and Water supplies:

- 1 x First Aid kit
- 8 x 4oz water bags
- 50 x Water purification tablets
- 100 x Seasickness tablets
- Survivor Manual

Airbus A340

Number of rafts – 6 (Yellow colour)
Type of Rafts – Raft/slide self-stabilising with locator lights
Pyrotechnics – Day/Night Flares
Automatic Survival Beacon (ASB) – Rescue 99
Emergency Medical and Water supplies:

- 1 x First Aid kit
- 8 x 4oz water bags
- 50 x Water purification tablets
- 100 x Seasickness tablets
- Survivor Manual

Abbreviation: CMC Crisis Management Centre

Responsibilities: Flight Dispatch Manager

Date:	01/04/08	Page	26 of 91	FDP-08
Authority:	MFD	Issue	2 Rev	01

	Operations Procedures Manual	FDP-09 Communication
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Title: Communication

Type: Flight Dispatch

Purpose: To be able to communicate with concerned departments, Airport Authorities and Aircraft effectively

Scope: Flight Dispatch, IOC, MCC, Commercial, Navigation Services & Aircraft Performance, Airport Agents and Authorities, Operating Crew, FALCON

References: OMA 2.4.2.2.4, 8.5.7.5

Procedure:

1. Communication with Gulf Air departments, Airport Agents & Authorities:

The communication media for sending and receiving important information that affects the operation should be via email and/or SITA. When there is an email and SITA communication failure, fax and telephone can be used depending on the type of information.

A. Payload Limitation:

If a given sector has unusual payload limitation due to WX or NOTAM, advise Load Control of the departure station and IOC for necessary action.

B. Aircraft MEL/CDL Heavy Penalties:

Aircraft MEL/CDL that have operational penalty will be advised by MCC to Flight Dispatch and Navigation & Aircraft Performance section.

If an aircraft MEL/CDL penalties cause a restriction to fly the given sector, Flight Dispatch shall advise MCC & IOC for necessary action.

If the penalty cause flying on a different routing that will increase the block time, the cost and may affect the booked load, advise the same to IOC and MCC for possible change of aircraft.

If change of aircraft is not possible and flight will operate under the same condition, Flight Dispatch should advise the payload penalty to Load Control of departure station.

C. Over Fuelled Aircraft

When an aircraft is over fuelled beyond the minimum sector fuel due to maintenance requirement or aircraft change due to last minute technical problem, the actual fuel in tank should be reported to Flight Dispatch by MCC.

Flight Dispatch will check the estimated booked load for the sector and compare against the maximum allowable payload based on the actual fuel in tank and advise the effect to MCC & IOC.

Date:	01/04/08	Page	27 of 91	FDP-09
Authority:	MFD	Issue	2	Rev 01

If it can not be defuelled and there is payload restriction, Flight Dispatch should advise the penalty to Load Control of departure station.

D. Destination WX:

When the reported Forecast and/or METAR for a flight is below minima during schedule time of arrival, IOC should be informed for necessary action.

If a decision has been taken by IOC to delay the flight, the new ETD will be reflected in Airops and the necessary correction for Flight Plan calculation and ATC plan should be done by the Dispatcher.

E. Arrival Traffic Delay:

When Flight Dispatch receives information for Arrival Traffic delay from various sources (Euro Control, Airport Authority, etc.) due to Strike or VIP Movement or some other reason, advise IOC.

If the delay is for specific time of period and it is valid for so many days, advise Navigation Service to update the timing in Mfs2 database.

If the possible delay time is not given, analyze the detail of the message and consult Fleet Office for approval to carry extra holding fuel.
Normally 30 to 60 minutes extra holding fuel is considered depending on the situation.

F. Sending Briefing Package

After the preparation of the Briefing Package for each flight, all BP should be sent to the respective airport agents with copy to 'flight.plan1' email for reference.

G. Filing ATC plan

ATC plan should be sent via SITA line using the given SITA and/or AFTN addresses. Copy to Airport agents should be send via the given email and/or SITA address. In case of SITA communication failure, copy of the ATC plan should be faxed to the respective ATC Briefing Office or ask the handling agent to forward the plan to them.

H. Reporting System Fault

All Communication and Equipment failure should be reported to GF IT on telephone 8500.

If a docket number or Incident Number is not issued immediately, another report should be filed via email to GF IT.

I. Reporting Mfs2 System & Data Fault

Any fault and/or error related to the Flight Planning system should be reported to FWZ, Navigation Service and GF IT depending on the cause of the problem.

Date:	01/04/08	Page	28 of 91	FDP-09
Authority:	MFD	Issue	2	Rev 01

J. Major System Problem

When there is a major system failure that may disrupt operation due to production of BP, it should be advised to the following party as practical as possible.

- VPFO
- VPIOC & IOC
- VP Training
- CPs
- Navigation Services & Aircraft Performance
- MCC
- Affected stations and flight
- GF IT
- FWZ

2. Communication with Aircraft:

A. Possible Reasons for Communicating with Aircraft:

- Weather update for significant en-route weather activity and destination/ 1st alternate when it is different from the forecast
- To pass OFP data to pilots on the ground
- Urgent NOTAM that affects operations of the flight
- En-route flight planning analysis when requested by pilots
- ETOPS flight watch

B. Contacting an Aircraft through Existing System

- VHF – Use Dispatch radio for calling aircraft within VHF range (Flights around Bahrain area)
- HF – Give the aircraft call sign and registration to FALCON to give a SELCAL to the flight.
- ACARS – If aircraft is fitted with ACARS system and it is serviceable, transmit your message via ACARS.
- SATCOM – If aircraft is fitted with SATCOM, use the normal telephone system and follow the instruction below on item 4.

C. SATCOM

To call the aircraft dial:

Date:	01/04/08	Page	29 of 91	FDP-09
Authority:	MFD	Issue	2	Rev 01

- 9 for an outside line
- 00 for International Dialling Code
- 331 49932333
- An automated voice will ask for a Personal Identification Number
- Dial 5407610 followed with the # sign
- An automated voice will ask for the Aircraft Identifier (AI)
- Dial the AI followed by the # sign
- At this point the crew should answer. Communicate using standard radio telephony in the same manner as an HF Radio patch
- When using SATCOM, log details in the Flight Dispatch Logbook

Aircraft Identifier Numbers

KA	34140071
KB	34140072
KC	34140073
KD	34140074
KE	34140104
KF	34140105
LG	34140117
LH	34140120
LI	34140121
LJ	34140122

Example

To call KC dial –

9 00 331 49932333, 5407610 #, 34140073 #

D. Contacting an Aircraft Through ATC

- This procedure is to be used when unable to contact aircraft in company radio and communication systems (ACARS/SATCOM)
- Check aircraft position from the flight plan to find out the FIR it is flying.
- Call the Area Control Centre (ACC) and request to call the flight and ask them to contact Flight Dispatch. (ATC telephone numbers are in ETOPS section of

FDMP and for the rest of FIR not listed there contact the local airports agent for assistant to get the phone number for the ACC.

E. Log detail of all operational communications with aircrafts.

Abbreviation:	ATC	Air Traffic control
	BP	Briefing Packages
	ETOPS	Extended Operations with Twin Engine Aircraft
	FDP	Flight Dispatch Procedure
	FWZ	flugwerkzeuge (Flight Planning Company)
	HF	High Frequency
	Mfs2	Modular Flight Planning System Version 2
	NOTAM	Notice to Air Men
	OFP	Operational Flight Plan
	SATCOM	Telephone Satellite Communication
	VHF	Very High Frequency

Responsibilities: Manager Flight Dispatch

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	Operations Procedures Manual	FDP-10 Flight Plan Input Sheets Preparation
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Title: **Flight Plan Input Sheets Preparation Program**

Type: Flight Dispatch

Purpose: To prepare the following day's Input sheets that displays all relevant information from Airops and AIMS.

Scope: Flight Dispatch, IOC, and Gulf Tech

References:

Procedure:

1. Introduction

The in-house developed program stored in Workstation 'D' is to be used for preparation of Input sheets. The assigned Flight Dispatcher doing rotation for next day must be able to open and run this program along with AIMS and AIROPS to download data required to be displayed on the Input Sheets.

Flight Plans inputs are categorized into the following 3 classes in Flight Dispatch for additional follow up and check during OFP preparation:

- Standard Flights (Non-ETOPS normal flights)
- Non-Standard Flights (Non-ETOPS flights with additional requirements)
 - o Special state requirement to over fly FIR
 - o Additional check for route availability due to multiple restrictions by NOTAM
 - o Area of Operation with: SIGWX (TS/TC)
 - o Volcanic Activity
- ETOPS (Flights beyond 60 minutes from en-route alternates)

2. Input sheets Preparation:

- A. Check with Operation Control if all planned scheduled and non-scheduled flights for next day are ready in the AIROPS system
- B. Open AIMS and run 'Report Generator'
Normally it will take about 2-3 minutes to load the data.
- C. Down load cockpit crew names for next day flights
- D. From Input sheets Program, select the date and time period required for all Input Sheets of next day to retrieve flights information from AIROPS. Normally it is for the following day starting 0001 UTC to 2400 UTC, but can be extended to the days after depending on the reason for Tankering and Crew convenience.
- E. Select the Tanker Box. Dispatcher must take extra care to check Fuel Index manually against the airports from the sheet that received from Fueling Section.

Date:	01/04/08	Page	33 of 91	FDP-10
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-10 Flight Plan Input Sheets Preparation
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- F. Add additional comment for each leg in the Remark Box when required.
- G. Select the ETOPS, Non-Standard and Standard flights separately from the given selection box.
- H. Print the Input sheets for the selected ETOPS, Non-Standard & Standard Flights separately.
- I. Double check the work against the flight list from AIROPS and add/remove inputs to match the list on the Daily Movement Plan for that day.
- J. Add/Correct Input's comments by hand whenever required.

Note: For full steps where and how to access the input sheet program, refer the description on '**Appendix A**' of this procedure

Abbreviations:

AIMS	Airline Information Management System
AIROPS	Airlines Operations Control System

Responsibilities: Flight Dispatch Manager

Date:	01/04/08	Page	34 of 91	FDP-10
Authority:	MFD	Issue	2	Rev 00

Date:	01/04/08	Page	35 of 91		FDP-10
Authority:	MFD	Issue	2	Rev	00

- 9) "Format Options" box appear. Click OK with out change.
- 10) Select "[C:]" drive.
- 11) Double click "Daily Flight_Airops"
- 12) Select "crewnames.xls" to change file name then click "Save".
- 13) Click "Yes" to overwrite existing file.
- 14) Open the same file (crewnames.xls) and do saving again in the same folder.
Click "Yes" to replace existing file with latest Excel Format.

(Now crew names are ready to be loaded from Input Sheet Program Main Menu)

B. Running Input sheets Program:

- 1) From the desktop, double click the shortcut to "DCFP Input.mde".
- 2) Enter password "cfpinput". The screen for Main Menu (OFP Input Preparation) will be displayed.
- 3) From top left corner of the screen, click "Upload Latest Schedule from AirOps"
- 4) Type date/time "from" in given format and Enter.
- 5) Type date/time "to" and Enter.
- 6) Click "OK". You are now back to the Main Menu screen.
- 7) Click "Convert Latest Schedule to Working DB".
- 8) Click "Confirm-Latest Schedule to DB".
- 9) Click "OK".
- 10) Click "Exit". Now the "From - To" box on the right corner should reflect the selected time period.
- 11) Click "Upload Crew Names".
- 12) Double click "Upload Crew Names".
- 13) Click "OK" then Exit.
- 14) Click "Open Schedule Already Loaded"

C. Using The Tankering Boxes:

- 1) Multi sector Tankering for full CFP input:
 - Click box "TB" on first sector.
 - Click box "TE" on last sector.
 - Input between TB and TE will be displayed in Tankering sequence later on the print out.
- 2) Multi sector Tankering followed by single "TEST" plan input :
 - Click box "TB" on first sector.
 - Click box "TE" on last sector.
 - Click box "Tb" and "Te" on last sector of the TEST plan.
 - This will initiate duplication of the "TEST" plan sector on the print out.
- 3) Multi sector Tankering followed by two "TEST" plan input:
 - Click box "TB" on first sector.
 - Click box "TE" on last sector.
 - Click box "Tb" on the first TEST plan input.
 - Click box "Te" on the second TEST plan input.

4) Adding a TEST blank input to a single input segment:

- Click box “TB”, “TE”, “BLKINPT” for the same segment.

5) Multi sector Tankering followed by blank TEST input :

- Click box “TB” on first sector.
- Click box “TE” on last sector.
- Click box “BLKINPT” on last sector.

D. For ETOPS and NON STANDARD flights click box under “ETOPS” or “N-STD”.

E. PRINT:

- 1) “Print All”: will give you full input sheet report on the screen in time (ETD) sequence including for NON-STD and ETOPS plan input.

Follow the following steps:

- Click-File; click-Print then click-OK.
- Click “close” button to go back to the “SCHEDULE” screen.

- 2) “Print N-STD Only” for NON-STD input sheet print out.

- 3) “Print ETOPS Only” for ETOPS input sheet print out.

When STD input followed by a TEST ETOPS/NON-STD input with “Tb/Te” marked in the box, on the print out for ETOPS/N-STD it will duplicate the TEST plan that you may discard as it is an extra copy.

EXIT to go back to “Main Menu”.

F. Comparing Latest data from AIROPS against data in the Inputs Program:

- 1) This is to see any change (Time/ACFT/Sector/Flt No) against the schedule loaded for input sheet preparation. Thus **KEEP RUNNING THE DATA BASE CONVERSION ONLY ONCE A DAY (Step B.7)**, so that a comparison of change between the new and the original input schedule can be done.

2) STEPS TO DO COMPARISON:

- Click “Upload Latest schedule from AIROPS”
- Enter date/time begin and end of schedule for the required period of time for comparison.
- Click “Compare Latest and Already Loaded Schedule”.
- Click “View the Latest Change Only”.

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	Operations Procedures Manual	FDP-11 OFP Request Procedure
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Title: **OFP Request Procedure**

Type: Flight Dispatch

Purpose: To ensure all Gulf Air Flights receive an OFP that is accurate and complies with company policy and CAR-OPS 1 requirements

Scope: Flight Dispatch, Navigation Services & Aircraft Performance Sections, Meteorological Office Centre, Load Control, IOC, Bahrain Maintenance Control, Operating Flights, MFS2 Flight Planning System

References: Operations Manual, CAR-OPS 1.1060, Aircraft Manuals, JEPPESEN Airway Manual, AIP, and NOTAM

Procedure:

Introduction:

The OFP is prepared by Flight Dispatch for all Gulf Air flights. The OFP is a regulatory document and shall satisfy the requirement of CAA-Bahrain and GF Operations Manual.

Flight Dispatcher must analyse all the factors and check all data that required for the preparation of flight plan before releasing the OFP for safe, legal and economical operations.

Contents included in the procedures:

1. Flight List Page
2. OFP Preparation Time
3. Checking Flight Plans Data
4. Operating Minima for Origin, Destination & Alternate Airports
5. Generating Flight Plans
6. Checking Flight Plans Output
7. Distribution of Briefing Package
8. ZFW Change after OFP Produced
9. Destination & Alternates Minima Guidance Table
10. Checklist for OFP Preparation

1. Flight List Page:

To access the list page, for all flights for the date, the Flight Planning Client needs to be opened. Each Flight Dispatcher should use his/her own allocated User ID and Password for accessing the mfs:2 system.

The User ID and password for log-in into the system is individual staff number and password (GF Network User Name and Password).

By opening the flight plan client, the List Page for all flights will be displayed. Data read from other systems (Airops, AIMS, PAX & CGO System and SAP-MEL/CDL, etc) that will be used to process flight plan calculation are available through interface. Missing data shall be checked by the Dispatcher from the source of the information to be updated in the flight planning module of mfs:2.

Date:	01/04/08	Page	39 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

Desk Assignment:

The Duty Supervisor will assign flights to each Work Station.

The Work Stations are named as:

- **A** : All Access - Supervisor Work Station
- **B** : Supervisor Work Station
- **C** : Front Desk Flight Dispatcher Work Station
- **D** : Rear Desk Flight Dispatcher Work Station

Supervisor sign-in access has permission to allocate flights to each station and display all flights assigned to each work stations. After allocation of flights to each work station, Other Dispatchers will only view and work on flights assigned to their station only.

Please use the 'Filter Flights' option for easy identification and list all flights that are required to be displayed in accordance with the chosen filtering criteria.

2. OFP Preparation Time

The later we plan the better the EZFW and WX information should be. However there are other considerations, such as the various Civil Aviation Authority requirements, i.e. for slot allocation in Europe and the need to give staff at out stations enough time to print and distribute the Briefing Packages to crew.

The following the recommendation timing to release Flight Plans:

- Flights that pass through European airspace 4-5 hours before ETD
- All other sectors, 3 to 4 hours, but not less than 2 hours before ETD to allow enough time for collection and distribution of documents at out stations.
- Tankering sectors could be released earlier depend on the timing of the first flight.

Note: Start preparation around 2 Hrs before OFP releasing time to have enough time for unforeseen disruption. The **Flight Page Data, Varient, NOTAM Page, MET, MAP** can be started early followed by late review of the MET, ZFW & Airops data.

3. Checking Flight Plans Data

The flight plan data are comprised of Aircraft Registration, Sector, STD, Crew Name, MEL, etc. These data are automatically collected from the different available systems that are integrated to the "mfs2" system.

Dispatcher should cross-check accuracy of these data by referring the source of the information. On the case of missing data due to system problem, Dispatcher should enter data manually it into the appropriate field for correct flight plan calculation.

Date:	01/04/08	Page	40 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

3.1 Missing Flight:

Do the following when a flight is missing from the List page or to add a new flight:

- Select "New Flight" dialogue from the "List Page" window
- Fill all the required field on the given window (From, To, STD, STA, etc.) and click "OK" to save it in the "Flight List" page

3.2 Edit Existing Flight:

Do the following to edit existing flights or add more sector when a flight divert and the same is not taken by the system automatically:

3.2.1 Edit Flight:

- Select "Edit Leg" dialogue in the "List Page" window
- Change the required field only (Date, Org, STD, Dest, STA, Callsign) if the change is required on the existing flight
- Click 'Rename'
- Click "Rename" to add suffix "D" or "A" on the flight number as required

Change of Call Sign mean, only the flight number on the flight plan will change for ATC communication purpose. This can be used to add the suffix "A" or "B" to avoid conflicting call sign in the Gulf ports and the suffix "D" for the case of delayed flight.

3.2.2 Adding New Leg to Existing Flight during Diversion:

- It can be created as a new flight (Refer 3.1 above), or
- Select "Edit Leg" from the "List" page
- Enter all the required field (Date, Org, STD, Dest, STA, Callsign)
- Click "Insert" button
- Select "Keep Original"
- Click "OK"

Note: This entry will copy the common data from the original flight leg; such as crew name, Reg, MEL & Load.

3.3 Selection of Aircraft Registration:

When the interface failed or tail assignment has not been done, use the Aircraft pull-down window in 'Flight Page' for selection of aircraft, or type the A/C Registration in the 'List Page' under 'ACReg' column.

3.4 Load:

The Passenger (Pax) and Freight (Frgt) figures are taken from Falcon and Cargo system automatically at a given time interval.

Date:	01/04/08	Page	41 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

When the load information taken through the interface is different from the load reported by the respective station, insert the station EZFW figure in the appropriate field, assuming that station report is more up to date.

When the “Pax/Frgt” information reads ZERO or during interface failure, load information should be checked and entered manually.

3.5 Aims:

To Edit or Add Capt/FO name, insert staff number of the operating cockpit crew in the appropriate field under 'Edit Crew' box and click 'OK'.

3.6 Aircraft MEL /CDL:

If an aircraft is under MEL/CDL and the item is loaded in “mfs2” system, then the MEL page tab will have RED color for the selected active flight.

Aircraft MEL/CDL will be loaded in the system by SRT. But when there is a last minute addition and/or deletion of MEL/CDL items, informed by either pilots or Maintenance Control, Dispatcher should take appropriate action in the MEL List page of “mfs2” system.

Dispatcher MEL entry is valid only for one time use. To make it active until it is cleared, update must be done by the appropriate section (SRT) or Navigation Services.

To Edit or Enter MEL data:

- Select “MEL” page
- To ignore MEL implication select “Ignore” button for the appropriate MEL list
- To add new MEL, click “ADD” button to open the MEL dialog page, then
- Select the appropriate MEL from the list and click “OK” to activate

Note: Flight Dispatcher shall cross check the MEL/CDL penalty against aircraft manual at all times, and any discrepancy should be reported to Navigation Services for corrective action.

4. Operating Minima for Origin, Destination & Alternate Airports

Reference: OMA 8.1.2.3.2.
 OMA 8.1.2.3.3
 OMA 8.1.2.3.4
 OMA 8.1.3.2
 JEPPESEN Airway Manuals
 Destinations/Alternates Listing Table

4.1 Take-Off Alternate:

- a) When performance or weather condition prevents a flight returning to the departure airport, a take off alternate must be nominated. The landing minimum for departure airport is the applicable minima for approach and landing.

- b) Weather at take of alternate must be at or above the applicable landing minimums, during the period commencing 1 hour before and ending 1 hour after the expected ETA at the airport.
- c) Maximum distance for take off alternate shall not exceed the following:
 - A320 = 376NM
 - A330 = 430NM
 - A340 = 875NM
 - B767 = 430NM
- d) Twin-engine aeroplanes are restricted to 1 hr flying time at single engine cruising speed. For ETOPS certified operation, this can be extended to the maximum authorized ETOPS diversion time with condition of crew qualification and ETOPS maintenance check for the aircraft.
- e) Normally one or more of the alternates nominated in the 'Destination/Alternates' listings will be within the required range for use as a take-off alternate. If not, a take-off alternate should be nominated. For ETOPS approved and qualified flight, the take-off alternates can be extended upto the maximum approved ETOPS threshold distance.
- f) On the absence of suitable alternate airport, flight will be forced to be delayed until the origin airport weather improves.

4.2 Destination Minima:

Destination airport is considered suitable, if the weather reports or forecast for the destination indicate that from 1 hr before to 1 hr after ETA, it is above the applicable minima as per OMA 8.1.2.3.4 item 1.

The applicable minima are:

- a) For precision approach, RVR/visibility specified in accordance with the applicable airport operating Minima (AOM). (Refer Concept of Minima in OMA - 8.1.3.2)
- b) For a non-precision approach, VIS/RVR and Ceiling at or above MDH.
- c) If the destination weather does not meet the above minima or nil weather for destination, two suitable alternates must be selected, and the most distant one to be nominated as primary alternate on the OFP for fuel calculation.

4.3 En-route Alternates Minima:

A destination alternate shall be specified for all flights prior to take-off unless it satisfies the conditions prescribed in OMA 8.1.2.3.3 item A.

The requirements for selection of at least one or two alternates is explained as follows

- a) At least one destination alternate to be specified on the OFP request provided destination weather is above the applicable minima.
- b) Two usable destination alternates to be nominated if the weather reports or forecast for the destination indicate that from 1 hour before to 1 hour after ETA, it is below the applicable minima or no weather information for destination.

c) For escape route flights, en-route alternate ports should be above en-route minima.

Planning Minima for Destination & En-route Alternate:

TYPE OF APPROACH	PLANNING MINIMA /RVR
CAT II & CAT III	CAT I (Note 1)
CAT I	Non-Precision (Notes 1 and 2)
Non-precision	Non-Precision (Notes 1 and 2) plus 200 ft/1,000m
Circling	Visual

Note 1: RVR

Note 2: The ceiling must be at or above MDH

Note: Refer to OMA for any minima update and JEPESSEN for AOM update.

5. Generating Flight Plans

5.1 Available Route:

If route is stored for selected leg, the available Routing menu will allow you to view the list of routes available for the chosen city-pair.

The system is set for auto calculation (STD-10 Hrs) by going to the Variant page – load all available routes and calculate.

Do the following to select the most convenient route for the day:

- a. Check availability of data:
 - Airops information (Reg, Leg, STD)
 - Load
 - MEL/CDL if reported
 - Tankering
 - NOTAM – CF (Company Fuel)
 - WX (Airports and En-route)
 - Alternate
- b. Go to “VARIANT” page
- c. Select “Cr All” then “OK” (to delete existing variants)
- d. Select “CP Comp” (City Pair Comparison)
- e. Select the best route. (the variant with ‘*’ is the active route)
- f. Go to “Flight” page
- g. Tick the “EROPS” box & recalculate

5.2 Creating and Editing Route:

This page will allow Dispatchers to create and/or edit and optimize routings for given city-pair and en-route flight planning when it is required. Selection of waypoints to join to departure and arrival ports; option to avoid routing; waypoints and FIR can be processed on this page.

Do the following to Creat and/or Edit Routing:

Date:	01/04/08	Page	44 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

- a) Select "Edit Rtg" button from the "Flight" page
- b) Under "Routing Definition" type "OPT" for Optimum selection of the route, or insert "AWY" & "Wpt" in the field
- c) Click "Optimize" for further Waypoint, FIR and Segment edition
- d) The field "Avoid Closed Airways" & "Avoid Restricted Areas" must be marked always
- e) Insert "FIR" and/or "Wpts" and/or "Segs" to be avoided if required
- f) Click "Optimize" for the optimizer to build the route according to the criteria you put.
- g) If route is not calculated and "No Way Found" displayed in the "Status" window, do the following:
 - Add waypoints at departure airport and select "Take all"; and/or
 - Add waypoints at arrival airport and select "Take all", then click "Optimize"
- h) Click "OK"
- i) Rename the route (Z1 ... Z99) and you may leave a note under the box "Via" for your easy identification of the route.
- j) Select "Save Private" to keep the route for this leg only
- k) Click "OK" to go back to "Flight" page
- l) If there are other listed routes in the "Variant" page do comparison and follow steps on 5.1 above starting from bullet "b".
- m) If this is the only best available route, select "EROPS" and recalculate on the "Flight" page

5.3 NOTAMS:

The NOTAM page provides NOTAM Briefing for selected flight or chain of flights. Additional NOTAMS can be added by the Dispatcher by selecting the four letter code for the airport and/or FIR if required. To view the NOTAM, the flight must always be calculated to get the routing for collection of valid NOTAMS for the route.

Make sure that the Flight Page status displayed 'MRWUC' before going to view NOTAM; otherwise you will receive an error message 'Not Calculated'.

Do the following to read NOTAM before flight plan calculation:

- Select NOTAM page
- Click "Direct"
- Enter airports name under Airports field
- Enter FIR name under FIR field
- Click "View"

For airport NOTAM that cause traffic delay during arrival time, Dispatcher shall consider extra holding fuel on the flight plan calculation, by selecting a time value in mfs2, depending on the duration of the delay.

- Use "CF" entry for required extra fuel

5.4 Weather:

By selecting the Met Page, you can access the actual METAR and Forecast Weather information for departure, en-route, destination and destination alternates and any other airports required to be included. The Dispatcher shall check the METAR/TAF of the

Date:	01/04/08	Page	45 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

respective airports to monitor the automatic selection of suitable alternates by the flight planning system.

The en-route WX information display by the system is only for those airports that give 60 minutes EROPS coverage, and any additional WX information must be added manually when required.

- Open “MET” page and click ‘Clr Req’
- Select “GENERATE” to list city pair ports. Add more airports if required.
- Click at the end of the airports list and press “ENTER” on your keyboard.
- Save the output

The SIG WX Chart shall be checked and analysed to make sure that each flight is routed via a safe en-route weather phenomena.

Applicable SIG-Chart valid for the route shall be generated to analyse the en-route weather. The following are area of operations and Name/Description of SIG Chart that are recommended to be used to analyse the WX:

Area of Operation	SIG Chart Name	Description
Europe-Gulf-Far East VV	PGZE06	Asia, Europe & N. Africa
Gulf-Middle East-Gulf	PGCE15	Asia & Europe

SIG WX Chart can be reviewed by overlying the route on the Map Display after calculation of the flight plan:

- Load the route in the “Flight Page” first
- Select “Map”
- Select “Projection Parameters” box
- Click “Select SIG-Chart”
- Select the appropriate chart in terms of area and time of the flight
- Click “OK” to analyse the WX

Note: Avoid significant WX activity area like FREQCB, Tropical Storm & Tropical Cyclone.

5.6 Alternate Airport Selection:

Take-off, En-route and Destination alternates that is selected by the system automatically shall be cross checked. When there is a need for the Dispatcher to alter these alternates to fulfil different requirements, the same can be selected manually by accessing the respective field.

Do the following to add unlisted alternate when required:

- Click at the “Alternate” field
- Press “Insert” button on the key board to open the auto search dialog box
- Enter distance radius for auto search of alternates
- Click “?” symbol to search
- Select the appropriate GF approved airport
- Click “OK”

5.7 Fuel Tankering:

Fuel tankering is indicated by either a plus (+) or minus (-) value of dollar for carrying 1000kg of fuel between departure and destination airports. The **FEco** is the indicator for Fuel tankering that will be displayed with a “+/-” cost sign for approximate saving. The ‘Target’ field should read ‘Auto’ and the system is set to calculate the tankering legs STD-10 hrs.

Whenever there is a fuel saving (+) between the first and consecutive stations, the system will automatically analyze the fuel required for the next sectors and display the remaining fuel at the next airport in the ‘remF’ field of the Flight Page.

The system is set to limit tankering to MTOW/MLDW-2000. Dispatcher should review the remaining fuel for various reasons; such as to avoid possible payload penalty and/or aircraft swap

5.8 Extra Fuel:

When there is a requirement to add extra fuel due to NOTAM or when the fuel in the aircraft tank is more than the minimum required, the additional fuel should be included on the flight plan calculation.

5.9 Take –off and Landing Weight Performance (TOW/LDW):

Mfs:2 will use in future, the Standard Computerized Aircraft Performance program (SCAP) and forecast or predicted weather data to determine the Max TOW/LDW calculation for system selected runway based on the wind report.

However, for the case of operation, like KTM and SAH, where LDW restriction is based on the worst TOW figure at the airport, a manual entry of LDW value is required before flight plan calculation.

- Open the ‘Flight Page’
- Go to ‘Aircraft Weights and Limits’ box
- Select ‘LDW’ box on the right column (left column is actual value)
- Enter the ‘LDW’ figure for the station and aircraft type as required to limit maximum landing weight.
- Calculate flight plan

6. Checking Flight Plans Output

Dispatcher should carry out a quick check on OFP such as:

- Load
- Tankering
- Extra Fuel/CF due to NOTAM
- MEL/CDL if applicable
- Routing
- En-route WX
- Alternates
- Wind Shear
- Flight Level
- Remarks (if required)

Date:	01/04/08	Page	47 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

7. Distribution of Briefing Package

The Briefing Packages (BP) that are required to be processed are:

- 1) **OFP** (It includes extra copy of summary plan for crew signature)
- 2) **MET**
- 3) **NOTAM**
- 4) **GRAPH** (Select Upper Wind, EROPS Circle, Alternates, Restrictions)
- 5) **SIG Chart**

Processing Briefing Packages (BP) from Communication Page:

- a) Select the packages one at a time from the "Msg Type" field and proceed to step '2' for each selection (OFP, SFP, MET, NOTAM, GRAPH & SIGCHRT)
- b) Select "ADD" button under the "BP" window (for SIGCHRT: select the appropriate chart that is valid for the route and time).
- c) Double click each package in the "PB" field and print (for BAH flights), if it is not assigned to "PSS"
- d) Select address from the address file (copy to flightplan1@gulfair.com is required for all flights for record)
- e) Click "Send".

Note: If the station has no email, all packages should be transmitted via SITA line except the GRAPH & SIG Chart which is required to be faxed.

8. ZFW Change after OFP Produced

- a) Flight Dispatch will release another OFP when for each fleet the ZFW for the sector vary more than the following figures:
 - A340/A330/B767 more than 5000 KGS
 - A320 more than 3000 KGS
 - For flights assigned to "PSS", If ZFW is more than 1000kg or less than 3000kg against the OFP, insert the new ZFW in mfs2 and recalculate OFP to have accurate figure for the Fuel Message.
- b) For multi-tankering sectors, check the effect of the new OFP on next leg, and change the full plan if the carried over fuel is affected.
- c) Put a Remark for the crew for change of flight plan. Advise station to avoid confusion.

9. Destination & Alternates Minima Guidance Table

- A. This table is produced to refer to weather minima for destinations and at least the first and second alternates when preparing the OFP.
- B. The table is applicable to existing navigation aids and Airport Operating Minima as published by Jeppesen.
- C. If a NOTAM is issued downgrading existing navigation aids, the Minima Table must be ignored and the Jeppesen Approach Charts should be used to determine the new minima.

Date:	01/04/08	Page	48 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

- D. The Jeppesen Manuals check list must be checked every week for revisions in the Terminal Section and changes must be brought to the attention of Manager Flight Dispatch for immediate corrective action.

10. CHECKLIST FOR OFP PREPARATION

1. Open the 'List' page for Desk Assignment
2. Select Work Station from 'Desk' pull down menu

Duty Supervisor will allocate flights for each work station.

- Select 'n/a' (not assigned) under 'Desk' pull down menu to filter un assigned flights
 - Point on a Flight from 'List' page and right-mouse click to see lists of functions
 - Point on 'Desk Assgn' and select the Work Station (B, C and/or D) to assign
 - Repeat the same for all un-assigned and reassignment of flights
3. Select a Flight from 'List' page by double clicking anywhere between Fpfx and Dst columns
 4. The Flight page opens. Check data accuracy as per following list and update as necessary:
 - Aircraft Registration
 - Crew Names
 - MEL/CDL (Open the MEL manual to check the correction/s)
 - Route
 - Pax/Cargo/ZFW
 - Target set to "Auto", for auto tankering and FEco (fuel Economy)
 - CF (Company Fuel – such as holding fuel at destination) if required
 5. Normally the system calculates the minimum cost flight plan for all sectors at ETD-10 hrs, unless there is error. 'C' (calculated) is added in the 'Status' field, if calculated:
 - Go to Variants page and check if flight plan is calculated for the available variants and the min cost is already activated. If not, go to step 6.
 - Select 'EROPS' in the 'Flight' page
 - Calculate
 - If FEco is (+) and no tankering, go to the next leg (>>) and calculate the flight plan and go back to the original leg (<<) and calculate
 - Check Route/Flight Level/Restrictions/error message/s
 - Click MAP, ETOPs circles and then Profile and check for any unusual routing/profile
 - Check NOTAM
 - Check MET (check View MET for graphical display)
 - Check SIG Chart
 - Check if additional fuel required by NOTAM
 - Recalculate if data revised
 - Continue with Procedure 7 (skip 6)

6. If 'C' is not displayed in the 'Status' field, and there are Routes in the field:

Date:	01/04/08	Page	49 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

- Go to 'Variant' page
 - ✓ Click 'Cr All'
 - ✓ Click 'CP Comp'
 - ✓ Activate the min cost plan
 - If there is/are error message/s, try to resolve the error message in consultation with the Supervisor or Nav Services.
 - If all avenues are exhausted and the problem is routing,
 - o find previous flight routing on the same sector
 - o Copy the route description
 - o Go to Flight Page
 - o Go to Edit RTNG
 - o Paste the route in FPL section and click "Use FPL"
 - o Type a valid RtNr and Save.
 - Open 'Flight' page
 - Follow procedure '5' above from the second bullet point
7. To prepare multiple sector plan for single crews BP on flights operated by same crew:
- 'NOTAM' page
 - ✓ Click 'Chain' to see list of flights assigned to the same pilot
 - Go to 'Flight' page
 - ✓ Calculate all flight plans by following procedure 3 to 6 above
 - Go to 'NOTAM'
 - ✓ Click 'Chain' and view to check the flights number
 - Go to 'MET' page to prepare WX report
 - ✓ 'Group Name' field displays the first leg
 - ✓ In the 'Region' box, double click the legs to add lists in the 'Request' field.
 - ✓ In the 'Report' box, select 'All'
 - ✓ In the 'Request' field, Enter to view the WX report
8. To collect the 'Briefing Packages' (BP), go to 'Communication' page and add the following:
- 8.1. OFP
 - 8.2. MET
 - 8.3. NOTAM
 - 8.4. GRAPH (Select Upper Wind and EROPS)
 - 8.5. SIGCHRT
9. Check and Select the appropriate receiving addresses and send it.
For flights out of BAH, print and send the BP to flightplan1@gulfair.com email, for reference.
10. For ATC plan, select 'File', check the address (add/delete as necessary) and transmit.

The following airports require the ATC plan to be filed to the local ATC Briefing Office (ZPZX address). Please remove all 'ZRZX' for the en-route FIR countries except for UAE to avoid message duplication:
OMAA, OMDB, OIIE, OIMM, OISS, OYSN, HECA, OPLA, VNKT, VIDP, VTBS.

	Operations Procedures Manual	FDP-11 OFP Request Procedure
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ATC plan should be filed for all schedule flights outside European departure airports, including for non-schedule operations.

Note:

Calculation Error:

Check CFMU Test for all ATC plan to/from Europe airspace. If there is error and may take time for corrective action by editing the route, go to previous day plan and use the routings for current flight plan calculation.

During Calculation, if there is rejection and unable to solve the problem, use routing from previous day flight and copy on the current flight for calculation.

MEL/CDL Correction:

If the appropriate MEL/CDL correction can not be actioned from the 'MEL' page, in mfs:2 system, use the following alternative way for correction of the following variables from the 'Flight' page, until corrective action is taken by Navigation Services.

- Fuel Degradation: Add extra fuel as 'CF' by the amount of the penalty
- Speed in Mach : On the 'Route' display window, select 'ETO/Mach' box
Enter the Min/Max Mach speed as required
- Flight Level : On the 'Route' display window, select 'ETO/Mach' box
Enter Min/Max FL as required to restrict Flight Level

Load Information:

We should continue to monitor the Pax/Frgt/DCS and Telex for load information and update in the respective field for accurate flight plan calculation.

Tankering:

- The system will perform auto calculation for all available sectors ETD-12 Hrs.
- Check if the Pax/Frgt figures are loaded for correct tankering
- If the system not calculated the next required tankering sector/s, go to the next flight and 'Calculate'
- First sector 'Target' field should be in 'AUTO'
- Check 'RemF', LDW, TOW & ZFW for all sectors to have a buffer of 2000kg from the maximum and adjust tankering as required
- Calculate and review OFP before filing.

At the end of the day we are responsible for all the information/data given to crew. The system is helping us to discharge our duties effectively and efficiently but the ultimate responsibility lies on Dispatcher and then Flight Crew.

Date:	01/04/08	Page	51 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

Abbreviation:

AOM	Airport Operating Minima
BP	Briefing Packages
CDL	Configuration Deviation List
CF	Company Fuel
CPT	Captain
DCS	Departure Control system
ETD	Estimated Time of Departure
EZFW	Estimated Zero Fuel Weight
FL	Flight Level
F/O	First Officer
Frt	Freight
FREQCB	Frequent Cumulonimbus
IATA	International Air Transport Association
LDW	Landing Weight
MEL	Minimum Equipment List
MFS2	Modular Flight Planning System Version 2
NOTAM	Notice To Air Men
OFP	Operational Flight Plan
Pax	Passenger
PSS	Pilot Self Service
SCAP	Standard Computerized Aircraft Performance
SFP	Short Flight Plan
STD	Schedule Time of Departure
TOW	Take-Off Weight
WX	Weather

Responsibilities: Manager Flight Dispatch, Manager Navigation Services & Aircraft Performance, Load Control Agents, Maintenance Control,

Date:	01/04/08	Page	52 of 91	FDP-11
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-12 ATS Flight Plan
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Title:	ATS Flight Plan
Type:	Flight Dispatch
Purpose:	To ensure that all Gulf Air Flights have a departure clearance from the respective ATC authorities before departure
Scope:	Flight Dispatch, IOC, Operating Flights, and Departure Airports
References:	OMA 8.1.9, CAR-OPS 1.300

Procedure:

1. Introduction:

It is a regulatory and safety requirement to submit an RPL or file ATS flight plan on daily basis by an operator to concerned ATC authority.

2. Responsibility for Daily Filing of ATS Plan:

- Flight Dispatch is responsible for filing ATS flight plans for all flights except for European departure airports.
- Departure from European ports will be filed by the respective handling agents, but when it is required to file ATS flight plan at any time, Flight Dispatch will take action accordingly.

3. Filing from Mfs2 & SITATEX:

- File ATS plan after calculation of flight plan and the rest of checking is complet.
- The following airports require the ATS plan to be filed to the local ATC Briefing Office (**BPZX** address). Please remove all '**ZZXX**' for the en-route FIR countries for UAE departures.

OMAA, OMDB, OIIE, OIMM, OISS, OYSN, HECA, OPLA, VNKT, VIDP, VTBS

- To file from Mfs2
 - From "Communication" Page select "File"
 - Select "FPL" for filing; "DLA" to re-file delay "FPL"; "CNL" to cancel filed "FPL".
 - Edit the address in the 'File To' field if required
 - Click "SEND"
- To file from SITATEX:
 - Copy the ATS plan section from the OFF
 - Paste it on SITATEX window
 - Add the ICAO airport address (for the ATC Office) at the top of the message body.

Date:	01/04/08	Page	53 of 91	FDP-12
Authority:	MFD	Issue	2	Rev 00

Example: **FF AAAAZPZX OBBIZKZX** (“AAAAZPZX” ICAO airport address field)
DDHHMM OBBIZKZX (“DDHHMM” date/time group. OBBIZKZX is Dispatch AFTN address)

The “ICAO” address “ZPZX” can be changed depend on the available address list in mfs2.

- Add SITA address ‘BAHAPYF BAHOKGF’ in the address field and transmit. BAHAPYF is the address for Bahrain ATC Communication Centre, which will switch SITA message to AFTN network and re-transmit.

4. Conflicting Call Sign

- When a flight operates with the same call sign to more than 2 airports in the Gulf, it may cause a conflict if it is operated by two aircraft in the same airspace. This can happen when there is disruption followed by aircraft swaps to recover the schedule.
- To prevent this, Dispatch Officers must allocate call signs to split the sectors in the Gulf airports for ATC purpose, if the same is not reflected in the mfs2 “List” page:

- a) On flights from out station to Gulf port to Gulf port inbound flights add the suffix ‘A’ on the last leg.

Example: GF002 LHR-BAH-AUH
 Will transit Bahrain and become:
 GF002A for BAH-AUH

- b) On flights from Gulf port to Gulf port to out station flights add the suffix ‘B’ on the first leg.

Example: GF007 AUH-BAH-LHR
 Will transit Bahrain and become:
 GF007B for AUH-BAH

- Put a note on the OFP for the crew

Note: The additional suffixes to be added on the call sign are only for ATC purposes not to jeopardize safety. Call signs for ground staff and commercial use remain unchanged.

Abbreviation:	ATC	Air Traffic Control
	ATS	Air Traffic Services
	OFP	Operational Flight Plan
	RPL	Repetitive Flight Plan
	SITATEX	Window Based SITA
	ICAO	International Civil Aviation Organization

Responsibilities: Flight Dispatch Manager

	Operations Procedures Manual	FDP-13 Non-Standard Flights
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Title: **Non-Standard Flights**

Type: Flight Dispatch

Purpose: To ensure the OFP is checked against special FIRs' requirements and route schemes by overflying states; area of TS/TC and Volcanic Activities; and airport restrictions.

Scope: Flight Dispatch, Navigation Services & Aircraft Performance Sections, Operating Flights, IOC,

References: OMB-A340/A330 Fleet Instruction No. 007/2005, Iraq AIP 1.8.1, 1.8.3.1, 1.8.3.4, Navigation Services, EUROCONTROL, OMC Section 11.48, OMC Section 11.77,

Procedure:

Introduction:

For Flight Dispatch purposes, OFP's are categorized into 3 types for documentation and preparation. They are:

- Standard Flights (Non-ETOPS normal flights)
- Non-Standard Flights (Non-ETOPS flights with additional requirements)
- ETOPS (Flights beyond 60 minutes from en-route alternates)

Factors Considered for Non-Standard Flights:

- Special state requirement to over fly FIR
- Additional check for route availability due to multiple restrictions by NOTAM
- Area of Operation with: SIGWX (Tropical Storm/Tropical Cyclone)
: Volcanic Activity

Contents included in the procedures:

- FD-13.1 Gulf to/from Europe Operations
- FD-13.2 Gulf to/from Far East Operations
- FD-13.3 Kathmandu Operations with B767
- FD-13.4 Sanaa Operations with A320
- FD-13.5 Khartoum Fuel Situation
- FD-13.6 Diversion Flights
- FD-13.7 Fuel Policy Change for A330/A340 on Approved Sectors
- FD-13.8 Organized Track System for SYD-SIN sector
- FD-13.9 RTOW Calculation Program

Date:	01/04/08	Page	55 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

	Operations Procedures Manual	FDP-13 Non-Standard Flights
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FD-13.1 Gulf to/from Europe Operations

A. Reason:

- Iraq airspace restriction
- Escape route en-route WX check when routed via IRAN
- CFMU validation check
- Monitoring Aeronautical Notification Message (ANM) for heavy delay

B. Objective:

Routings across Europe region are required to be checked in Eurocontrol site before ATS plan is filed. In addition, due to the automated system generation of departure slot timing and enroute and/or airport traffic delay, Brussels's messages should be monitored closely for necessary action to avoid extensive departure and/or arrival delay.

C. Do the following:

- Select the best route. (Refer FDP-11)
- Do CFMU check
- If route is rejected, select another best route, or "Edit" the route
- Report the problem to Mfs2 & Navigation Services
- When extensive departure delay (more than 1 hour) is noticed due to ATC slot, check for alternative routing if the case is enroute traffic congestion at given FIR and liase with IOC for necessary action.
- When arrival delay is received by Eurocontrol, consider extra holding fuel on the flight plan calculation and advice crew.

Note: ATC slot message are revised contineously and often improve significantly. Liase with Handling Agent for more accurate information.

Date:	01/04/08	Page	56 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

	Operations Procedures Manual	FDP-13 Non-Standard Flights
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FD-13.2 Gulf to/from Far East Operations

A. Reason:

Because of known seasonal activities of Tropical Storm (TS)/Tropical Cyclone (TC) and Active Volcanic Area, flight planning in this area needs special attention for safety reason.

B. Objective:

To monitor TS/TC/Volcanic Activity in these regions and make sure OFP routes avoid these danger areas.

C. Procedures:

- a) When there is a known development of Tropical Storm (TS) or activity of Tropical cyclone (TC), or Volcanic Eruption in the area of operations, routes have to be thoroughly checked.
- b) Plot the area of activity on JEPESSEN chart for detailed view and follow-up of the TS/TC movement
- c) Analyze the routes on the SIG Chart map
- d) If Best Route for the day is affected, choose another route from the list that avoids affected area. If there is no routings, use "Edit" to creat new rout.
- e) Pass detailed information of the activity and action taken to operating crew and leave a note on the OFP.

Date:	01/04/08	Page	57 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

FD-13.3 Kathmandu Operations with B767

A. Reasons:

Kathmandu is one of Gulf Air's destinations where aircraft performance is limited due high significant terrain around the airport and has an airport elevation of 4390 feet AMSL. A takeoff alternate (Dhaka-DAC) is stored in the flight-planning database, to be displayed automatically when a flight plan is generated for any fleet type, but the requirement for the takeoff alternate is only for operations with twin-engine aircraft.

B. Objective:

To ensure aircraft weight is not exceeding the Performance Limited Landing Weight at Kathmandu, and Take-Off Alternate weather is monitored for B767 operation.

C. Procedures to Determine B767 Landing Weight at Kathmandu:

- Check or find out from KTM airport/Met Office, the forecast temperature during schedule arrival time.
- Determine MLGW by reading MTOW from the 'RTOW Charts - RWY 02 Flaps 5' for temperature not more than 29°. If however the forecast temperature is more than 29°, then MLGW 130,000kg will apply.
- If forecast temperature is not available, the maximum statistical temperature for that month and arrival time will be used to determine MLGW.

	KTM Maximum Temperature Record			
Local Time	0800 – 1030	1030 – 1230	1700 -1930	1930 - 2030
UTC	0230 - 0500	0500 - 0700	1130-1400	1400-1500
Jan	10	21	19	11
Feb	20	22	18	15
Mar	20	25	25	20
Apr	26	30	27	24
May	29	31	29	28
Jun	31	31	28	27
Jul	30	31	29	26
Aug	31	31	30	29
Sep	30	30	28	24
Oct	28	29	26	21
Nov	24	26	20	17
Dec	21	21	17	12

Table1. Kathmandu maximum temperature expected each month for the given time period. (Gray areas indicate temp > 29°C, there for 130,000kg to be used)

Example: Forecast landing Temp 25°C and assuming ETA KTM 14:30 UTC October.

Date:	01/04/08	Page	58 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

	Operations Procedures Manual	FDP-13 Non-Standard Flights
---	---------------------------------------	--

From table, Statistical temp= 21°: MTOW at 25°C = 134,700kg.
Thus 134,700 will be the planned landing weight and the OFP request will have a limitation of **LDW=134.7**

D. Procedures for Kathmandu Departure -Takeoff Alternate

- a) Due to engine out considerations, all B767 operations from KTM are required to have a take-off alternate nominated. VGZR-DAC has been stored in the flight planning system as the take-off alternate for KTM to Gulf sectors.
- b) No Dispatcher input required unless DAC weather is not suitable.
- c) Use CCU if DAC is below minima.
- d) If CCU and DAC are not suitable for take-off alternate, an alternate beyond 1-hour range is to be nominated if it satisfies the ETOPS qualification requirements in OMA 8.1.2.3.2.
- e) Delhi (DEL) is more than 430nms away and can be nominated for B767 takeoff alternate if it satisfies the above ETOPS requirements.
- f) Select "**DepAltn**" box to enter takeoff alternate.

Date:	01/04/08	Page	59 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

	Operations Procedures Manual	FDP-13 Non-Standard Flights
---	---------------------------------------	--

FD-13.4 Sanaa Operations with A320

A. Reason:

Sanaa is one of the destinations where there is performance limitation on our fleet due to high elevation and obstructions due to mountains around the airport. The A320 schedule operations can uplift limited payload from Sanaa due weight, altitude and temperature factors.

B. Objective:

To ensure aircraft weight is not exceeding the performance limitation at Sanaa airport at the time of planning the OFP.

C. Procedures:

- a) Request SAH airport to send forecast temperature and wind for time of arrival and departure.
- b) From SAH MET report, check RWY to be used and analyse the TOW from the RTOW charts.
- c) Flights to SAH (Tankering sector):
 - If there is forecast temperature for arrival time, check SAH TOW for RWY 18 for the reported temperature to limit LDW.
 - If no temperature information, restrict the maximum landing weight to 62,000KGS. i.e. LDW 62.0
- d) Flights from SAH:
 - Check the forecast temperature.
 - Analyse the RTOW charts and take QNH correction.
 - Enter '**TOW**' in the appropriate field.
 - Advise payload penalty to SAH airport and IOC
 - Consider late departure from SAH if payload penalty is high and temperature is expected to drop the next couple of hours. Discuss this with IOC.
 - If unable to get temperature information and the payload seems likely to exceed the TOW, advise crew that no forecast temperature available to check OFP TOW.

Date:	01/04/08	Page	60 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

	Operations Procedures Manual	FDP-13 Non-Standard Flights
---	---------------------------------------	--

FD-13.5 Khartoum Fuel Availability:

A. Reason:

Availability of fuel at (KRT) is not reliable. Tankering or partial uplift of fuel for return sector maybe required depending on the availability (including amount) of fuel.

B. Objective:

To check the fuel situation for each flight of the day and plan the fuel uplift and allowable payload for the flight and advise payload restrictions to airport.

C. Procedure:

- a) Send a telex to KRT to check fuel availability status for the flight
- b) If KRT arrange a fuel that is not sufficient for return sector, run a TEST OFP for Gulf-KRT-Gulf to check the Ramp Fuel for both sectors
- c) Determin the FOD and enter Target fuel on Gulf-KRT leg to have sufficient fuel to cover the shortage from KRT, even at the expense of payload. Consider Taxi-in fuel on the FOD.
- d) If there is payload penalty or aircraft fuel tank capacity limitation for A320 operation, recalculate OFP with Cost Index entry at "MINF" for minimu fuel calculation.
- e) If there is no fuel in KRT and the round trip fuel plan will cause heavy payload penalty, discuss the matter with IOC to consider a tech stop for refuelling.
- f) Advise any payload penalty to load control staff and put a comment on the OFP for the crew.

Date:	01/04/08	Page	61 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

FD-13.6 Diversion Flights

A. Reason:

Unscheduled operation requires a rapid action to release OFP and file ATS Plan to the ATC unit.

B. Objective:

OFP, ATS Plan and RTOW Charts (if not available in VOL 3) need to be prepared and sent to the crew.

C. Procedure:

a) Destination Diversion

- Check route availability in the MFS2 system. If it is not available, "Edit" route.
- Check Origin/Destination/Alternates WX and NOTAM.
- Obtain AZFW from the crew or Load Control of departure's airport
- If the AZFW is exceeding the calculated OFP ZFW due to TOW/LDW limitation, check alternative way by reviewing the alternate listing. This can happen when the sector is short and distant alternate is nominated.
- Request the handling agent address to release OFP if the airport is not one of Gulf Air destination. Call the destination's airport staff or the closest GF airport's handling agent to find out the address. If Aircraft ACARS is serviceable, send OFP via ACARS.
- File ATS Plan to the local ATC unit.

b) En-route Diversion

- If the diversion is to an online Gulf Air destination airport, the OFP can be released to the available address for that airport. If it is an off line airport, try to find out the address from the nearest Gulf Air airport handling agent or the ATC centre to send the OFP. Crew can be also contacted to check with ground staff to pass available address. If the aircraft has a serviceable ACARS printer, OFP can be sent directly to the crew.
- If there are no RTOW Charts in VOL 3 for that aircraft type, call Performance Section to prepare charts
- Obtain AZFW of the flight.
- If the diversion leg is not picked from Airops automatically, the additional sector is required to be entered manually either using the "Edit" or "Creat" leg entry. (Refer FDP11-3.2.2)
- Check route availability, otherwise "Edit" routing
- Check Origin/Destination/Alternates WX and NOTAM.
- File ATS Plan.

c) En-route Flight Plan Calculation:

- Click "Flt Follow" from "Detail" page
- Put actual data (position, time, fuel & AZFW) in the fields and click "Insert", if there is no automatic updates from the aircraft
- Click "OK" to go back to "Detail" page

Date:	01/04/08	Page	62 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

	Operations Procedures Manual	FDP-13 Non-Standard Flights
---	---------------------------------------	--

- Click “Edit Rtg” for route change only; or
Click “Chg Dest” then ‘Edit Rtg’ for destination change. Review the alternate for the new destination.
- Click “Calc” to get fuel requirement from the position selected
- Review the calculated fuel; the “EXTRA” field will show negative value if the fuel is insufficient.

Date:	01/04/08	Page	63 of 91	FDP-13
Authority:	MFD	Issue	2 Rev	02

FD-13.7 Fuel Policy Change for A330 / A340 on Approved Sectors:

- Gulf Air has approval from CAA-Bahrain to reduce the contingency fuel amount from 5% to 3% for sectors listed in the table below.
- The MFS2 Flight Planning System is capable of identifying sectors and equipment approved for 3% contingency fuel calculation.
- Refer to OMA 8.1.7.3.3 for list of sectors & fleet for 3 % contingency fuel

(3% calculation of contingency fuel)

SECTORS	A / C TYPE
GLF-MNL-GLF	A340
GLF-BKK-GLF	A340
GLF-KUL-GLF	A340
GLF-SIN-SYD	A340
SYD-SIN-GLF	A340
GLF-CGK-GLF	B767

SECTORS	A / C TYPE
GLF-DUB-GLF	A330 / A340
GLF-LHR-GLF	A330 / A340
GLF-CDG-GLF	A330 / A340
GLF-FRA-GLF	A330 / A340

Table 1. Sectors and Fleet type approved to carry 3% contingency fuel as of Apr/08.

	Operations Procedures Manual	FDP-13 Non-Standard Flights
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FDP 13.8 RTOW Calculation Program and LDW Check

Introduction:

The Airbus and Boeing aircraft performance software (Octopus/WINPEP/BPS/MRK7J) in Navigation Services and Aircraft Performance can be accessed from Flight Dispatch PC through the network as well as from Aircraft through ACARS system.

When there is snow/ice forecast on the WX report, the same have to be reported to the Chief Pilot and Operation Control for necessary coordination.

Procedure:

A. RTOW Calculation:

The RTOW calculation program is mainly designed to calculate allowable MTOW, FLEX or Assumed temperature for actual weight and corresponding speed when aircraft is operating under MEL and/or Contaminated RWY condition that may result in payload limitation.

Crew may contact Flight Dispatch through available communication system and pass the following to get MTOW or FLEX/assumed temperature for actual weight and the corresponding speed (V1/VR/V2).

- Aircraft registration
- Aerodrome IATA code
- RWY designation
- SHIFT value (if any)
- QNH
- OAT
- Wind
- Actual TOW (if provided)
- MEL inoperative item (if any)
- RWY contamination (if any)

Dispatcher shall enter the above data into the system for calculation and the result to be passed to the crew through available communication system (ACARS, SITA, Voice).

A. Landing Weight Check:

Refer FDP-13.4 for Landig Weight determination to Sanaa operation.

Refer FDP-13.5 for B767 Landing Weight determination to Kathmandu operation.

Refer FPCOM 2.03.10 for Airbuses aircraft CONTAMINATED/WET LDW analyses

Refer Vol 3 Mains Performance Calculation - Landing Weight Calculation page L3 to L7 for B767 CONTAMINATED/WET LDW analyses.

Contact Navigation & Aircraft Performnace section when it is required.

Date:	01/04/08	Page	65 of 91	FDP-13
Authority:	MFD	Issue	2	Rev 02

Abbreviations:	AFTN	Aeronautical Fixed Telecommunication Network
	ACARS	Airborne Communications Addressing and Reporting System
	AIP	Aeronautical Information Publication
	ATS	Air Traffic Services
	CCU	Calcutta
	CDR	Conditional Route
	CRAM	Conditional Route Availability Message
	DAC	Dhaka
	ETA	Estimated Time of arrival
	ETOPS	Extended Operations for Twin Engine Aircraft
	FIR	Flight Information Region
	FL	Flight Level
	FOD	Fuel Over Head destination
	KRT	Khartoum
	KTM	Kathmandu
	LDW	Landing Weight
	MEL	Minimum Equipment List
	METAR	Meteorological Report for Aerodrome
	MTOW	Maximum Take-Off Weight
	OAT	Outside Air Temperature
	OFP	Operational Flight Plan
	QNH	Altitude above sea level based on airport pressure
	RAMCC	Regional Air Movement Control Centre
	RF	Ramp Fuel
	RTOW	Regulated Take-Off Weight
	RWY	Runway
	SIGWX	Significant Weather
	TC	Tropical Cyclone
	TOW	Take-Off weight
	TS	Tropical Storm

Responsibilities: Flight Dispatch Manager, Navigation Services, Operating Crew

	Operations Procedures Manual	FDP-14 Fuel Tankering Procedures
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Title:	Fuel Tankering Procedures
Type:	Flight Dispatch
Purpose:	To uplift additional fuel from lower fuel cost airports.
Scope:	Flight Dispatch, Operating Flight, Load Control, IOC, Bahrain Maintenance Control
References:	OMA 2.4.2.2.3.3, 8.1.10.10

Procedure:

1. Introductions:

Gulf Air has a policy to uplift fuel from lower fuel cost airports when there is a positive saving of 1USC per USG. The Flight Planning System computes the calculation and comparison analysis, when the Flight Dispatcher adds the additional sectors.

The following procedures shall be followed when the departure airport fuel price is lower than the arrival airport.

2. Fuel Tankering General

- A. When the rotation sequence has an aircraft with a lengthy stopover, discuss the situation with the Shift Supervisor and consider the likelihood of the aircraft being re-assigned. This may influence your decision as whether to tanker or how much it is required to uplift.
- B. For flights that have a lengthy stopover prior to the next departure at the arrival airport, exercise caution before tankering. Check the destinations of other flights departing from the arrival airport. Consideration shall be given to the possibility of aircraft changes, which could require de-fuelling. Discuss with IOC for further input.
- C. For aircraft going to maintenance, ask Maintenance Control for advice.
- D. When tankering for 3 sectors with 2 Short Flight Plan summary and the full plans for the next sectors are to be done later by a new shift, the information must be brought to the attention of the Dispatcher doing the full plans.
- E. Put aside the summary plans for shift hand-over or leave a note in the Log entry regarding the amount of fuel already on board. If the sector sequence is changed, this amount may be more fuel than required for the new sector and a "BlockF" entry would be required with Remark on OFP to explain why there is additional fuel on board.

Date:	01/04/08	Page	67 of 91	FDP-14
Authority:	MFD	Issue	2	Rev 00

- F. Consider limiting the tankering to less than Max Landing Weight by using 'LDW' entry in mfs2. This will build fuel up to the specified landing weight of the aircraft or reduce payload.

Example: For an A320, "**LDW 600**" will limit the fuel build up to that landing weight, leaving a good buffer in case the aircraft is reassigned.

3. Payload Considerations to Special Destinations

3.1. Gulf to Indian Sub-Continent Airports Tankering

- A. Due to the high fuel price at these airports, tankering on these sectors will provide fuel savings, if payload is not affected.
- B. Caution shall be exercised when tankering for sectors from Gulf airport to Gulf airport then to Indian Sub Continent airports.

Example: BAH-AUH-DEL-AUH or;
BAH-AUH-MCT-BOM-AUH

- C. In the above example, if too much fuel is uplifted from BAH there could be a payload penalty when departing from AUH or MCT to the Sub-Continent.
- D. Examine each case separately and discuss with the Shift Supervisor if necessary.
- E. Often there is a late increase in payload, so consider adding a buffer to the system payload and monitor the EZFW from the airport.

3.2. Gulf to CAI Tankering

- A. From past experience during the Teacher's Charter flights and peak seasons, the actual payload for flights to CAI is more than the system booked load.
- B. When the fuel price at Cairo is expensive and it is worth uplifting fuel from the Gulf airports for return sector, extra care must be taken on planning multi sector inputs.

Example: BAH-AUH-CAI-AUH. Maximum fuel uplift from BAH may have payload restriction from AUH due extra carry over fuel for return sector, CAI-AUH.

- C. Check with load control to get EZFW; or add a buffer to the booked load.

Note: When excess baggage is expected, you may use 130KGS for each passenger and baggage weight for the sector.

Abbreviations:	EZFW:	Estimated Zero Fuel Weight
	LDW:	Landing Weight
	OFPS:	Operational Flight Plans
	PLD:	Payload
	USC:	US Cents
	USG:	US Gallons
	EZFW:	Estimated Zero Fuel weight

Responsibilities: Flight Dispatch

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	Operations Procedures Manual	FDP-15 Handling Payload Restrictions
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Title:	Handling Payload Restrictions
Type:	Flight Dispatch
Purpose:	To plan for unusual payload restrictions to affected airports, and IOC and other concerned parties.
Scope:	Flight Dispatch, Operating Flight, IOC, Maintenance Control Centre, and Aircraft Performance Sections
References:	

1. Introductions:

Due to various reasons, a sector load can be restricted and may cause last minute offloading of passengers and/or baggage. If the payload restriction is noted prior to departure time, early release of this information to Load Control and other concerned sections will reduce inconvenience.

2. Procedures to Handle Payload restrictions

A. Dispatcher must monitor factors that may cause payload penalties. Some of these are:

- Destination/Alternate weather
- En-route Tropical Storm Warning
- Route restrictions
- Aircraft MEL/CDL
- Fuel non availability
- WAT limitation (Weight, Altitude, Temperature). Etc.

B. When such warnings are received:

- Analyze the effect of the penalty for the given sectors
- If it affects payload, check the booked load or get the EZFW from Load Control
- Run a TEST OFP to determine allowable payload for the sectors
- If payload is affected, inform the penalty to Load Control of that airport and IOC for further action.
- If the penalty is high and it is due to aircraft MEL/CDL, check if the defect can be rectified or an aircraft can be changed.

Date:	01/04/08	Page	71 of 91	FDP-15
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-15 Handling Payload Restrictions
---	---------------------------------------	---

- If the aircraft can not be fixed and planned to be dispatched under existing MEL/CDL performance penalty and payload is affected as a result, advice Performance Sections to produce special RTOW Charts.

Abbreviations:	CDL	Configuration Deviation List
	EZFW	Estimated Zero Fuel Weight
	MEL	Minimum Equipment List
	RTOW	Regulated Take-Off Weight

Responsibilities: Manager Flight Dispatch, Maintenance Control, Navigation Service and Aircraft Performance Section

Date:	01/04/08	Page	72 of 91	FDP-15
Authority:	MFD	Issue	2	Rev
				00

	Operations Procedures Manual	FDP-16 MEL/CDL Affecting Operations
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Title:	MEL/CDL Affecting Operations
Type:	Flight Dispatch
Purpose:	To take appropriate action for existing MEL/CDL affecting performance of aircraft when advised by Maintenance Control
Scope:	Flight Dispatch, Operating Flight, Navigation Services and Aircraft Performance Section, IOC, Maintenance Control Centre
References:	OMA Section 8.6, CAR-OPS 1.025, MEL/CDL Manuals,

Procedure:

1. Introduction:

Aircraft are certified by the manufacturer and approved by DGCA-Oman for operation with certain systems and/or components to be in operative. These systems and components are all listed under the MEL/CDL of the aircraft to despatch flights with condition as prescribed in the manuals.

2. MEL/CDL Handling Procedure.

- A. Flight Dispatch receives a notice from Maintenance Control Centre regarding an aircraft under MEL/CDL.
- B. The message is received through SITA or email with detail containing aircraft registration, defect type, MEL/CDL reference and additional relevant information if any.
- C. Flight Dispatch will refer to the MEL/CDL manuals to check if there is any performance penalty or any other operational effects.
- D. If there is restriction on the area of operation and/or routes assigned to operate, check for any other alternative. If there is no alternative, advise Maintenance Control and IOC for further action. (See Note below and procedures on FDP17)
- E. If there is a penalty, check its effect from departure airports and subsequent sectors, and apply it on the calculation of OFP. Advise crew on the flight plan Remark Section for the action taken on the OFP.
- F. If MEL/CDL is not loaded or required to be removed in mfs2, add or remove MEL/CDL manually, for that flight, and advise MCC or Navigation Services for corrective action.
- G. If there is a significant penalty, which may affect payload, advise Aircraft Performance Section, to produce RTOW Charts with the inoperative item for required airports.
- H. Advise payload penalty to Maintenance Control Centre, IOC and Load Control of affected stations

Date:	01/04/08	Page	73 of 91	FDP-16
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-16 MEL/CDL Affecting Operations
---	---------------------------------------	--

- I. Keep a note on the White Board regarding the MEL/CDL for easy follow up and hand over to next shift staff.

2.1 RNP-10 Non-Compliance Procedure

- A. In the event of an aircraft operating under MEL for navigation equipment failure or degradation of the RNP-10 standard, the Dispatcher must check whether the route is allowable or not.
- B. Check the En-Routes Charts on JEPPESEN airway manual, to check if the selected routes FL on the OFP are RNP-10 route.
- C. If the routes are non-RNP 10, the defect is allowable and the only amendment required is add a remark "RNP-10 EQUIPMENT UNDER MEL" item 18 of of the ATS flight plan.
- D. If the given routes or area of operation is RNP-10, the aircraft shall avoid the routes or airspace.
 - Check with ATC if there is a lower flight level below the indicated Lower Limit Flight Altitude for RNP. JEPPESEN charts do not always indicate the Minimum Flight Altitude available for the route, but the full detail is available in AIP. Check with ATC to verify.
 - Assuming that a lower level for the route is available and approved; plan a new OFP at the approved lower Flight Level by avoiding the RNP airspace.
 - Put a note on the OFP for the crew that Lower Flight Level is approved by ATC to operate with RNP-10 inoperative.

Example: GF059/BOM-MCT/A40ED FMGC inoperative;

- As per MEL, aircraft can not fly on RNP-10 route or airspace;
- Best route is: VABB.BIM02B.SUGID.N571.PARAR.N767.MCT;
- From JEPPESEN 7/8 High/Low En-route Chart dated 1 Feb 08: the segment SUGID.N571.PARAR is RNP10 from FL280
- Request BOM ATC to fly at FL260 on airway N571 due aircraft not fit for RNP route.
- Once approved release the Flight Plan by restricting the maximum flight level below FL280.
- Insert a remark 'RMK' on the ATS flight plan and OFP

- E. If ATC rejects the request or there is no available lower level below the RNP 10 Flight level for the given routes, avoid using the airspace. Check for an alternative route, or build a Non-RNP10 route in the system and release the OFP accordingly.

Date:	01/04/08	Page	74 of 91	FDP-16
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-16 MEL/CDL Affecting Operations
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- F. If there is no alternative route for Non-RNP10 flight, aircraft must be fixed at departure airport

2.2 RVSM Procedure

- A. The letter '**W**' must be shown in item 10 of the ATS plan to indicate compliance for RVSM airspace.
- B. Flights operating with unserviceable RVSM equipment shall be planned on routes and levels avoiding RVSM airspace.

2.3 TCAS Inoperative Procedures:

- A. General:

A TCAS II system is required to be installed in EU member states and others regions. When aircraft is under MEL with TCAS inoperative in EU states, exemption can be arranged either by direct contact to EUROCONTROL Centre or in oordination with London GF Operation Office. The same can be arranged by direct contact to the ATC authority for others region; however, this may be refused.

- B. TCAS Inoperative Procedures:

- Check the white board or telex for MEL message from Maintenance Control Centre (MCC)
- Refer MEL manual for TCAS inoperative
- Liase with Chief Pilot (CP) approval due to restriction by BAH CAA. (UAE and OMAN ATC strongly object operation with TCAS inop)
- If exemption has been approved by CP and the enroute states, put the following on Item 18 of the ATS plan:

RMK/TCAS UNDER MEL

Note: Aircraft with MEL affecting route or area of operation should be rectified before departure due to operational restrictions that may occur.

Date:	01/04/08	Page	75 of 91	FDP-16
Authority:	MFD	Issue	2	Rev 00

	Operations Procedures Manual	FDP-16 MEL/CDL Affecting Operations
---	---------------------------------------	--

Abbreviations:	ATS	Air Traffic services
	CDL	Configuration Deviation List
	CP	Chief Pilot
	EUR	Europe
	FL	Flight Level
	FYI	For Your Information
	INOP	In-Operative
	MCC	Maintenance Contro Centre
	MEL	Minimum Equipment List
	OFF	Operational Flight Plan
	REF	Reference
	RTOW	Regulated Take-Off Weight
	RNP	Required Navigational Performance
	RVSM	Reduced Vertical Separation Minima
	TCAS	Traffic Alert and Avoidance System

Responsibilities: Manager Flight Dispatch, Bahrain Maintenance Control Centre, Manager Navigation Services & Aircraft Performance, IOC, Operating Crew

Date:	01/04/08	Page	76 of 91	FDP-16
Authority:	MFD	Issue	2 Rev	00

	Operations Procedures Manual	FDP-17 High Speed & Maximum Range OFF
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Title:	High Speed & Maximum Range OFF
Type:	Flight Dispatch
Purpose:	To change the optimum speed selected by the Flight Planning System to meet company requirements.
Scope:	Flight Dispatch, Operating Flight, IOC, and Crew Control
References:	OMA Section 8.1.10.2.1, 8.1.10.3

Procedure:

1. Introduction:

The standard flight plan speed is selected by the Flight Planning System to meet the criteria of low operating cost, however during certain circumstances it may be necessary to reduce the flight time by increasing the chosen Cost Index (CI) of the OFP. There may also be occasions where reducing the CI of the OFP is necessary to improve payload uplift on sectors limited by aircraft performance or fuel tank capacity.

2. High Speed OFF Procedure:

- A. High Speed Flight Plans may be requested by IOC, or Crew Control, or Operating Crew for the following reasons:
 - Crew duty time limitation.
 - Schedule recovery after extended delay.
 - Airport operational hour restriction, etc.
- B. Factors to be considered before planning:
 - Aircraft MEL/CDL restrictions
 - En-route fixed speed/Mach technique restriction
 - En-route weather (such as Turbulence, Shear Rate)
- C. Procedure for selecting High Speed flight plan:
 The Dispatcher can select either the Cost Control or ETA Control button in order for the system to pick higher Cost Index (CI) to meet new reduced flight time.

Cost Control – High Speed

- Cost Index field show “Econ” by default. Do not change it
- EROPS box must be selected – “on”
- Click ‘Opt Dly’ from ‘Flight’ page
- Click ‘DlyMgmt’ button
- Select ‘Cost Control’
- Enter new arrival time after the delay in ‘ETA Late’ field
- Enter a cost of USD70/min in the ‘Cost/minute’ field if it is not available
- Select ‘OK’

Date:	01/04/08	Page	77 of 91	FDP-17
Authority:	MFD	Issue	2	Rev 00

- Calc and check the new CI, Time, Flight Cost for comparison

If the new calculated time, by using the 'Cost Control' method, is not meeting the requirement, and the increase of the speed is mandatory regardless of the cost, use the following technique:

ETA Control – High Speed

- Cruise Index is showing "Econ" by default. Do not change it.
- Select 'Opt Delay' from "Flight" page
- Select 'DlyMgmt'
- Select 'ETA Control'
- Select "Target" time
- Click "OK" and calculate
- Check the extra cost and CI selected for necessary adjustment

Note: If the speed on the calculated plan is close to Maximum Operating Speed (Vmm/Vmo), adjust the target time to reduce the speed to get practical speed that may be used by the crew.

B767/A330/A340 Vmm is 0.86 and as a guide do not exceed 0.84, and for A320 Vmm is 0.82 and not to exceed 0.80, unless the situation is operationally demanding.

3. Maximum Range Speed OFF

- A. This procedure is to be used to improve payload uplift for flights on sectors where there is off-loading of revenue due to aircraft fuel tank capacity limitation.
- B. Affected flights are long sector operations with A320 fleet and operations to airports (e.g. KRT) where there is fuel problem. If payload is restricted due to aircraft fuel tank capacity limitation and computer selected Cost Index is high, changing the speed to low Cost Index will improve the payload.
- C. Factors to be considered before planning:
 - Aircraft MEL/CDL
 - En-route fixed speed/Mach technique restriction
- D. Select 'MINF' in the 'Flight Page' of the flight planning system and calculate.
- E. Advise the reason to the crew if the CI is different from the normal 'ECON' speed..

Abbreviations:	CDL	Configuration Deviation List
	CI	Cost Index
	MEL	Minimum Equipment List
	MINF	Minimum Fuel
	MINT	Minimum Time
	Vmm	Maximum Mach speed
	Vmo	Maximum Operating indicated speed
	OFF	Operational Flight Plan

Date:	01/04/08	Page	78 of 91	FDP-17
Authority:	MFD	Issue	2 Rev	00

	Operations Procedures Manual	FDP-17 High Speed & Maximum Range OFF
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WX Weather
ZFW Zero Fuel Weight

Responsibilities: Manager Flight Dispatch, IOC, and Crew Control

Date:	01/04/08	Page	79 of 91	FDP-17
Authority:	MFD	Issue	2	Rev
				00

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	Operations Procedures Manual	FDP-18 ETOPS Operations
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Title:	ETOPS Operations
Type:	Flight Dispatch
Purpose:	To ensure all approved Gulf Air ETOPS flights are dispatched and monitored in accordance with the Operations Manual.
Scope:	Flight Dispatch, Operating Flights, Navigation Services, Maintenance Control Centre, and IOC
References:	OMA Section 8.5, CAR-OPS 1.245/1.246, and JAR-OPS 1.245/1.246

Procedure:

1. Introduction:

Aviation regulations restrict two engine aircraft to flying on routes beyond 60 minutes at the approved one-engine inoperative cruise speed, under standard atmospheric condition in still air, from an adequate airport, for Performance Class A aeroplanes. All existing B767 fleet are satisfying the requirements for ETOPS Operations and approved by DGCA-Oman.

2. Adequate Airports:

All adequate airports for ETOPS en-route alternates' selection shall satisfy the criteria listed in section 8.5.2.1 of Operations Manual - Part A.

3. ETOPS En-route Alternate:

A suitable airport, which has been nominated and indicated on OFP for particular ETOPS flight at planning stage, the minimum forecast weather conditions shall satisfy the table in OMA Section 8.5.6.1 for a valid period commencing one hour before the earliest ETA up to one hour after the latest ETA at the airport.

4. Gulf Air ETOPS Fleet Facts

Flight Dispatch Officer must be fully aware of Gulf Air ETOPS fleet and approvals by Bahrain-CAA and shall fully comply with it on the preparation of Flight Plan.

Dispatcher must be vigilant on the use of the flight planning keywords that may override the approved ETOPS Diversion Distance or speed.

- Gulf Air is currently approved and operates ETOPS for all B767ER fleet.
(Refer OMA 8.5.3.1)
- Baharain-CAA approval for One-Engine-Inoperative Cruise Speed for B767ER is:

330 KIAS / M.80
(Refer OMA 8.5.4.2)

Date:	01/04/08	Page	81 of 91	FDP-18
Authority:	MFD	Issue	2	Rev 00

- As per CAR-OPS 1 requirement, the applicable specific aircraft Reference Gross Weight at threshold point from departures airport is:

170,000kgs.

(Refer OMA 8.5.4.3)

- Maximum Diversion Distance at the approved speed for B767ER is:

B767	
75 minutes	533 nm
60 minutes	430 nm

(Refer OMA 8.5.4.4)

- ETOPS Fuel Policy:

The OFP fuel calculation is in compliance with OMA 8.5.4.5.

- ETOPS Areas of Operation:

(Refer OMA 8.5.5)

5. MFS2 ETOPS OFP

Do not use any entry to change the diversion distance set in mfs2 unless it is required.

The flight planning system will reject the plan if any part of the route is outside the approval maximum diversion distance of one of the ETOPS alternate nominated.

In addition to the standard OFP information, the ETOPS OFP has the following detail:

ERA - ADJ = ETOPS ADJUSTMENT: additional fuel required to satisfy the requirements of ETOPS critical fuel scenario. For each critical point considered in the ETOPS INFORMATION section of an ETOPS OFP, the flight-planning computer crosschecks the fuel remaining at the critical point and the fuel required to proceed to the applicable suitable alternates. If any of the fuel required figures is greater than the fuel remaining, then additional fuel equivalent to the deficit is added to the RAMP FUEL figure, and shown in the ETOPS ADJ line.

EA: ETOPS Alternates are listed along with the Intermediate Airports on the Remark section

E.ENT: ETOPS ENTRY point - the point on the aircraft's outbound route, which is one-hour flying time at the approved one-engine-inoperative speed under standard conditions in still air from a suitable airport.

GOXXXX: "XXXX" is the diversion airport from the given equi time point for the flight to GO in case diversion is required.

C-EXT: ETOPS Exit point - the point on the aircraft's final leg of an ETOPS segment and entering the 60 minutes rule.

CRITICAL FUEL SCENARIO: It contains the earliest and latest arrival times at the nominated ETOPS alternate ports. The minimum ETOPS required fuel and others related information (TRK/MOCA/ etc) are also contained in this field.

6. Aircraft ETOPS Status

The Dispatch Officer must check the aircraft technical status before planning for ETOPS. Any MEL/CDL shall be checked from the daily Status Report that is published by MCC every 12 hours. The MEL manual shall be referred for any existing defect to decide whether the flight can be dispatched on ETOPS or Non-ETOPS route.

Any MEL/CDL performance correction or penalty must be checked and applied on the calculation of the Flight Plan.

7. ETOPS En-route Alternates Weather Minima and ERA Selection

A. ETOPS EN-route Alternates Weather Minima:

Dispatcher must check that the selected ETOPS en-route alternates at planning stage, 1 hour before the earliest arrival time to 1 hour after the latest arrival time, should satisfy the minima in OMA 8.5.6.1.

That is:

Approach Facility Configuration	Alternate Airport Weather Minima	
	Ceiling	Visibility
At least 1 NAVAID Providing precision or non-precision app	Add 400ft to the DH or MDH	Add 1500m with authorised landing minima
At least 2 NAVAID providing precision or non-precision App for 2 separate RWYS	Add 200ft to the higher of the 2 DHs or MDHs	Add 800m to the higher of the 2 landing minima

B. ETOPS En route Alternats Airport Selection

The aircraft authorized Threshold Time (THT) and approved Maximum Diversion Time (MDT) is set in the flight planning system and can be referred on the "EROPS" page.

The "EROPS" check box in the "Flight" page should be activated always for the system to check adequacy of en-route diversion airports within the given threshold time, and suitability check with in the given maximum diversion time.

The possible en-route ports for given city pair are stored in the navigation data city pair file. However if there is no enough coverage, the calculation will be rejected and an error message "Exceeding maximum diversion time ..." will be displayed. Until Navigation Service department add the required en-route ports, Dispatcher should check and add approved airports to have coverage and be legal for

dispatching the flight. Refer to “OMC” Chapter 11 for list of authorized airports that can be used by Gulf Air.

Note: If the available ETOPS alternates are not suitable, plan a Non-ETOPS route.

8. ETOPS OFP Preparation

- A. Check aircraft status for any MEL/CDL to determine ETOPS capability
- B. Prepare OFP (Refer FDP-11)
- C. Check weather for ETOPS en-route alternates
- D. Check NOTAM to determine adequacy for ETOPS operation
- E. Change "EROPS" airports when it is required due to NOTAM, WX and/or Operational reason as follows:
 - a. Go to "EROPS" page
 - b. Review ETOPS alternate
 - c. If change is required, remove the airport that is not required from the "EROPS" port list and Calculate
 - d. If adding airport is required, type the IATA or ICAO code in the EROPS airport list field
 - e. If searching for new airport is required, follow the following procedure:
 - Go to “EROPS” page
 - Go to “Routing” box and select waypoint and check range from en-route port
 - Edit the search radius in “Search” box; Distance shall not exceed the maximum ‘THT’ or ‘MDT’ as approved for the aircraft
 - Select approved airport within range and double click to select it
 - Click “Calc” to check EROPS coverage; OR click “Map” and mark the “EROPS” box to view diagram.
- F. Release OFP

9. Flight Watch Procedure

- A. Monitor that the aircraft sends ETOPS departure message (within one hour of departure), via RADIO, ACARS or SITA message from agent incontact with the flight. Otherwise attempt to contact the aircraft via available means of communication.
 - Record the details (ATD, ZFW, Rump Fuel, ETA Entry/Exit, etc.)
 - Read back the message to the crew for confrmation
 - Compare OFP and crew figures and alert the crew if there are significant differences
- B. Monitor the weather and NOTAM for the noiminated ETOPS alternates to ensure compliance with Adequate / Suitable criteria. ETOPS planning minima do not apply in flight; instead weather condition must be above the normal Landing Minima.
- C. Inform crew any new relevant information received during the flight
- D. Monitor receipt of ETOPS exit message and record details. If not received, attempt to contact the aircraft via HF/ACARS.

Date:	01/04/08	Page	84 of 91	FDP-18
Authority:	MFD	Issue	2	Rev 00

- E. If unable to contact, check the aircraft position from the OFP detail to call the closest ATC centre. Request the ATC centre to contact the aircraft and ask to call Bahrain Flight Dispatch. ATC contact numbers are listed at the beginning of each 'ERA' selection table. Or, if there is another GF flight around the area that can be contacted, the message for contacting the ETOPS flight can be transmitted.

10.ETOPS Weather Minima Table

Refer to 'APPENDIX- B' in FDMP - Section 12.

Warning: The JEPPESEN approach plate for the airport to be referred to analyse the minima, when the airport navigation aid is unserviceable and/or down graded and/or wind component is not permitting on the selected RWY direction.

11. Air Traffic Control Centre Contacts

AIR TRAFFIC CONTROL CENTRE CONTACTS	
FIR REGIONS	ACC TELEPHONES
MUSCAT	(968) 24-519550/519551 (TWR)
MUMBAI	(91) 22 – 2682 8004/2682 8002 (FIC)
CHENNAI	(91) 44 – 2256 1515 (Ask for ACC)
KOLKATA	(91) 33 – 2550 2998/2511 9520
COLOMBO	(94) 1 – 252455
YANGON	(95) 1 - 666539/663814 (TWR)
KULA LUMPUR	(60) 3 - 7847 3573/8787 4230 (TWR)
JAKARTA	(62) 21 - 5506176/5506102 (TWR)

Abbreviations:

ACARS	Airborne Communications Addressing and Reporting System
AFM	Airplane Flight Manual
AMSL	Above Mean Sea Level
BRS	Best Route Selection
CDL	Configuration Deviation List
DH	Decision Height
ERA	ETOPS En-Route Alternate
ETA	Estimated Time of Arrival
ETP	Equal Time Point
FOB	Fuel on Board
HF	High frequency
ISA	International Standard Atmosphere
MDH	Minimum Descent Altitude
MEL	Minimum Equipment List
MSA	Minimum Safe Altitude
MTOW	Maximum Regulated Take-Off weight
NAVAID	Navigational Aid
NOTAM	Notice to Air Men
OFP	Operational Flight Plan
RFF	Rescue and Fire Fighting
SIGWX	Significant Weather
TAFOR	Terminal Area Forecast

Responsibilities: Manager Flight Dispatch, Manager Navigation Services and Aircraft Performance, Bahrain Maintenance Control Centre

	Operations Procedures Manual	FDP-19 Pilot Self Service
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Title:	Pilot Self Service
Type:	Flight Dispatch
Purpose:	To enable operating cockpit crew to retrieve briefing package for assigned flights.
Scope:	Flight Dispatch, Operating Flights, Navigation Services, FWZ, GF-IT
References:	OMA Section 8.1.10.6

Procedure:

The PSS is a service to let the pilot access mfs:2 system to retrieve the briefing package assigned to them.

Procedure to assign BP to PSS:

- Complete the BP preparation by sending to the proper address and file ATS plan.
- Activate the 'PSS' from the 'Flight' page by marking the box to enable access to the system.
- Flight should be unselected, by activating another flight, to avoid locking when the pilot access via PSS.
- View flights assigned to 'PSS' to check:
 - go to 'List' page
 - click 'Filter' field, and
 - select 'PSS'
- Flight Dispatch will assist crew to get BP through PSS when required. Use the PSS handout prepared for pilots to refer to page required for explanation
- DCS ZFW update:
 - Check DCS ZFW around 2 hrs before ETD
 - If DCS ZFW is more than 1000kg or less than 3000kg against the OFP, insert the new ZFW in mfs2 and recalculate OFP
 - If DCS ZFW is not ready until crew report, brief crew to plan fuel as s/by
 - Print and attach all DCS ZFW information for verification
 - Signed copy of OFP to be attached and filed
- If crew unable to print BP due to various reason, Flight Dispatcher will print from his/her station and hand over to crew

Date:	01/04/08	Page	87 of 91	FDP-19
Authority:	MFD	Issue	2	Rev 00

Note: Do not assign flight to PSS if OFP calculation is affected by one of the following:

- ❖ Manual FL change due to Wind Shear
- ❖ High Speed plan
- ❖ Route restriction
- ❖ NOTAM that requires urgent attention (Fuel, Airport/Airspace Closure, Strike, etc)
- ❖ Significant WX information
- ❖ Mfs2 system failure

Abbreviation:

BP	Briefing Package
DCS	Departure Control System
FL	Flight Level
FWZ	Flugwerkzeuge
OFP	Operational Flight Plan
Mfs2	Modular Flight Planning System
NOTAM	Notic to Air Men
PSS	Pilot Self Service
WX	Weather
ZFW	Zero Fuel Weight

Responsibilities: Manager Flight Dispatch, Manager Navigation Services and Aircraft Performance, FWZ, Operating Crew

	Operations Procedures Manual	FDP-20 Mfs2 Warning
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Title:	Mfs2 Warning
Type:	Flight Dispatch
Purpose:	To ensure all warning received from the flight planning module is monitored
Scope:	Flight Dispatch, Operating Flight, FWZ
References:	Flight Dispatch Manual

Procedure:

Warning and Flight Dispatcher Action:

The system is set to provide warning at different level of the flight before departure until the aircraft lands. Dispatcher should monitor the pop-ups and the color indicator for the warning for necessary quick action.

Warning can be identified as a pop-up messages or change of the flight color into RED.

To see the warning: right-mouse click the flight and select 'Show Warnings' to display the warning specific to the active flight.

Action required after verifying the warning:

A. LOADCHG Warning:

- When there is a Load/ZFW warning, Dispatcher should verify the information and re-calculate the flight plan. The revised OFP that is released to the station should be communicated to the handling agent for correct OFP delivery.
- If there is a change of route on the new OFP and there is enough time to replace it with previous issued OFP, original ATS flight plan should be cancelled followed by re-filing the new plan on the next couple of minutes with remark on Item 18 of the AFP: 'RMK/RTE CHG'.

B. OFPSTAT Warning:

- Check if OFP is not released, and if not
- Prepare OFP and the rest of the BP and release it.
- File ATS plan

File ATS plan for ATC not filed message if it is not filed before or there is a change that is required to advise the ATC.

C. ACCHG/ Sector Change Warning:

- Recalculate OFP and release it along the rest of BP
- File ATS flight plan

D. STD Change Warning:

- New ETD should be analysed against latest WX information and a new OFP should be prepared if a new WX is updated.

Date:	01/04/08	Page	89 of 90	FDP-20
Authority:	MFD	Issue	2	Rev 00

- NOTAM should be checked for new ETD due to possible shift of time window for route restriction. This can be carried out by recalculation of flight plan with the new ETD to verify en-route restriction.
- Check ATC plan filing

E. MET Warning:

- Dispatcher should analyse the WX
- If WX is affecting and aircraft is on the ground, new OFP will be calculated and information will be passed to crew
- If it is after airborne:
 - ✓ Get aircraft position and actual information (AZFW/FUEL/FL)
 - ✓ Dest WX: En-route OFP to be calculated with distant alternate
 - ✓ Altn WX: New alternate to be nominated and en-route OFP to be calculated for fuel analyses.
 - ✓ WX information and possible plan to be passed to the crew

F. NOTAM Warning:

- If Destination airport is closed before departure, advise IOC and discuss the possibilities of actions required. If it is decided to depart on time with extra holding fuel, flight plan calculation should cater the minimum required extra holding fuel to avoid short of fuel.
- If Destination airport is closed after departure, advise the flight and IOC for next possible action (to continue or divert depend on the remaining fuel and/or reason of closure)
- If destination alternate is closed, nominate another airport.
- If en-route alternate is closed, change the alternate airport. If this is not possible, route should be changed to have the minimum ETOPS/Non-ETOPS en-route diversion range.

Note: Aircraft can be contacted after airborne through the following possible means:

- ACARS
- HF/VHF
- The nearest Area Control Centre (Contact the respective airport manager for ATC contact number)
- Other GF aircraft around the area of operation

Abbreviation:

ACARS	Airborne Communications Addressing and Reporting System
AZFW	Actual Zero Fuel Weight
FL	Flight Level
HF	High Frequency
Mfs2	Modular Flight Planning System Version 2
OFP	Operational Flight Plan
WX	Weather
VHF	Very High Frequency

Responsibilities: Manager Flight Dispatch, FWZ

Date:	01/04/08	Page	90 of 90	FDP-20
Authority:	MFD	Issue	2	Rev 00